Harmonic Analysis

— *EYNTKA* Series —

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September 8, 2026

Contents

0.0.1 Reading the numbering A star marks *bonus material*: a chapter, section or subsection that belongs in the book but that nothing later depends on. Skipping it costs nothing downstream. The star sits on the number, as in §2.7⋆; in divisions written before this convention it sits at the front of the title instead, as in **Uniform Spaces*. The two mean the same thing.

Introduction

Fourier's claim, in 1807, was that an arbitrary function on an interval could be written as a sum of sines and cosines. Lagrange did not believe it, and the committee that reviewed the memoir was right to hesitate: as stated the claim is false, and the century that followed was largely spent finding out in what sense it is true. What survived is better than what was claimed. Decomposing a function into frequencies turned out to be the observation that translation-invariant operators are all diagonal in the same basis. Differentiation, convolution, and the heat and wave propagators commute with translation, and the Fourier transform is precisely the change of coordinates that makes every one of them multiplication by a function. That is why an equation becomes an algebraic identity when transformed, and it is the reason this subject sits underneath analysis rather than beside it.

Two prerequisites are used and not rebuilt. The invariant measure that makes $\int_G f\, d\mu$ meaningful is Haar measure, constructed in *Real Analysis* [14]. The topological-vector-space foundations of distribution theory, meaning test functions, the distributional derivative and the $W^{k,p}$ scale, are *Functional Analysis* [13]. This book uses only the Fourier-facing layer of that theory and cites the rest.

Goal

Chapter 1 develops the transform on $\mathbb{R}^n$ and the series on $\mathbb{T}^n$ together, since they are the same construction over two different groups, and settles the convergence question the subject opened with: in what sense a Fourier series represents the function it came from, and why L^2 is the setting in which the answer is cleanest.

Chapter 2 widens the domain. The transform is badly behaved on L^p for $p > 2$ and undefined on functions that do not decay, and the repair is to enlarge the class of objects rather than restrict the class of functions: the Schwartz space is stable under the transform, so its dual, the tempered distributions, carries the transform automatically. Everything that follows lives there. Singular integrals and Calderón-Zygmund theory answer which operators remain bounded on L^p once the kernel fails to be integrable, the Sobolev scale H^s recasts differentiability as decay of the transform, and the closing section applies both to partial differential equations, which is where the subject's classical motivation ends up.

Throughout Part I the group underneath is $\mathbb{R}^n$ under translation, and nothing is used about it except its translations. Chapter 3 therefore replaces it. On an arbitrary locally compact abelian group the frequencies ξ become the characters, the dual group $\hat{G}$ collects them, and Haar measure gives the integral the transform is defined by. The structural theorems of the first part hold unchanged: inversion, Plancherel, and Pontryagin duality $\hat{\hat{G}} \cong G$, which says that passing to frequencies loses nothing. The classical theory is the case $G = \mathbb{R}^n$, and $\mathbb{T}$, $\mathbb{Z}$ and $\mathbb{Q}_p$ are the same theorem at other groups.

What the reader will be able to do

By the end of this book, the reader should be able to:

1. Move a problem between the spatial and frequency sides deliberately, knowing which function space the transform is being taken on and what is true there (L^1, L^2, $\mathcal{S}$, or $\mathcal{S}'$) rather than transforming formally and hoping.
2. Recognize a singular integral operator, and know what has to be checked for it to be bounded on L^p: the Calderón-Zygmund decomposition, weak-type $(1, 1)$, and interpolation.
3. Use the H^s scale as a statement about decay of the transform, and read a Sobolev embedding or an elliptic regularity estimate as a fact about frequencies rather than derivatives.
4. State and use Pontryagin duality, and recover the classical inversion and Plancherel theorems as the case $G = \mathbb{R}^n$, including recognizing $\mathbb{T}$, $\mathbb{Z}$, and $\mathbb{Q}_p$ as instances of the same theorem.

Omitted Material

- **The non-abelian theory.** Everything in Part II assumes G abelian, which is exactly what makes the characters one-dimensional and the dual a group. Dropping commutativity replaces characters by higher-dimensional unitary representations and the dual group by a space with no group structure. That theory, comprising the group C^*-algebra, Gelfand-Raikov, direct integrals and the abstract Plancherel theorem, is *Unitary Representation Theory* [10], for which this part is the abelian model. Folland's *A Course in Abstract Harmonic Analysis* covers both sides in one volume.
- **The foundations of distribution theory.** Test functions, the distributional derivative, convolution, and $W^{k,p}$ are developed in *Functional Analysis* [13], which owns the Fréchet and LF machinery they rest on. Only the tempered layer is built here.
- **The theory of PDE proper.** The transform is applied to differential equations here as an illustration of what it makes possible. Existence, uniqueness and regularity theory are *Linear Elliptic and Parabolic PDE* [4] and *Nonlinear Wave and Dispersive PDE* [8]. The phase-space refinement, in which the frequency decomposition is localized in space as well, is *Microlocal Analysis* [7].

- **Time-frequency analysis and wavelets.** Windowed transforms, Gabor frames and wavelet bases answer a question this book does not ask, namely how to localize in space and frequency simultaneously, and they have narrow application for a first reading of the classical theory. Daubechies's *Ten Lectures on Wavelets* is the standard reference.

- **Automorphic forms and the trace formula.** Harmonic analysis on the quotient of a group by an arithmetic subgroup is where this subject meets number theory. It needs the non-abelian theory first, and it is taken up in *Local Langlands for* $GL_2(F)$ [5].

Part I

Fourier Analysis and Hard Analysis on $\mathbb{R}^n$

The first part is Fourier analysis in its classical, Euclidean form together with the hard analysis it powers. It opens with the Fourier transform and Fourier series on $\mathbb{R}^n$ and $\mathbb{T}^n$ and their convergence theory, extends the transform to the tempered distributions on which it acts most naturally, and develops the operators this makes available: singular integrals and Calderón-Zygmund theory, the Sobolev scale, and applications to partial differential equations. Throughout, the group underneath is $\mathbb{R}^n$ under translation; the abstract theory of part II is what remains when that group is replaced by an arbitrary locally compact abelian one. The foundational theory of distributions as a topological vector space is developed in [13]; this part uses only its Fourier-facing layer.

1

Fourier Analysis

In the 19th century, Joseph Fourier was trying to find out how to model the movement of heat through solid bodies, in particular model how heat seems to dissipate and homogenise within its medium. For simplicity, let's say this body was a rode, and we can represent any point on the rode with the interval $[a, b]$. If $u : [0, t] \times [a, b] \to \mathbb{R}$ is a function that at any given point $c \in [a, b]$ and at any given time $t_0 \in [0, t]$ give you the temperature, then Fourier devised that the function u must respect the following differential equation:

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$$

which can be interpreted that in any movement in time for some point $c \in [a, b]$, the temperature at c will equal to the *average* of its neighbor (see 3Blue1Brown's video on Fourier analysis for an excellent visualization of this). After choosing the value at $u|_{t_0=0}$ which represent the initial distribution of heat in the rode $[a, b]$, as t_0 changes $u|_{t_0}$ represents the temperature of the rod at every point at time t_0 . Working with this equation, Fourier saw that trigonometric identities are a solution for u given the restraint (with suitable scaling[1]). The next insight Fourier had was that a huge amount of functions can be interpreted as an infinite sum of trigonometric functions, and even better, he found how to find the appropriate coefficients for the trigonometric functions! In particular, this family of equations is *dense* in the space of all L^2 functions on a compact interval (or even better, they form an orthonormal basis), and since the PDE in question is *linear*, we can use the Dominated convergence theorem to show that the resulting sequence can mean that $u(0, \cdot)$ can be any L^2-integrable function (or really, simply integrable since the domain is compact). This is known as the *Fourier Series*, and going form a function to its Fourier series is called the *Fourier Transform.*

This was the beginning of Fourier Analysis, also known as Harmonic Analysis[2]. Since then, it was

[1] In particular, some care must be given to the boundary conditions, but this is something that would be covered in more detail in a proper class in PDE's

[2] Sometimes, Harmonic Analysis is used as a generalization of Fourier Analysis in the case of non-abelian groups, something which will be made more clear in this chapter

asked what spaces of functions admit a Fourier transform, can we go back and forth between the Fourier series and the original function, and can we generalize the domain of the functions (this will be pushed to be compact abelian group, and more generally locally compact groups). The great advantage of being able to find the space of functions that have a Fourier transform is that (under suitable conditions) derivatives work really well with infinite series, trigonometric functions have some of the simple derivatives (they are periodic of order 4), and the Fourier Transform allows us to transfer our problem into a potentially nice space (similarly to how in Representation Theory, it is much easier to find the representation of the monster group to study it rather than directly study it).

These days, Harmonic Analysis has found a large range of applications, from solving my different types of PDE's, to having connections to algebraic number theory (the pursuit of this connection is known as *Langlands Program*). In this chapter, we will cover the elementary theory required to understand the theory of Fourier analysis.

1.1 Basic Definitions for Fourier Analysis

In this chapter, we will always work with $\mathbb{R}^n$ with the Lebesgue measure, denoted m. If $E \subseteq \mathbb{R}^n$ is measurable, $L^p(E)$ (or $L^p(E, m)$) is the set of all L^p functions from E to $\mathbb{R}$, $C^k(E)$ is the set of all k-times continuously differentiable functions from E to $\mathbb{R}$, $C^\infty(E)$ (the set of all smooth functions form E to $\mathbb{R}$) is equal to $\cap_{k=1}^\infty C^k(E)$, and $C_c^\infty(E)$ is the set of all smooth function whose compact support is in E. If $E = \mathbb{R}^n$, we will often drop the domain of the function space, so:

$$L^p = L(\mathbb{R}^n), \qquad C^\infty = C^\infty(\mathbb{R}^n) \qquad C_c^\infty = C_c^\infty(\mathbb{R}^n)$$

The inner product and norm on $\mathbb{R}^n$ is the usual dot product and euclidean norm:

$$x \cdot y = \sum_i^n x_i y_i \qquad \|x\| = |x| = \sqrt{x \cdot x}$$

Derivative play a prominent role in this chapter, and so we establish important notation. We will assume the standard basis and write:

$$\partial_j = \frac{\partial}{\partial x^j}$$

and for higher order mixed derivatives, we will use a multi-index, which is an ordered n-tuple $\alpha = (\alpha_1, ..., \alpha_n)$. As usual:

$$|\alpha| = \sum_1^n \alpha_i \quad \alpha! = \prod_1^n \alpha_i! \quad \partial^\alpha = \left(\frac{\partial}{\partial x_i}\right)^{\alpha_1} \cdots \left(\frac{\partial}{\partial x_n}\right)^{\alpha_n}$$

and for any $x = (x_1, x_2, ..., x_n) \in \mathbb{R}^n$:

$$x^\alpha = \prod_1^n x_i^{\alpha_i}$$

Be aware that $|\alpha|$ and $|x|$ do not means the same thing; the meaning should be clear from context. For reference, the product rule on a multi-index expands to:

$$\partial^\alpha(fg) = \sum_{\beta+\gamma=\alpha} \frac{\alpha!}{\beta!\gamma!} (\partial^\beta f)(\partial^\gamma g)$$

A common function we'll use a lot in C_c^∞ is:

$$\psi(x) = \eta(1 - |x|^2) = \begin{cases} \exp((|x|^2 - 1)^{-1})) & |x| < 1 \\ 0 & |x| \geq 1 \end{cases}$$

where $\eta(t) = e^{1/t}\chi_{(0,\infty)}(t)$. Another common subset of C^∞ that will be of use is the following which ensures that a function and all its derivatives decay faster than polynomial time:

Definition 1.1.1: Schwartz Space

The set $\mathcal{S} \subseteq C^\infty$ is the set of all smooth functions and all their derivatives which vanish faster than any power of $|x|$, that is, for any $N \in \mathbb{N}$ and any multi-index α, if we define:

$$\|f\|_{(N,\alpha)} = \sup_{x \in \mathbb{R}^n} (1 + |x|)^N |\partial^\alpha f(x)|$$

Then:

$$\mathcal{S} = \left\{ f \in C^\infty \;\middle|\; \|f\|_{(N,\alpha)} < \infty \text{ for all } N, \alpha \right\}$$

Example 1.1.2: Schwartz Functions

1. Naturally, $C_c^\infty \subseteq \mathcal{S}$
2. $f_\alpha(x) = x^\alpha e^{-|x|^2}$ for any multi-index α is in $\mathcal{S}$

Note that if $f \in \mathcal{S}$, then $\alpha f \in L^p$ for all α and all $p \in [1, \infty]$ since $|\partial^\alpha f(x)| \leq C_N(1 + |x|)^{-N}$ for all $N \in \mathbb{N}$, and $(1 + |x|)^{-N} \in L^p$, for all $N > n/p$

Proposition 1.1.3: $\mathcal{S}$ is a Fréchet Space

The Schwartz space $\mathcal{S}$ is a Fréchet space with the topology defined by the norms $\|\cdot\|_{(N,\alpha)}$

***Proof*:**
All is easy to prove with the possible exception of completeness. Let $\{f_k\}$ be a cauchy sequence in $\mathcal{S}$ so that $\|f_i - f_j\|_{(N,\alpha)} \to 0$ for all N, α. In particular, for each α the sequence $\{\partial^\alpha f_k\}$ converges uniformly to some g_α. Letting e_i denote the standard ith basis vector, we get:

$$f_k(x + te_i) - f_k(x) = \int_0^t \partial_i f_k(x + se_i) ds$$

Letting $k \to \infty$, we get:

$$g_0(x + te_i) - g_0(x) = \int_0^t g_{e_i}(x + se_i) ds$$

and the fundamental theorem of calculus shows that $g_{e_i} = \partial_i g_0$, and so by induction we get $g_\alpha = \partial^\alpha g_0$ for all α. From this, it is now easy to verify that $\|f_k - g_0\|_{(N,\alpha)} \to 0$ for all α and $N \in \mathbb{N}$.

A useful characterization of $\mathcal{S}$ is also the following:

Proposition 1.1.4: Characterizing Schwartz Functions

If $f \in C^\infty$, Then $f \in \mathcal{S}$ if and only if $x^\beta \partial^\alpha f$ is bounded for all multi-indices α, β, if and only if $\partial^\alpha(x^\beta f)$ is bounded for all multi-indices α, β.

***Proof*:**
If $f \in \mathcal{S}$, then naturally $|x^\beta| \leq (1+|x|)^N$ for $\beta \leq N$. Conversely, notice that $\sum_1^n |x_i|^N$ is strictly positive on the unit sphere $|x| = 1$ (which is compact), and hence has a positive minimum, say δ. It follows that $\sum_1^n |x_i|^N \geq \delta |x|^N$ for all x since both sides are homogeneous of degree N, and hence:

$$(1+|x|)^N \leq 2^N(1+|x|^N) \leq 2^N \left[1 + \delta^{-1} \sum_1^n \left|x_i^N\right|\right] \leq 2^N \delta^{-1} \sum_{|\beta| \leq N} |x^\beta|$$

Giving the first equivalence. For the second equivalence, notice that for every α, $\partial^\alpha(x^\beta f)$ is a linear combination of terms of the form $x^\gamma \partial^\delta f$ and vice versa, by the product rule, completing the proof.

Next we define a periodic function to be a function for which $f(x+k) = f(x)$ for all $x \in \mathbb{R}^n$ and $k \in \mathbb{Z}^n$. Without loss of generality the period of the functions will be of size 1. Thus, we can consider the space on which we define periodic functions to be $\mathbb{R}^n/\mathbb{Z}^n \cong (\mathbb{R}/\mathbb{Z})^n \equiv \mathbb{T}^n = S^1 \times S^1 \times \cdots \times S^1$, i.e. we can think of defining period functions on an n-torus (note that $\mathbb{T}^1 = S^1$, but $\mathbb{T}^2 \neq S^2$). Naturally, $\mathbb{T}^n$ is a compact Hausdorff space. It can be identified with all points $z = (z_1, z_2, ..., z_n) \in (\mathbb{C}^n)^*$ with $|z_i| = 1$ via the map

$$(x_1, x_2, ..., x_n) \mapsto (e^{2\pi i x_1}, ..., e^{2\pi i x_n})$$

For measure-theoretic purposes, we will sometimes identify $\mathbb{T}^n$ with the unit cube:

$$Q = \left[\frac{-1}{2}, \frac{1}{2}\right)^n$$

and when we talk about the Lebesgue measure on $\mathbb{T}^n$, we will mean the induced measure from Q. This means that $m(\mathbb{T}^n) = 1$. We will think of functions on $\mathbb{T}^n$ as either periodic functions on $\mathbb{R}^n$ or as functions on Q (the context will make it clear which we pick).

Exercise 1.1.1

1. Is the sum of two periodic functions periodic?

1.1.1 Convolution

The convolution $f * g$, Young's convolution inequality, translation-continuity in L^p, the smoothing lemma, approximate identities (good kernels), and the density of smooth functions in L^p are developed as foundational L^p operations in the functional-analysis part, [13, Convolution and Approximate Identities]. Here we record only the fact special to the Schwartz class, which the Fourier theory below uses:

Proposition 1.1.5: Convolution in Schwartz Space

Let $f, g \in \mathcal{S}$. Then $f * g \in \mathcal{S}$

***Proof*:**
First, $f * g \in C^\infty$ by [13, convolution preserves smoothness]. Next, we establish the following for the coming chain of inequalities:

$$1 + |x| \overset{\triangle\text{ineq.}}{\leq} 1 + |x - y| + |y| \leq (1 + |x - y|)(1 + |y|)$$

and so:

$$\begin{aligned}(1 + |x|)^N |\partial^\alpha (f * g)(x)| &\leq \int (1 + |x - y|)^N |\partial^\alpha f(x - y)| (1 + |y|)^N |g(y)| dy \\ &= \|f\|_{(N,\alpha)} \|g\|_{(N+n+1,\alpha)} \int (1 + |y|)^{-n-1} dy\end{aligned}$$

which is certainly finite

1.2 Fourier Transform

Consider again the heat equation on a compact interval:

$$\frac{du}{dt} = \frac{\partial^2 u}{\partial x^2} \tag{1.1}$$

where u is a solution. If u represents a rod that is at time $t = 0$ half hot on the left and half cold on the right, with the two halves in contact at the midpoint, then as we let time move forward, the temperature will equalize due to diffusion. The shape goes from a step function, to a smoothed profile resembling the error function $\operatorname{erf}(x)$, and as time $\to \infty$ it converges to a constant function representing the average temperature. These are all functions with distinctly different behaviours.

Now, assume that the heat of the rod at $t = 0$ has the shape of $\sin(x)$ or $\cos(x)$. Then as we let time propagate these functions dampen uniformly, notably *they are always the same function*, simply scaled. This is essentially the behaviour of *eigenvectors*, namely as time moves forward we are scaling by a factor. To have eigenvectors, we need a linear transformation between vector spaces. Let S_t be the *time evolution operator* that takes in a state f_{t_0} (a function on the domain that is a t_0-slice of a solution $u(x, t_0)$) and outputs f_{t_0+t}, in particular $S_t : \mathscr{P}(X) \to \mathscr{P}(X)$ where $\mathscr{P}(X)$ is the state space over X and S_t is the evolve by t operator:

$$S_t(u(-, t_0)) = u(-, t_0 + t)$$

Or, viewed as an (abelian) monoid action[3]:

$$S : [0, \infty) \times \mathscr{P}(X) \to \mathscr{P}(X) \qquad t \cdot f_{t_0} = f_{t_0+t}$$

[3]The heat equation is well-posed forward in time but ill-posed backward: reversing diffusion amplifies noise catastrophically (failing Hadamard's conditions for well-posedness; see [4]). Hence S_t forms a monoid $[0, \infty)$ rather than a group—time evolution is irreversible.

Note that it isn't "obvious" what type of space $\mathscr{P}(X)$ is, we will assume that this is a Hilbert space; see [4] for the justification of this assumption. At minimum it is certainly the case that any (smooth) L^2-integrable function is a valid starting state that can then evolve according to the heat equation. If the reader can accept that there is a reasonable sense in which integrable functions can be differentiated as shall be demonstrated in [13, Distribution Theory], the reader can let $\mathscr{P}(X) = L^2([0,1])$.

The choice of L^2 is not arbitrary. As we saw in the preceding section, L^2 is the unique L^p space where the duality map J_p is linear, the geometry is flat, and vectors and covectors are canonically identified. The Fourier transform exploits this: it is an *isometry* of L^2 (Plancherel's theorem), which is possible precisely because the "rotation" from position coordinates to frequency coordinates preserves the inner product. On L^p for $p \neq 2$, the Fourier transform does not stay within the same space; instead, the Hausdorff-Young inequality gives

$$\|\hat{f}\|_{L^q} \leq C\|f\|_{L^p}, \qquad \tfrac{1}{p} + \tfrac{1}{q} = 1, \quad 1 \leq p \leq 2$$

mapping L^p into its dual L^q—reflecting the nonlinear duality of those spaces. Only at $p = 2$ does the transform remain an endomorphism, because only there is $V \cong V^*$ linearly.

Now, certainly S_t is linear as the heat equation is linear and homogeneous[4]. We can thus ask if it has eigenvectors, that is functions $f_r \in \mathscr{P}(X)$ where

$$S_t f_r(x) = \lambda_t f_r(x)$$

If we let t vary, we can define a function $\lambda(t)$ dependent on the time. Given a compact domain $X = [a,b]$, by direct computations we can check that $\sin(kx)$ is an eigenvector, where this is indeed a state as it is a t-slice of:

$$u(x,t) = e^{-k^2 t}\sin(kx)$$

For example at $t = 0$:

$$S_t(\sin(kx)) = e^{-k^2 t}\sin(kx)$$

With the assumption that $\mathscr{P}$ is a Hilbert space, if these eigenfunctions form an orthonormal basis, we can decompose a function into a "weighted sum" of these well-behaved functions with respect to time. For the rest of this section we shall take for granted $\{\sin(kx)\}_{k\in\mathbb{N}}$ form an orthonormal basis for S_t (proven in theorem 1.2.4) after appropriate normalization depending on the domain.

It must be pointed out that the same argument works for $\cos(kx)$:

$$S_t(\cos(kx)) = e^{-k^2 t}\cos(kx)$$

and these *also* form an orthonormal basis that *also* diagonalizes S_t. These two sets are individually complete orthogonal bases for $L^2([0,1])$ (with appropriate boundary conditions), so their union is *overcomplete*—it contains "twice as many" basis elements as needed. Moreover, the combined set is not even orthogonal:

$$\int_0^1 \sin(\pi x)\cos(2\pi x)dx = \frac{-2}{3\pi} \neq 0$$

More fundamentally, as we shall see, neither set alone diagonalizes the *translation* operator—they only block-diagonalize it—and the passage to complex exponentials will be forced by the requirement of full diagonalization.

[4]It is also continuous and normal, but for that we would need to choose whether we consider $\mathscr{P}$ as a Banach space or a distribution space; intuitively the heat equation dissipates energy so $\|S_t f\| \leq \|f\|$

Thus, the question shifts into one about finding a *natural* eigenbasis for S_t. A-priori, it is not necessarily the case that these are the only eigenbases. What we need is for the eigenbasis to "respect" some inherent geometry of the space. Phrased differently, the eigenbasis needs to be *invariant* with respect to some operator(s), where the operators represent some symmetry of the space. To get a better grasp of invariance of operators, we may start looking at other PDEs and considering diagonalizing the operator S_t with respect to the solutions of those PDEs.

For example, if we solve the Heat equation on a sphere:

$$\frac{\partial u}{\partial t} = \Delta_{\text{sphere}} u$$

the operator can still be diagonalized (see [4]). What we shall want is for an eigenbasis that respects the spherical geometry, namely for it to be invariant to an $SO(3)$ action. Such eigenfunctions do exist, they are the *Spherical Harmonics* $Y_l^m(\theta, \varphi)$, and they are invariant to rotation[5]

$$\Delta(f \circ R) = (\Delta f) \circ R$$

To give another example that is famous in physics, consider the Quantum Harmonic Oscillator (QHO):

$$i\frac{\partial \psi}{\partial t} = -\frac{\partial^2 \psi}{\partial x^2} + x^2 \psi$$

This PDE is diagonalizable (see [4]), but it is *not* diagonalizable by the Fourier basis e^{ikx}. The reason, in the language of our framework, is that the potential $V(x) = x^2$ *breaks translation invariance*: since $x^2 \neq (x+a)^2$, the QHO Hamiltonian $H = -\partial_x^2 + x^2$ does not commute with the translation operator τ_a. By the general principle we are developing—different symmetries produce different eigenbases—the translation eigenbasis cannot diagonalize the QHO.

Instead it was shown to be diagonalized (uniquely) by the *Hermite Functions* (a product of the Hermite polynomials H_n with a Gaussian):

$$\varphi_n(x) = H_n(x) e^{-x^2/2}$$

The relevant symmetry is the *Heisenberg-Weyl algebra* generated by the ladder operators $a = \frac{1}{\sqrt{2}}(x + \partial_x)$ and $a^\dagger = \frac{1}{\sqrt{2}}(x - \partial_x)$, whose eigenbasis is precisely the Hermite functions.

A revealing computation shows *why* the Fourier basis fails here. If we attempt to apply the Fourier transform, the $-\partial_x^2$ term becomes multiplication by k^2 in frequency space, but the x^2 multiplication term becomes a Laplacian $-\partial_k^2$ in k-space. The result is the *identical* PDE in frequency space:

$$i\frac{\partial \hat{\psi}}{\partial t} = k^2 \hat{\psi} - \frac{\partial^2 \hat{\psi}}{\partial k^2}$$

The Fourier transform swaps the PDE for a copy of itself without solving it. In physics, this reflects the fact that the QHO Hamiltonian is *invariant* under the Fourier transform—it treats position and momentum symmetrically—creating a fascinating link between abstract harmonic analysis and the concrete geometry of quantum mechanics, a connection to be explored in another book.

[5] Important technicality: $SO(3)$ is not abelian, hence it cannot be that *all* rotation operators simultaneously commute. The best solution is to pick a "maximal set" of commuting symmetries: usually the Total Angular Momentum (L^2, representing the amount of rotation invariance) and Azimuthal Symmetry (L_z, rotation around the z-axis). This choice of a maximal commuting family is the analytic form of selecting a *maximal torus* (Cartan subalgebra) in the Lie algebra $\mathfrak{so}(3)$.

So what is the invariance we want to impose for the operator S_t with respect to the heat equation? In the example of the sphere, we had the group $SO(3)$. A compact interval has no inherent invariance simply because it is not a group. However, by gluing the end points of the interval we get a 1-*torus* (a circle)[6]. We now have the rotation group operation on $\mathbb{T}^1$, or equivalently the translation operation on $\mathbb{R}/\mathbb{Z}$[7]. This does shift the geometry and hence the inherent dynamics of our PDEs; the problem of solving a PDE on a compact interval will be a subset of solving PDEs on a torus as shall be addressed shortly. Let us first simply consider S_t on $L^2(\mathbb{T}^1)$ and see how it interacts with the translation operator. Since we are working on the torus, we now have that $\sin(2\pi kx)$ and $\cos(2\pi lx)$ *are* orthogonal (where the 2π-scaling was to ensure a full period of each lies within the torus), and $\{\sin(kx), \cos(lx)\}_{k,l\in\mathbb{N}}$ is an orthonormal basis (after appropriate scaling and stretching). Now, note that sin and cos bases are *not* translation invariant, for example:

$$\tau_a(\sin(kx)) = \sin(k(x+a)) = \sin(kx)\cos(ka) + \cos(kx)\sin(ka)$$

and so they are not eigenvectors of τ_a. However, they still *block-diagonalize* the translation operator τ_a with blocks of size 2:

$$\begin{pmatrix} \cos(ka) & \sin(ka) \\ -\sin(ka) & \cos(ka) \end{pmatrix}$$

If we look at these blocks, we shall see that they are all rotation matrices by an angle ka; in other words the imposition is that we did not have enough eigenvalues. By extending the codomain to $\mathbb{C}$, we can readily verify that:

$$\tau_a(e^{iz}) = e^{ia} \cdot e^{iz}$$

and so $\{e^{ikz}\}_{k\in\mathbb{Z}}$ is our desired eigenbasis for τ_a. Now, the question remains whether translations of these functions commute with S_t, namely:

$$S_t(\tau_a(f)) = \tau_a(S_t(f)) \qquad \forall a \in \mathbb{T}^1$$

and indeed:

$$\begin{aligned}
\tau_a(S_t(e^{ikx})) &= \tau_a(e^{-k^2t}e^{ikx}) \\
&= e^{-k^2t}e^{ik(x+a)} \\
&= e^{-k^2t}e^{ika}e^{ikx} \\
&= e^{ika}S_t(e^{ikx}) \\
&= S_t(e^{ika}e^{ikx}) && S_t \text{ is linear} \\
&= S_t(\tau_a(e^{ikx}))
\end{aligned}$$

What this means physically is that the heat equation dissipates the same way no matter where you are on a ring-shaped rod. If the rod was half wood and half metal, then this translation invariance shall fail.

This principle is the spectral-theoretic form of *Noether's theorem*: continuous symmetries of a physical system correspond to conservation laws. Translation invariance corresponds to conservation of *momentum*, and the Fourier decomposition $f = \sum \hat{f}(k)e^{ikx}$ is precisely the decomposition into

[6] More generally, any rectangular domain is homeomorphic to $[0,1]^n$ which then can be glued to become $\mathbb{T}^n = (S^1)^n$, and so this same translation structure appears in higher dimensions

[7] Really $\mathbb{C}/\mathbb{Z}$ as we shall justify shortly

states of definite momentum k. Each Fourier mode evolves independently—its amplitude changes but its frequency is preserved—because momentum is conserved by any translation-invariant dynamics.

Now, notice that as $\tau_a \circ S_t = S_t \circ \tau_a$ for all $a \in \mathbb{T}^1$ (or in commutator bracket notation $[\tau_a, S_t] = 0$ for all a, t), the eigenvectors of τ_a are the same eigenvectors as S_t. The eigenvectors of S_t are in general hard to compute, but the eigenvectors of τ_a are *precisely* the functions e^{ikx}. In fact, any (linear) operator L where $\tau_a \circ L = L \circ \tau_a$ will have the same eigenvectors. Thus, we are actually finding eigenvectors of the translation operators τ_a and then finding transformations that commute with translation. Note that any Linear Differential Operator with Constant Coefficients

$$L = a_n \frac{d^n}{dx^n} + \cdots + a_1 \frac{d}{dx} + a_0$$

and any convolution operator:

$$L(f) = f * g = \int f(y)g(x-y)dy$$

will commute with translation. This is also why the Heat Equation, Wave Equation, and Schrödinger Equation can all be solved (or transformed) using the Fourier transform.

From Local to Global: The Exponential Map

We have established that the Fourier basis functions e^{ikx} are the eigenfunctions of the "global" spatial translation operator τ_a. However, the reader likely recognizes these functions primarily as the solutions to the differential equation $f' = \lambda f$. This connection between the *global* translation τ_a and the *local* derivative $D = \frac{d}{dx}$ is a central theme of Lie Theory, and it provides the structural explanation for *why* exponentials are the natural Fourier basis.

In [13, Distribution Theory], it shall be shown that it makes sense to infinitely differentiate integrable functions. For now, we rely on the intuition that any continuous function can be approximated by a smooth function (and locally by a polynomial).

Consider the Taylor expansion of a function f around a point x:

$$f(z) = f(x) + f'(x)(z-x) + \frac{f''(x)}{2!}(z-x)^2 + \dots$$

If we wish to evaluate f at a shifted point $x + y$, the Taylor series gives:

$$f(x+y) = f(x) + yf'(x) + \frac{y^2}{2!}f''(x) + \dots$$

Notice that we can re-write this using the differentiation operator $D = \frac{d}{dx}$:

$$f(x+y) = \left(I + yD + \frac{y^2 D^2}{2!} + \dots \right) f(x)$$

Recognizing the series inside the parenthesis as the definition of the matrix exponential, we arrive at the compact operator identity:

$$\tau_y = e^{yD}$$

This equation states that *differentiation generates translation.*

Now, consider the spectral consequences. If an operator A has an eigenvector v with eigenvalue λ ($Av = \lambda v$), then the operator exponential e^A acts on v as:

$$e^A v = \left(I + A + \frac{A^2}{2!} + \dots\right) v = \left(1 + \lambda + \frac{\lambda^2}{2!} + \dots\right) v = e^\lambda v$$

Applying this to our spatial operators: since $f(x) = e^{ikx}$ is an eigenfunction of the derivative D with eigenvalue ik (i.e., $De^{ikx} = ike^{ikx}$), it must automatically be an eigenfunction of the translation operator $\tau_y = e^{yD}$ with eigenvalue e^{iyk}.

This confirms our earlier finding: the Fourier basis e^{ikx} simultaneously diagonalizes the derivative (and by extension the Laplacian $\Delta = D^2$) and the translation group.

Theorem 1.2.1: Eigenvector Mapping Lemma

Any eigenvector with eigenvalue λ of an operator A is an eigenvector of the operator e^A with eigenvalue e^λ.

***Proof*:**
Follows from the power series definition of the matrix exponential shown above.

Remark. The full *Spectral Mapping Theorem* is the stronger statement $\sigma(e^A) = e^{\sigma(A)}$ for the entire spectrum, not just eigenvalues. The eigenvector case stated above is the special case sufficient for our purposes.

To see this explicitly, observe the action of these operators on the basis function $f_k(x) = e^{ikx}$:

- *Derivative:* $Df_k = (ik)f_k$ (Eigenvalue ik)
- *Laplacian:* $\Delta f_k = D^2 f_k = (ik)^2 f_k = -k^2 f_k$ (Eigenvalue $-k^2$)
- *Translation:* $\tau_a f_k = e^{aD} f_k = e^{a(ik)} f_k$ (Eigenvalue e^{ika})

In the language of representation theory, the differentiation operator is the *Lie Algebra* generator $\mathfrak{g}$, and the translation operator is the *Lie Group* element $e^{\mathfrak{g}}$. The Fourier basis provides the unique coordinate system where the group action and its generator are both simple scalings.

The Fourier Variable as a Cotangent Coordinate. The eigenvalue list above reveals a deeper geometric structure that connects directly to the duality theory presented in chapter [13, Hilbert Space]. Observe the pattern: the derivative $D = \frac{d}{dx}$ is a *tangent* operation (it acts on the spatial/position variable x), yet its eigenvalue ik is a scalar that labels the *frequency* k. In physics, x is position and k is momentum—and these are canonically dual variables, related as tangent and cotangent coordinates.

The Fourier transform makes this duality explicit. Under the transform:

- The derivative $\frac{d}{dx}$ (a tangent operation) becomes *multiplication by ik* (a cotangent scalar).
- Multiplication by x (a tangent/position operation) becomes $-i\frac{d}{dk}$ (a cotangent derivative).

The Fourier transform *exchanges tangent and cotangent roles.* In the language of symplectic geometry, it is a canonical transformation swapping position and momentum coordinates on the phase space $T^*\mathbb{R}$. Where the duality map J_p from chapter [13, Hilbert Space] provides a *pointwise, infinitesimal* passage from tangent to cotangent data (via the gradient of the energy), the Fourier transform achieves this passage *globally* via integration against the characters e^{ikx}.

This perspective also explains the QHO's Fourier invariance noted above. The QHO Hamiltonian $H = -\partial_x^2 + x^2$ maps under the Fourier transform to $k^2 - \partial_k^2 = -\partial_k^2 + k^2$: the identical operator with x replaced by k. This is precisely because H treats position and momentum *symmetrically*—it sits on the diagonal of the tangent-cotangent duality, which is why the Fourier transform (which swaps the two) leaves it invariant.

Boundary Problem

Having understood why the Fourier basis is structurally natural, we now examine two important practical consequences: how boundary conditions on an interval arise from the symmetry of the torus, and how the Fourier transform decouples partial differential equations.

The immediate point that must be clarified when switching to $\mathbb{T}^1$ is that this shifts the dynamics of any PDE we are solving, as now points at the boundary are affected by their neighbourhood. However, for PDEs that will respect the group structure of $\mathbb{T}^1$, the effects at the boundary can be "cancelled out" in two separate ways that shall recover the original two bases $\{\sin(kx)\}_{k\in\mathbb{Z}}$ and $\{\cos(kx)\}_{k\in\mathbb{Z}}$ we saw before (again, appropriately normalized). The key insight is that the interval $[0,1]$ can be viewed as a *sub-domain* of the torus $\mathbb{T}^1$ (identified with $[-1,1]$) subject to a symmetry constraint. While the torus possesses full translation invariance, the interval possesses *reflection invariance.* By imposing a symmetry on the torus, we create "virtual boundaries" at the fixed points of the reflection ($x = 0$ and $x = 1$). This gives rise to two distinct orthonormal bases for $L^2([0,1])$, each corresponding to a different physical boundary condition:

- *Odd Symmetry (Dirichlet Conditions):* Consider the subspace of functions on $\mathbb{T}^1$ that are *odd*, satisfying $u(-x) = -u(x)$. At the fixed point $x = 0$, this symmetry forces $u(0) = -u(0)$, implying $u(0) = 0$. Physically, the heat flowing from the right is perfectly cancelled by the "negative heat" flowing from the left, creating a virtual cold wall. The eigenfunctions respecting this symmetry are precisely the sine functions:

  $$\sin(n\pi x) = \frac{e^{in\pi x} - e^{-in\pi x}}{2i}$$

  Restricting these functions to $[0,1]$ yields the *Sine Basis*, which diagonalizes the Laplacian subject to $u(0) = u(1) = 0$.

- *Even Symmetry (Neumann Conditions):* Consider the subspace of functions on $\mathbb{T}^1$ that are *even*, satisfying $u(-x) = u(x)$. At the fixed point, the derivative must be odd, forcing $u'(0) = 0$. Physically, this represents an insulated boundary where heat reflects back without loss. The eigenfunctions respecting this symmetry are the cosine functions:

  $$\cos(n\pi x) = \frac{e^{in\pi x} + e^{-in\pi x}}{2}$$

  Restricting these to $[0,1]$ yields the *Cosine Basis*, which diagonalizes the Laplacian subject to $u'(0) = u'(1) = 0$.

This leads to a result in Harmonic Analysis known as *half-range expansions*: while sines and cosines *together* form a basis for the full torus, *either* set alone forms a complete orthonormal basis for the interval $[0, 1]$.

Decoupling PDE

To understand the consequences better, take a solution $u : \mathbb{T}^1 \times [0, \infty) \to \mathbb{C}$. Then, think of each (space) neighborhood in the usual heat equation as influencing its infinitesimal neighbourhood, namely the $\frac{\partial^2 u}{\partial x^2}$ term determines how each neighbouring point affects each other as time propagates. We can think of this as being uncountably many dependent equations.

Let us now change the "basis" of our state space from the position basis $\{\delta_x\}_{x\in\mathbb{T}^1}$ to the eigenbasis of the spatial operator $\{e^{ikx}\}_{k\in\mathbb{Z}}$. By taking the Fourier transform, i.e. transforming to the translation-invariant orthonormal basis, we change a solution to:

$$u(x, t) : \mathbb{T}^1 \times [0, \infty) \to \mathbb{C} \qquad \xmapsto{\mathcal{F}_x} \qquad \hat{u}(k, t) : \mathbb{Z} \times [0, \infty) \to \mathbb{C}$$

where the $\mathbb{Z}$ is the index for the eigenbasis $\{f_k\}_{k\in\mathbb{Z}}$. Then if $\hat{u}_k$ is the function whose input is the index of the eigenvector and output is its coefficient, we see that as $\hat{u}_k$ propagates through time, $\hat{u}_k(t)$ is only dependent on the original function up to scaling. In fact we shall show that the heat equation PDE decomposes into the equations:

$$\frac{d\hat{u}_k}{dt} = -k^2 \hat{u}_k \qquad \forall k \in \mathbb{Z}$$

where now we have *countably many decoupled* equations.

Non-Compact Domain

Notice that $\sin, \cos, e^z$ are not L^2-integrable on $\mathbb{R}^n$ (resp. $\mathbb{C}^n$), hence these cannot form an orthonormal basis on $L^2(\mathbb{R}^n)$. Thus, the initial foray into Fourier will focus on functions with domain $\mathbb{T}^n$. In section 1.2.2 we shall show how to generalize the tools to the non-compact case of Euclidean space $\mathbb{R}^n$; the passage to a general locally compact abelian group is the subject of part II.

The structural reason for this shift is illuminated by *Pontryagin duality* (section 3.2). The characters of a group G themselves form a group $\hat{G}$ (the *Pontryagin dual*) under pointwise multiplication, and the Fourier transform is a map $L^2(G) \to L^2(\hat{G})$. For the compact group $G = \mathbb{T}^1$, the dual group is discrete: $\hat{G} = \mathbb{Z}$ (indexed by frequency k), and the Fourier transform produces a *discrete* sequence of coefficients $\hat{f}(k)$. Conversely, for the discrete group $G = \mathbb{Z}$, the dual is compact: $\hat{G} = \mathbb{T}^1$. And remarkably, $G = \mathbb{R}$ is *self-dual*: $\hat{\mathbb{R}} \cong \mathbb{R}$, so the Fourier transform maps functions of a continuous variable to functions of a continuous variable, replacing the discrete sum $\sum \hat{f}(k)e^{ikx}$ with the continuous integral $\int \hat{f}(\xi)e^{2\pi i\xi x}\, d\xi$. Pontryagin duality ($\hat{\hat{G}} \cong G$) is then the abstract statement that the Fourier transform is involutive.

Generalizing using Representation Theory

In the preceding discussion, our motivation was to solve a specific time-dependent PDE (the heat equation). We discovered that the key to solving it lay not in the time evolution S_t itself, but in the underlying symmetry of the spatial domain (translation invariance). The basis functions e^{ikx} worked because they diagonalized the translation operator τ_a, which we then understood via the exponential map $\tau_a = e^{aD}$ as the passage from the Lie algebra (differentiation) to the Lie group (translation).

We now abstract this principle. We no longer need a specific "time" equation; instead, we look for the basis that diagonalizes the action of the symmetry group itself. By finding the irreducible representations of the symmetry group, we construct a "universal" eigenbasis that will automatically diagonalize *any* linear operator respecting that symmetry (be it the Laplacian, the Heat Operator, or others).

Recall the representation theory of finite groups. If G is a finite group, we can represent G on a vector space V (over a field k) by taking a homomorphism $\rho : G \to \mathrm{GL}(V)$.

$$G \curvearrowright V \qquad \rho(g)v = g \cdot v$$

As we are not interested in any arithmetic limitations, we take $k = \mathbb{C}$. Then by Maschke's theorem (see [12, chapter 24]) the induced $\mathbb{C}[G]$-module structure on V is semi-simple, meaning:

$$V \cong_G \bigoplus_i V_i$$

where V_i are irreducible representations. Given the basis for the above isomorphism, any $g \in G$ will give a matrix representation $\rho(g)$ which respects this block-diagonal decomposition:

$$\rho(g) = \begin{bmatrix} \mathbf{M}_1(g) & 0 & 0 & 0 \\ 0 & \mathbf{M}_2(g) & 0 & 0 \\ 0 & 0 & \mathbf{M}_3(g) & 0 \\ 0 & 0 & 0 & \dots \end{bmatrix}$$

where $\mathbf{M}_i(g)$ represents the matrix block acting on the subspace V_i. Crucially, if G is *abelian*, Schur's Lemma implies that *all* irreducible representations are 1-dimensional. Hence, the matrices become scalars:

$$\rho(g) = \begin{bmatrix} \lambda_1(g) & 0 & 0 & 0 \\ 0 & \lambda_2(g) & 0 & 0 \\ 0 & 0 & \lambda_3(g) & 0 \\ 0 & 0 & 0 & \dots \end{bmatrix}$$

Since G is abelian, the operators $\{\rho(g)\}_{g \in G}$ commute, and the construction above shows they are *simultaneously diagonalizable* by a single basis.

Now, we generalize this to function spaces. While for a finite group the vector space of functions $\mathbb{C}[G]$ is finite-dimensional, equipping it with an inner product structure is essential to define unitarity and orthogonality. A finite group G carries a natural unique measure (the counting measure), which we can normalize to be a probability measure (the Haar measure $\mu(g) = 1/|G|$). This allows us to define the Hilbert space $L^2(G)$ with inner product:

$$\langle f, h \rangle = \frac{1}{|G|} \sum_{x \in G} f(x)\overline{h(x)}$$

We define the *Right Regular Representation*[8] of G on this space:

$$G \curvearrowright L^2(G) \qquad (R_g \cdot f)(x) = f(xg)$$

[8] or, Left Regular Representations $(L_g \cdot f)(x) = f(g^{-1}x)$

We verify that this action is unitary (and hence can be diagonalized with an orthonormal basis). Since right-multiplication by g is a bijection on the group G (it permutes the elements), the sum over x is identical to the sum over $y = xg$:

$$\begin{aligned}\langle R_g f, R_g h\rangle &= \frac{1}{|G|}\sum_{x\in G} f(xg)\overline{h(xg)}\\ &= \frac{1}{|G|}\sum_{y\in G} f(y)\overline{h(y)}\\ &= \langle f, h\rangle\end{aligned}$$

Thus, the operators R_g are unitary. Since G is abelian, the operators $\{R_g\}_{g\in G}$ form a commuting family of unitary operators. By the spectral theorem for unitary operators, there exists an orthonormal basis $\mathcal{B}$ of $L^2(G)$ that simultaneously diagonalizes the action. If $\varphi \in \mathcal{B}$ is such a basis vector, then:

$$R_g(\varphi) = \chi_\varphi(g)\varphi$$

where $\chi_\varphi(g) \in \mathbb{C}$ is the eigenvalue. Since the representation is a homomorphism ($R_{gh} = R_g R_h$), the eigenvalues must satisfy the multiplicative property $\chi_\varphi(gh) = \chi_\varphi(g)\chi_\varphi(h)$. Moreover, since R_g is unitary, $|\chi_\varphi(g)| = 1$ and hence the codomain is S^1. These homomorphisms $\chi : G \to S^1$ are precisely the *characters* of the group. Thus,

> the group-theoretic eigenbasis for $L^2(G)$ is simply the set of characters!

We shall explore this more in section 3.2.

1.2.2 Advanced: Operators Commuting with Translations We focused on finding the basis that diagonalizes translations (R_g). But what are the operators that commute with all translations?

$$T \circ R_g = R_g \circ T \qquad \forall g \in G$$

A fundamental result in Harmonic Analysis states that any such linear operator T is a *Convolution Operator*. That is, there exists a function $k \in L^1(G)$ (the kernel) such that $Tf = f * k$.
The deeper algebraic structure is this: the algebra of all translation-invariant operators on $L^2(G)$ is isomorphic to the *group algebra* $L^1(G)$ acting by convolution. The Fourier transform simultaneously diagonalizes this entire algebra—it is the *Gelfand transform* of the commutative Banach algebra $L^1(G)$, mapping the group algebra to its spectrum (the space of characters). This is the structural reason why the Fourier transform converts convolution to pointwise multiplication: it maps the group algebra to its maximal ideal space, where every element acts as a scalar.
This explains why the Heat Operator, the Laplacian, and other physical operators can all be written as convolutions: they are simply the most general linear maps that respect the symmetry of the space.

The Infinite Compact Case

In the case where the domain is infinite (like the torus $\mathbb{T}^n$), we must be more careful. First, let us stick to compact domains so that we generalize the finite case directly. Functions defined on a rectangular domain with periodic boundary conditions are equivalent to functions on the torus $\mathbb{T}^n = (S^1)^n$. Our state space is the Hilbert space $L^2(\mathbb{T}^n, \mathbb{C})$.

We want to act by translation by $\mathbb{R}^n$, but since the domain is periodic, this descends to a faithful action of the compact group $\mathbb{R}^n/\mathbb{Z}^n \cong \mathbb{T}^n$. It is easy to check that translation is unitary ($\langle \tau_y f, \tau_y g\rangle = \langle f, g\rangle$) using the translation invariance of the Lebesgue measure.

We now invoke the *Peter-Weyl Theorem*, which is the generalization of Maschke's theorem to compact topological groups (see [11]). It states that the unitary representation of a compact group on $L^2(G)$ decomposes into a Hilbert direct sum of finite-dimensional irreducible representations:

$$L^2(G) \cong_G \widehat{\bigoplus_i} V_i$$

Since $\mathbb{T}^n$ is abelian, all irreducible representations V_i are *one-dimensional*. Being one-dimensional and unitary, they are given by characters $\chi : \mathbb{T}^n \to S^1$.

Thus, the Peter-Weyl theorem guarantees that the characters of $\mathbb{T}^n$ form a complete orthonormal basis for $L^2(\mathbb{T}^n)$. We have recovered our Fourier basis purely from the algebraic structure of the domain, without reference to any specific differential equation!

1.2.1 Fourier Transform on n-Torus

Finally, we look at the details and formally define the Fourier transform. Let us first formalize the eigenvectors of the translation operator:

Theorem 1.2.3: Eigenvector of Translation Functions

Let φ be a measurable function on $\mathbb{R}^n$ (resp. $\mathbb{T}^n$) such that $\varphi(x+y) = \varphi(x)\varphi(y)$ and $|\varphi(x)| = 1$. Then there exists a $\xi \in \mathbb{R}^n$ (resp. $\xi \in \mathbb{T}^n$) such that $\varphi(x) = e^{2\pi i \xi \cdot x}$ where $\xi \cdot x$ is the dot product.

***Proof*:**
We prove it for $\mathbb{R}$, the result naturally generalizing for $\mathbb{R}^n$ and automatically working for $\mathbb{T}^n$. Let $a \in \mathbb{R}$ such that $\int_0^a \varphi(t)dt \neq 0$. Such an a exists, or else by the Lebesgue Differentiation Theorem $\varphi = 0$ a.e. Setting $A = \left(\int_0^a \varphi(t)dt\right)^{-1}$, we get:

$$\varphi(x) = \varphi(x)AA^{-1} = A\int_0^a \varphi(x)\varphi(t)dx = A\int_0^a \varphi(x+t)dt = A\int_x^{x+a} \varphi(t)dt$$

Thus, φ must be continuous, being the integral of a locally integrable function. But that makes the function inside the integral continuous, and so φ is in fact C^1 (in fact C^∞, but we will not need this fact). Now, notice that:

$$\varphi'(x) = A[\varphi(x+a) - \varphi(x)] = B\varphi(x)$$

where $B = A(\varphi(a) - 1)$. This is a simple ODE, which can be solved to show we get an exponential (and is unique by the uniqueness of a solution to an ODE). If you don't know about ODEs, then notice that

$$\frac{d}{dx}(e^{-Bx}\varphi(x)) = 0$$

so $e^{-Bx}\varphi(x)$ is constant. Since $\varphi(0) = 1$, we have $\varphi(x) = e^{Bx}$, and since $|\varphi(x)| = 1$, B must be purely imaginary meaning $B = 2\pi i\xi$ for some $\xi \in \mathbb{R}$, completing the proof for $\mathbb{R}$. For $\mathbb{T}$, we would consider φ to be periodic with period 1 and $e^{2\pi i\xi} = 1$ if and only if $\xi \in \mathbb{Z}$

For the n-dimensional case, let $e_1, e_2, ..., e_n$ be the standard basis for $\mathbb{R}^n$ and define the coordiante functions $\varphi_i(t) = \varphi(te_i)$. Naturally, each φ_i satisifes the property $\varphi_i(t+s) = \varphi_i(t)\varphi_i(s)$ on $\mathbb{R}$, and so $\varphi_i(t) = e^{2\pi i\xi_i t}$, and so:

$$\varphi(x) = \varphi\left(\sum_1^n x_i e_i\right) = \prod_1^n \varphi_i(x_i) = e^{2\pi i\xi\cdot x}$$

completing the proof.

Hence, a function satisfying this translation invariance is of the form $f(x) = ce^{2\pi i\xi\cdot x}$ for some constant $c \in \mathbb{R}$. With the classification of these functions, we will want to use these to decompose reasonably arbitrary functions in $\mathbb{R}^n$ and $\mathbb{T}^n$ (i.e. we will show they form a dense set). In the case of L^2 with $\mathbb{T}^n$, we get a very precise result:

Theorem 1.2.4: Orthonormality Of Translation Functions

Let $E_k(x) = e^{2\pi ik\cdot x}$. Then the set $\{E_k \mid k \in \mathbb{Z}^n\}$ is an orthonormal basis for $L^2(\mathbb{T}^n)$

***Proof*:**
For orthonormality, notice that:

$$\int_0^1 e^{2\pi ikt}dt = \begin{cases} 1 & k = 0 \\ 0 & \text{otherwise} \end{cases}$$

using this, notice that if $n \neq m$, then:

$$\begin{aligned} \langle E_a, E_b\rangle &= \int_0^1 e^{2\pi iat}\overline{e^{2\pi ibt}}dt \\ &= \vdots \\ &= \frac{1}{2\pi i(a-b)}(e^{2\pi i(a-b)} - e^0) \\ &= \frac{1}{2\pi i(a-b)}(1-1) \\ &= 0 \end{aligned}$$

and if $n = m$, then we get

$$\int_0^1 e^{2\pi ikt}e^{2\pi i-kt}dt = \int_0^1 dt = 1$$

For density, since $E_xE_y = E_{x+y}$, the set of finte linear combinations of the E_k's form an algebra which clearly seperates points on $\mathbb{T}^n$, and $E_0 = 1$, $\overline{E_k} = E_{-k}$. Since $\mathbb{T}^n$ compact, by the Stone-Weiestrass theorem, this algebra is dense in $C(\mathbb{T}^n)$ in the uniform norm and hence in the L^2 norm, and $C(\mathbb{T}^n)$ is dense in $L^2(\mathbb{T}^n)$, and so $\{E_k\}$ is a orthonormal basis for $L^2(\mathbb{T}^n)$, as we sought to show.

Using this, we can now define the Fourier transform:

Definition 1.2.5: Fourier Transform And Fourier Series on Torus

Let $f \in L^2(\mathbb{T}^n)$, $\langle -, - \rangle$ be the induced inner product, and E_k the orthonormal basis from theorem 1.2.4. Then the *Fourier Transform* $\hat{f} : \mathbb{Z}^n \to \mathbb{C}$ is the function defined as:

$$\hat{f}(k) = \langle f, E_k \rangle = \int_{\mathbb{T}^n} f(x) e^{-2\pi i k \cdot x} dx$$

The map $f \mapsto \hat{f}$ is called the *Fourier transform*. The representation

$$f = \sum_{k \in \mathbb{Z}^n} \hat{f}(k) E_k$$

is called the *Fourier series* of f.

Example 1.2.6: Discrete Spectrum Examples (Torus)

1. *The Pure Tone (Finite Spectrum):* Let $f(x) = \sin(2\pi m x)$ for some integer m. Using Euler's identity:
$$f(x) = \frac{e^{2\pi i m x} - e^{-2\pi i m x}}{2i}$$
The Fourier transform is a pair of spikes: $\hat{f}(m) = \frac{1}{2i}$, $\hat{f}(-m) = -\frac{1}{2i}$, and $\hat{f}(k) = 0$ everywhere else. A smooth, oscillating function on the torus has a *finite* number of non-zero coefficients.

2. *The Square Wave (Algebraic Decay):* Consider the periodic function equal to $+1$ on $[0, 1/2)$ and -1 on $[1/2, 1)$.
$$\hat{f}(k) = \frac{1 - (-1)^k}{\pi i k} \quad \text{for } k \neq 0$$
This gives non-zero values only for odd k, decaying as $1/k$. Discontinuities in the function lead to coefficients that decay slowly (algebraically). This slow decay causes the partial sums to oscillate violently near the jump (Gibbs Phenomenon).

3. *The Periodized Gaussian (Jacobi Theta Function):* Instead of a single Gaussian (which doesn't live on a torus), we sum infinitely many copies:
$$f(x) = \sum_{n \in \mathbb{Z}} e^{-\pi(x-n)^2}$$
Remarkably, the Fourier series coefficients are sampled from the Gaussian itself:
$$\hat{f}(k) = e^{-\pi k^2}$$
This connects local geometry to global geometry. Even though the function wraps around the circle, its coefficients still decay super-fast (exponentially), reflecting the smoothness of the function.

4. *The Dirac Comb (Flat Spectrum):* Let $f = \delta_0$ be the Dirac delta distribution at the origin (periodized).
$$\hat{f}(k) = \int_{\mathbb{T}} \delta_0(x) e^{-2\pi i k x} dx = 1$$
Extreme localization in space (a single point) leads to extreme delocalization in frequency (all frequencies contribute equally).

By the definition of an orthonormal basis, the Fourier series of f converges to f in the L^2-norm. In the next sections, we will address point-wise convergence. As this is simply a conversion of basis:

Theorem 1.2.7: Parseval's Identity On $\mathbb{T}^n$

$$\|\hat{f}\|_2 = \|f\|_2$$

***Proof*:**
Theorem 1.2.4 shows that $L^2(\mathbb{T}^n)$ surjects onto $\ell^2(\mathbb{Z}^n)$ and forms an orthonormal basis, hence by [13, Orthonormal Basis Condition]
$$\|\hat{f}\|_2 = \|f\|_2$$

Let us take some time concretely interpreting $\hat{f}(k)$. First, $\hat{f}(0)$ represents the "average total energy" in the system:
$$\hat{f}(0) = \int_{\mathbb{T}^n} f(x) dx$$
where as $m(\mathbb{T}^n) = 1$ the normalization is hidden. If we visualized this, we can think of it as the center of mass of the graph of a compact function wrapped around the circle. Then $\hat{f}(n)$ for $n \in \mathbb{Z}^k$ is that same graph wrapped around in each cardinal direction n_i number of times, and then taking its average:
$$\hat{f}(n) = \int_{\mathbb{T}^n} f(x) e^{2\pi i x \cdot n} dx$$

Now, let's say that instead of trying to take the formal approach we've done, we motivate the coefficient formula geometrically. Consider the inner product as a projection that will project onto one of the axis given by one of the terms in the series:
$$f = \cdots + c_{-1} e^{-1 \cdot 2\pi i t} + c_0 e^{0 \cdot 2\pi i t} + c_1 e^{1 \cdot 2\pi i t} + c_2 e^{2 \cdot 2\pi i t} + \cdots$$

How do we find the coefficients? First, by the DCT:
$$\begin{aligned}\int f(t) dt &= \int \left(\cdots + c_{-1} e^{-1 \cdot 2\pi i t} + c_0 e^{0 \cdot 2\pi i t} + c_1 e^{1 \cdot 2\pi i t} + c_2 e^{2 \cdot 2\pi i t}\right) dt \\ &= \cdots + \int c_{-1} e^{-1 \cdot 2\pi i t} dt + \int c_0 e^{0 \cdot 2\pi i t} dt + \int c_1 e^{1 \cdot 2\pi i t} dt + \int c_2 e^{2 \cdot 2\pi i t} dt + \cdots\end{aligned}$$

Now, notice that all terms except the one containing c_0 are disappear, since the rotation of by the exponential will always guarantee to make the integrals average to zero! Thus, c_0 can be thought of as the average mass point of f. To find the other terms, we simply have to multiply f at the beginning by $e^{k2\pi it}$

$$\int f(t)e^{-k2\pi it}dt$$

and then we have shifted over all of our terms except the kth term, which we can now find the integral of! In fact, notice we have just recovered:

$$\hat{f}(k) = \int f(t)e^{-k2\pi it}dt$$

giving us a geometric intuition for this formula!

Now, this gives the Fourier transform on $L^2(\mathbb{T}^n, \mathbb{C})$, but we often want to work with different L^p spaces. First, the definition of the Fourier transform $\hat{f}(k)$ makes sense if $f \in L^1(\mathbb{T}^n)$ since:

$$|\hat{f}(\xi)| = \left| \int f(x)e^{2\pi ix\cdot\xi}dx \right| \leq \int \left| f(x)e^{2\pi ix\cdot\xi} \right| dx = \int |f(x)| = \|f\|_1$$

Hence for each $\xi \in \mathbb{Z}^n$, if $f \in L^1(\mathbb{T}^n)$ then $|\hat{f}(\xi)| \leq \|f\|_1$ showing us that $\hat{f}$ must be essentially bounded:

$$\|\hat{f}\|_\infty$$

so the Fourier series would then extend to a norm-decreasing map from $L^1(\mathbb{T}^n)$ to $\ell^\infty(\mathbb{Z}^n)$. In particular, this shows us that there is a sort of duality between the speed of decay of a function f and its Fourier transform $\hat{f}$. The nature of this duality is made explicit in the following theorem:

Theorem 1.2.8: Hausdorff-Young Inequality over $\mathbb{T}^n$

Let $1 \leq p \leq 2$ and q be the conjugate exponent of p so that $\frac{1}{p} + \frac{1}{q} = 1$. If $f \in L^p(\mathbb{T}^n)$, then

$$\hat{f} \in \ell^q(\mathbb{Z}^n) \qquad \text{and} \qquad \|\hat{f}\|_q \leq \|f\|_p$$

***Proof*:**

Since $\|\hat{f}\|_\infty \leq \|f\|_1$ and $\|\hat{f}\|_2 = \|f\|_2$ for $f \in L^1$ and $f \in L^2$, this follows from the Riesz-Thorin interpolation theorem ([13, Riesz-Thorin Interpolation Theorem])

Example 1.2.9: Advanced: The Theta Function and Modularity

1. *The Heat Kernel Trace:* Recall example 1.2.6.3, where we periodized the Gaussian to get $K_t(x) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 t}e^{2\pi inx}$. If we evaluate this at the identity ($x = 0$), we obtain the classical *Theta Function* $\theta(it)$:

$$\theta(it) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 t}$$

In the language of Number Theory, this is a modular form (specifically of weight 1/2). The fact that the Fourier transform of a Gaussian is a Gaussian translates directly into the *functional equation* of the Theta function via the Poisson Summation Formula:

$$\theta(t^{-1}) = \sqrt{t}\theta(t)$$

This identity is the "analytic engine" that proves the analytic continuation of the Riemann Zeta function $\zeta(s)$.

2. *Gauss Sums (Arithmetic Fourier Transform):* If we replace our field $\mathbb{C}$ with a finite field $\mathbb{F}_p$ and our torus with the additive group of the field, the Fourier Transform becomes the *Gauss Sum*:
$$g(a) = \sum_{x \in \mathbb{F}_p} \chi(x) e^{2\pi i a x/p}$$
Just as $\int |f|^2 = \int |\hat{f}|^2$ (Plancherel), we have $|g(a)| = \sqrt{p}$. The "square root cancellation" of Fourier coefficients in analytic number theory is the discrete analogue of the decay rates we study in analysis.

1.2.2 Fourier Transform on Euclidean Space

We often want to work with $L^2(\mathbb{R}^n)$. The problem is that $e^{2\pi i k}$ is *not* an element of $L^2(\mathbb{R}^n)$ as its integral is infinite. To generalize the domain to $\mathbb{R}^n$, we need to shift perspectives. Recall that all unitary representation of $\mathbb{T}^n$ are given by functions in:

$$\mathrm{Hom}(\mathbb{T}^n, S^1) \cong \mathbb{Z}^n$$

This is the *Pontryagin dual* of $\mathbb{T}^n$ and represents all the characters. We can generalize this and choose a locally compact (topological) groups and consider the Pontryagin dual $\mathrm{Hom}(G, S^1)$. As shall be shown in chapter 3, there shall always be a unique invariant Borel measure defined on any locally compact group up to scaling (ex. on $\mathbb{R}$ it is the Lebesgue measure and on $\mathbb{Z}$ it is the counting measure). It can be shown that $\mathrm{Hom}(G, S^1)$ is also a locally compact group and hence has a unique measure. When G is compact, $\mathrm{Hom}(G, S^1)$ is *discrete* and the natural Haar measure is the counting measure. If G is not compact but locally compact, $\mathrm{Hom}(G, S^1)$ is *continuous*, most importantly:

$$\mathrm{Hom}(\mathbb{R}^d, S^1) \cong \mathbb{R}^d \qquad \text{given by} \qquad (\xi \mapsto e^{2\pi i x \cdot \xi}) \mapsto x$$

Notice that if G is compact and abelian, $\mathrm{Hom}(G, S^1)$ represents all the characters. Hence if G is *locally compact*, then $\hat{G} = \mathrm{Hom}(G, S^1)$, can interpreted as the generalization of characters.

1.2.10 Advanced: The Notion of Points If you have seen Grothendieck's notion of the *Functor of Points*, then the "right" way to see points is not as elements of a set, but as *morphisms* from a test object. In algebraic geometry, a point of a scheme X is determined by the morphisms $T \to X$. In harmonic analysis, this philosophy manifests as *Tannaka-Krein Duality.* We can reconstruct a group G solely from its category of representations $\mathrm{Rep}(G)$. A "geometric point" $g \in G$ corresponds precisely to a tensor automorphism of the fiber functor (the forgetful functor to vector spaces). Thus, the Fourier Transform is simply the mechanism that translates between the geometric perspective (points as elements) and the spectral perspective (points as representations), proving they are dual aspects of the same underlying structure.

We can thus generalize the orthonormal basis interpretation of the Fourier transform to

$$\mathcal{F} : L^2(G, \mathbb{C}) \to L^2(\hat{G}, \mathbb{C})$$

We have not proven it is well-defined, namely since $\hat{G}$ need not be an orthonormal basis anymore. For example if $f \in L^2(\mathbb{R}^d)$, it is not necessarily the case that $f \in L^1(\mathbb{R}^d)$ however as we've defined it so far we integrate against $e^{2\pi i x\cdot\xi}$:

$$\hat{f}(\xi) = \int_{\mathbb{R}^n} f(x)e^{2\pi i x\cdot\xi}dx$$

Simply take $(1+|x|)^{-1} \in L^2(\mathbb{R})$ which not in $L^1(\mathbb{R})$. However, $\mathcal{F}$ shall still be a well-defined map between L^2 spaces as shall be shown in upcoming sections. Note that in the case of $G = \mathbb{T}^n$ this is exactly what we had before:

$$\mathcal{F} : L^2(\mathbb{T}^n, \mathbb{C}) \to \ell^2(\mathbb{Z}^n, \mathbb{C})$$

There is a lot of details that are brushed off for now; this is all covered in chapter 3.

Now, if $f \in L^1$, $\hat{f}(\xi)$ will certainly be bounded (by $\|f\|_1$), and hence it is essentially bounded:

$$\|\mathcal{F}(f)\|_\infty < \infty$$

We thus have our first definition of Fourier transform on $\mathbb{R}^d$:

Definition 1.2.11: Fourier Transform on L^1

Let $f \in L^1(\mathbb{R}^n)$. Then the *Fourier Transform* $\hat{f} : \mathbb{R}^n \to \mathbb{C}$ is the function defined as:

$$\hat{f}(\xi) = \int_{\mathbb{R}^n} f(x)e^{-2\pi i\xi\cdot x}dx$$

The map $f \mapsto \hat{f}$ is called the *Fourier transform*.

As it turns out, $\mathcal{F}(f)$ is also continuous, hence it is not only essentially bounded (in L^∞) but *actually* bounded (in BC):

Theorem 1.2.12: Riemann-Lebesgue Lemma

$$\mathcal{F}(L^1(\mathbb{R}^n)) \subseteq \mathrm{BC}(\mathbb{R}^n)$$

***Proof*:**

Let $f \in L^1(\mathbb{R}^n)$ and consider $\{\xi_k\}_{k=1}^\infty \subset \mathbb{R}^n$ such that $\xi_k \to \xi$. It suffices to show $\lim_{k\to\infty} \hat{f}(\xi_k) = \hat{f}(\xi)$.

By the definition of the Fourier transform,

$$\hat{f}(\xi_k) = \int_{\mathbb{R}^n} f(x)e^{-2\pi i\xi_k\cdot x}\,dx.$$

Let $F_k(x) = f(x)e^{-2\pi i\xi_k\cdot x}$. Since the complex exponential is continuous, we have pointwise convergence for every $x \in \mathbb{R}^n$:

$$\lim_{k\to\infty} F_k(x) = f(x)\lim_{k\to\infty} e^{-2\pi i\xi_k\cdot x} = f(x)e^{-2\pi i\xi\cdot x}.$$

Next, for all k and x, the integrand is bounded by the magnitude of f:

$$|F_k(x)| = \left|f(x)e^{-2\pi i\xi_k\cdot x}\right| = |f(x)|\cdot 1 = |f(x)|.$$

Since $f \in L^1(\mathbb{R}^n)$, $|f|$ dominates it. Thus, by the DCT:

$$\lim_{k\to\infty} \int_{\mathbb{R}^n} F_k(x)\,dx = \int_{\mathbb{R}^n} \lim_{k\to\infty} F_k(x)\,dx = \int_{\mathbb{R}^n} f(x)e^{-2\pi i\xi\cdot x}\,dx = \hat{f}(\xi).$$

showing $\hat{f}$ is continuous, as we sought to show.

Proposition 1.2.13: Properties Of Fourier Transform

Let $f, g \in L^1(\mathbb{R}^n)$. Then:

1. $\widehat{(\tau_y f)}(\xi) = e^{-2\pi i\xi\cdot y}\hat{f}(\xi)$ and $\tau_\eta(\hat{f}) = \widehat{h}$ where $h(x) = e^{2\pi i\eta\cdot x}f(x)$
2. If T is an invertible linear transformation on $\mathbb{R}^n$ and $S = (T^*)^{-1}$ is its inverse transpose, Then $\widehat{(f\circ T)} = |\det(T)|^{-1}\hat{f}\circ S$. In particular, if T is a rotation, then $\widehat{(f\circ T)} = \hat{f}\circ T$. Furthermore, if $T(x) = t^{-1}x$ (for $t > 0$), then $\widehat{(f\circ T)}(\xi) = t^n\hat{f}(t\xi)$.

   Note: This implies the Heisenberg Uncertainty Principle qualitatively: dilating (widening) f results in contracting (narrowing) $\hat{f}$.
3. Convolution becomes multiplication after the Fourier Transform:
$$\widehat{(f * g)} = \hat{f}\hat{g}$$
4. (*Conjugate Symmetry*) if f is real-valued, then $\hat{f}$ (which is still complex-valued) satisfies the Hermitian symmetry condition:
$$\hat{f}(-\xi) = \overline{\hat{f}(\xi)}$$
   This implies the information in the codomain is 'half' redundant.
5. If $x^\alpha f \in L^1$ for $|\alpha| \le k$ then $\hat{f} \in C^k$ and
$$\partial^\alpha\hat{f}(\xi) = [\widehat{(-2\pi ix)^\alpha f}](\xi)$$
6. if $f \in C^k$, $\partial^\alpha f \in L^1$ for $|\alpha| \le k$, and $\partial^\alpha f \in C_0$, for $|\alpha| \le k-1$, then
$$\widehat{(\partial^\alpha f)}(\xi) = (2\pi i\xi)^\alpha\hat{f}(\xi)$$
7. upgraded Riemann-Lebesgue lemma:
$$\mathcal{F}(L^1(\mathbb{R}^n)) \subseteq C_0(\mathbb{R}^n)$$

***Proof*:**

1. Computing directly:
$$\widehat{(\tau_y f)}(\xi) = \int f(x-y)e^{-2\pi i\xi\cdot x}dx$$

Let $z = x - y$, then $dx = dz$ and $x = z + y$:

$$= \int f(z)e^{-2\pi i\xi\cdot(z+y)}dz = e^{-2\pi i\xi\cdot y}\int f(z)e^{-2\pi i\xi\cdot z}dz = e^{-2\pi i\xi\cdot y}\hat{f}(\xi)$$

For the second part, let $h(x) = e^{2\pi i\eta\cdot x}f(x)$.

$$\hat{h}(\xi) = \int e^{2\pi i\eta\cdot x}f(x)e^{-2\pi i\xi\cdot x}dx = \int f(x)e^{-2\pi i(\xi-\eta)\cdot x}dx = \hat{f}(\xi - \eta) = \tau_\eta(\hat{f})(\xi)$$

2. Apply theorem [14, Linear Transformation on Integrability] (Change of Variables). Let $y = Tx$, then $x = T^{-1}y$ and $dx = |\det T|^{-1}dy$:

$$\begin{aligned}\widehat{(f\circ T)}(\xi) &= \int f(Tx)\exp(-2\pi i\xi\cdot x)dx \\ &= |\det T|^{-1}\int f(y)\exp(-2\pi i\xi\cdot T^{-1}y)dy\end{aligned}$$

Using the definition of the adjoint (transpose), $\xi\cdot T^{-1}y = (T^{-1})^*\xi\cdot y = S\xi\cdot y$:

$$\begin{aligned}&= |\det T|^{-1}\int f(y)\exp(-2\pi iS\xi\cdot y)dy \\ &= |\det T|^{-1}\hat{f}(S\xi)\end{aligned}$$

3. By Fubini's Theorem (justified since $f, g \in L^1$):

$$\begin{aligned}\widehat{(f*g)}(\xi) &= \int\left(\int f(x-y)g(y)dy\right)e^{-2\pi i\xi\cdot x}dx \\ &= \int g(y)\left(\int f(x-y)e^{-2\pi i\xi\cdot x}dx\right)dy\end{aligned}$$

Applying the translation property (proven in 1) to the inner integral where f is translated by y:

$$\begin{aligned}&= \int g(y)\left(e^{-2\pi i\xi\cdot y}\hat{f}(\xi)\right)dy \\ &= \hat{f}(\xi)\int g(y)e^{-2\pi i\xi\cdot y}dy \\ &= \hat{f}(\xi)\hat{g}(\xi)\end{aligned}$$

4. exercise.

5. By Differentiating under the integral sign (justified by the Dominated Convergence Theorem and the assumption $x^\alpha f \in L^1$):

$$\partial^\alpha\hat{f}(\xi) = \partial_\xi^\alpha\int f(x)e^{-2\pi i\xi\cdot x}dx = \int f(x)\partial_\xi^\alpha(e^{-2\pi i\xi\cdot x})dx = \int f(x)(-2\pi ix)^\alpha e^{-2\pi i\xi\cdot x}dx$$

6. Assuming first that $n = 1$ and $k = 1$. Since $f \in C_0$, f vanishes at boundaries. We integrate by parts:
$$\widehat{f'}(\xi) = \int_{-\infty}^{\infty} f'(x)e^{-2\pi i\xi x}dx$$
Let $u = e^{-2\pi i\xi x}$ and $dv = f'(x)dx$. Then $du = -2\pi i\xi e^{-2\pi i\xi x}dx$ and $v = f(x)$.
$$= \left[f(x)e^{-2\pi i\xi x}\right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} f(x)(-2\pi i\xi)e^{-2\pi i\xi x}dx$$
The boundary term is 0 because $f \in C_0$.
$$= 2\pi i\xi \int_{-\infty}^{\infty} f(x)e^{-2\pi i\xi x}dx = 2\pi i\xi\hat{f}(\xi)$$
For $n > 1$, we compute $\widehat{(\partial_j f)}$ by integrating by parts with respect to the j-th variable x_j. The boundary terms vanish because $f \in C_0(\mathbb{R}^n)$. The general case follows by induction on $|\alpha|$.

7. By the previous part, if $f \in C_c^1$ (Compactly supported C^1 functions), then $\widehat{\partial_j f}(\xi) = 2\pi i\xi_j\hat{f}(\xi)$. Since $\widehat{\partial_j f}$ is bounded (as the Fourier transform of an L^1 function), $|\xi_j\hat{f}(\xi)|$ is bounded. This implies $\hat{f}(\xi)$ decays as $|\xi| \to \infty$, so $\hat{f} \in C_0$.

   By a similar argument to [13, smooth functions are dense in L^p], C_c^1 is dense in L^1. Let $f \in L^1$ and $f_n \in C_c^1$ such that $f_n \to f$ in L^1. The Fourier transform is a bounded linear map from L^1 to L^∞ (specifically $\|\hat{g}\|_\infty \leq \|g\|_1$), so:
$$\|\hat{f}_n - \hat{f}\|_\infty \leq \|f_n - f\|_1 \to 0$$
   Thus $\hat{f}_n \to \hat{f}$ uniformly. Since C_0 is closed under the uniform norm (a uniform limit of functions vanishing at infinity also vanishes at infinity), we conclude $\hat{f} \in C_0$.

Notice that (5) and (6) give an interesting relation between the smoothness of f and the decay of $\hat{f}$: the smoother f is, the faster $\hat{f}$ decays (and vice-versa).

Proposition 1.2.14: Regularity and Decay Duality

The smoothness of a function determines the decay of its Fourier transform, and the decay of a function determines the smoothness of its Fourier transform. In particular:

1. *Decay ⇒ Smoothness:* If $(1+|x|)^k f(x) \in L^1(\mathbb{R}^n)$ for some integer $k \geq 1$, then $\hat{f} \in C^k(\mathbb{R}^n)$. Specifically, differentiation converts to multiplication by a polynomial in the spatial domain:
$$\partial^\alpha \hat{f}(\xi) = \int_{\mathbb{R}^n} (-2\pi i x)^\alpha f(x) e^{-2\pi i \xi \cdot x}\, dx$$
for all multi-indices $|\alpha| \leq k$.

2. *Smoothness ⇒ Decay:* If $f \in C^k(\mathbb{R}^n)$ and $\partial^\alpha f \in L^1(\mathbb{R}^n)$ for all $|\alpha| \leq k$ (and f vanishes at infinity), then $\hat{f}$ decays rapidly:
$$|\hat{f}(\xi)| \leq \frac{C}{(1+|\xi|)^k}$$
for some constant C.

***Proof*:**

1. **(Decay ⇒ Smoothness):** Formally differentiating:
$$\frac{\partial}{\partial \xi_j} \hat{f}(\xi) = \frac{\partial}{\partial \xi_j} \int_{\mathbb{R}^n} f(x) e^{-2\pi i \xi \cdot x}\, dx = \int_{\mathbb{R}^n} f(x) \frac{\partial}{\partial \xi_j} \left(e^{-2\pi i \xi \cdot x}\right) dx.$$
Calculating the derivative of the exponential gives $(-2\pi i x_j) e^{-2\pi i \xi \cdot x}$. Thus the integrand becomes $(-2\pi i x_j) f(x) e^{-2\pi i \xi \cdot x}$. To justify this operation rigorously, we need a dominating function. Observe that:
$$\left|(-2\pi i x_j) f(x) e^{-2\pi i \xi \cdot x}\right| = 2\pi |x_j| |f(x)| \leq 2\pi |x| |f(x)|.$$
By our hypothesis, $(1+|x|)^k f \in L^1$, which implies $|x| f \in L^1$. Thus, the integrand is dominated by an integrable function, and the Dominated Convergence Theorem allows differentiation. Repeating this for higher derivatives up to order k yields the result.

2. **(Smoothness ⇒ Decay):** We use the property that $\widehat{\partial^\alpha f}(\xi) = (2\pi i \xi)^\alpha \hat{f}(\xi)$. Since $\partial^\alpha f \in L^1$, by the standard Riemann-Lebesgue Lemma (or simply the boundedness of the Fourier transform), we know that its transform is bounded:
$$\|\widehat{\partial^\alpha f}\|_\infty \leq \|\partial^\alpha f\|_1 < \infty.$$
Substituting the identity, we have:
$$|(2\pi i \xi)^\alpha \hat{f}(\xi)| \leq \|\partial^\alpha f\|_1.$$
This implies that for any multi-index α with $|\alpha| \leq k$, the product $|\xi^\alpha \hat{f}(\xi)|$ is bounded. Since a function dominated by $1/|\xi|^k$ is equivalent to being bounded when multiplied by $|\xi|^k$, we conclude:
$$|\hat{f}(\xi)| \leq C(1+|\xi|)^{-k}.$$

Corollary 1.2.15: Paley-Wiener Theorem

If $f \in C_c^\infty(\mathbb{R}^n)$ has compact support (i.e., there exists $R > 0$ such that $f(x) = 0$ for $|x| > R$), then its Fourier transform $\hat{f}(\xi)$ extends to a holomorphic (complex analytic) function on $\mathbb{C}^n$.

***Proof*:**
Let $z \in \mathbb{C}^n$. Define the complex extension of the Fourier transform as:

$$F(z) = \int_{|x|\leq R} f(x)e^{-2\pi i x\cdot z}\,dx$$

Since f has compact support, the integration range is effectively bounded by the ball $B_R(0)$. To show $F(z)$ is holomorphic, it suffices to show it is complex differentiable with respect to each component z_j. We compute the partial derivative by differentiating under the integral sign:

$$\frac{\partial}{\partial z_j}F(z) = \int_{|x|\leq R} f(x)\frac{\partial}{\partial z_j}\left(e^{-2\pi i x\cdot z}\right)\,dx = \int_{|x|\leq R} f(x)(-2\pi i x_j)e^{-2\pi i x\cdot z}\,dx$$

To justify integral converges, let $z = \xi + i\eta$. The magnitude of the exponential term is:

$$\left|e^{-2\pi i x\cdot(\xi+i\eta)}\right| = \left|e^{-2\pi i x\cdot\xi}\right|\left|e^{2\pi x\cdot\eta}\right| = e^{2\pi x\cdot\eta}$$

Since the domain of integration is bounded ($|x| \leq R$), this term is bounded by $e^{2\pi R|\eta|}$. Furthermore, f and x_j are bounded on this domain. Thus, the integrand is bounded for any fixed z, and the integral is well-defined.

Since the derivative exists at every point $z \in \mathbb{C}^n$, the function $F(z)$ is entire (holomorphic everywhere). Consequently, the Fourier transform $\hat{f}(\xi)$ is simply the restriction of this entire function to the real subspace $\mathbb{R}^n$, implying that $\hat{f}$ is real-analytic.

Notice that $C_c^\infty(\mathbb{R}^n)$ does not map into compact spaces, since an analytic function cannot be non-zero and bounded. The closest equivalent is Schwartz space:

Corollary 1.2.16: Fourier Transform On Schwartz Space

The map $\mathcal{F}$ maps the Schwartz space $\mathcal{S}$ into itself

***Proof*:**
If $f \in \mathcal{S}$, then $x^\alpha\partial^\beta f \in L^1 \cap C_0$ for all α, β, so by proposition 1.2.13, $\hat{f}$ is C^∞ and

$$(\widehat{x^\alpha\partial^\beta f}) = (-1)^{|\alpha|}(2\pi i)^{|\beta|-|\alpha|}\partial^\alpha(\xi^\beta\hat{f}).$$

Thus $\partial^\alpha(\xi^\beta\hat{f})$ is bounded for all α, β, whence $\hat{f} \in \mathcal{S}$. Moreover, since $\int(1+|x|)^{-n-1}dx < \infty$,

$$\|(\widehat{x^\alpha\partial^\beta f})\|_u \leq \|x^\alpha\partial^\beta f\|_1 \leq C\|(1+|x|)^{n+1}x^\alpha\partial^\beta f\|_u.$$

It then follows that $\|\hat{f}\|_{(N,\beta)} \leq C_{N,\beta}\sum_{|\gamma|\leq|\beta|}\|f\|_{(N+n+1,\gamma)}$ by proposition 1.1.4, so the Fourier transform is continuous on $\mathcal{S}$.

1.2.17 Distribution Theory and Compactly Supported In distribution theory ([13, Distribution Theory]), it shall be shown that $C(\mathbb{R}^n)$ can be recovered by integrating against smooth functions with compact support. Thus, C_c^∞ would be ideal space for the Fourier transform *if* they mapped back to themselves. However, by the boundedness argument above it is not the case. *However*, in over totally disconnected topological groups, such as those present in the representation theory of absolute galois groups, the equivalent of Schwartz space *is* C_c^∞; see [5] for more details.

Example 1.2.18: Continuous Spectrum Examples (Euclidean)

1. *The Box Function (The Sinc Function):* Let $f(x) = \mathbb{1}_{[-1/2,1/2]}(x)$ (The indicator function of width 1).
$$\hat{f}(\xi) = \int_{-1/2}^{1/2} 1 \cdot e^{-2\pi i \xi x} dx = \frac{e^{-\pi i \xi} - e^{\pi i \xi}}{-2\pi i \xi} = \frac{\sin(\pi \xi)}{\pi \xi} \equiv \operatorname{sinc}(\xi)$$
Unlike the Square Wave on the torus (which had discrete coefficients $1/k$), this has a *continuous* spectrum that oscillates and decays as $1/\xi$. The compact support in space ($f = 0$ outside the box) forces the transform to extend infinitely (it cannot be compactly supported).

2. *The Two-Sided Exponential (Lorentzian):* Let $f(x) = e^{-2\pi |x|}$. This is a continuous L^1 function with a "cusp" at $x = 0$.
$$\hat{f}(\xi) = \frac{1}{\pi} \frac{1}{1+\xi^2}$$
This is the Cauchy (or Lorentzian) distribution. Notice the duality: Exponential decay in space leads to Polynomial ($1/\xi^2$) decay in frequency. The lack of smoothness at $x = 0$ (the cusp) prevents the frequency decay from being exponential.

3. *The Gaussian (Self-Dual):* Let $f(x) = e^{-\pi x^2}$. As proven in theorem 1.3.1:
$$\hat{f}(\xi) = e^{-\pi \xi^2}$$
The Gaussian is the unique eigenfunction of the Fourier Transform. It is the perfect balance point of the Uncertainty Principle—it is equally localized in both space and frequency.

4. *The Shifted Function (Modulation):* Let $f(x)$ be a standard Gaussian centered at $x = 100$ rather than 0: $g(x) = f(x - 100)$.
$$\hat{g}(\xi) = e^{-2\pi i (100)\xi} \hat{f}(\xi) = e^{-2\pi i 100 \xi} e^{-\pi \xi^2}$$
In Euclidean space, shifting a function far away does not change the "energy envelope" of its spectrum (it's still a Gaussian), but it causes the complex phase to wind rapidly.

Example 1.2.19: Advanced uses of Fourier Transform

1. *The Poincaré Bundle:* In our definition of the Fourier Transform $\hat{f}(\xi) = \int f(x) e^{-2\pi i \xi \cdot x} dx$, the kernel $K(x, \xi) = e^{-2\pi i \xi \cdot x}$ plays the central role. In Algebraic Geometry, if A is an abelian variety (say a complex torus V/Λ), its dual $\hat{A}$ parametrizes line bundles on A. The universal

line bundle $\mathcal{P}$ on $A \times \hat{A}$ (the Poincaré bundle) is the geometric geometrization of the kernel $e^{-2\pi i \xi \cdot x}$. The Fourier transform generalizes to the *Fourier-Mukai Transform*, an equivalence of derived categories of coherent sheaves:

$$\Phi_{\mathcal{P}}(\mathcal{F}^\bullet) = \mathbf{R}\pi_{2,*}(\pi_1^* \mathcal{F}^\bullet \otimes \mathcal{P})$$

Here, the integration $\int dx$ is replaced by the pushforward $\mathbf{R}\pi_{2,*}$, and the product is the tensor product.

2. *The Spectrum of the Primes (Riemann's Explicit Formula):* You may have heard that the zeros of the Riemann Zeta function constitute the "spectrum" of the prime numbers. This is literal, not metaphorical. Consider the Chebyshev function $\psi(x) = \sum_{p^k \leq x} \log p$, which counts primes with a logarithmic weight. Riemann's Explicit Formula links this step function to the zeros $\rho = \frac{1}{2} + i\gamma$ of $\zeta(s)$:

$$\psi(x) = x - \sum_{\rho} \frac{x^\rho}{\rho} - \log(2\pi) - \frac{1}{2}\log(1 - x^{-2})$$

If we make the logarithmic change of variables $x = e^u$, the sum over zeros becomes:

$$\sum_{\rho} \frac{e^{u(\frac{1}{2}+i\gamma)}}{\frac{1}{2} + i\gamma} = e^{u/2} \sum_{\gamma} \left(\frac{1}{\frac{1}{2} + i\gamma} \right) e^{i\gamma u}$$

This is a Fourier series (or integral) in the variable u!

- The "time" Domain signal is the distribution of primes (spikes at $u = \log p$).
- The "Frequency" Domain is the set of imaginary parts of zeta zeros (γ).

Just as a musical chord is "recovered" by summing its constituent frequencies, the location of prime numbers is recovered by summing the waves defined by the Riemann Zeros. This duality is an instance of Fourier Duality on the ideles.

3. *Pontryagin Duality as a Moduli Problem:* We defined $\hat{G} = \text{Hom}(G, S^1)$. For a number theorist, we can view the frequency domain $\hat{G}$ as a *moduli space* of rank-1 unitary local systems on G.

- If $G = \mathbb{R}$, the moduli space is $\mathbb{R}$ (Euclidean Fourier Transform).
- If $G = S^1$, the moduli space is $\mathbb{Z}$ (Fourier Series).
- If $G = \mathbb{Z}$, the moduli space is S^1 (Discrete Time Fourier Transform).

The "Duality" implies that the moduli space of the moduli space recovers the original space.

4. *The Circle Method (Diophantine Equations):* How many integer solutions are there to $n_1^k + \cdots + n_s^k = N$? (Waring's Problem). This is a combinatorial problem that Fourier Analysis turns into an analytic one. Construct the generating function (a trigonometric polynomial):

$$f(\alpha) = \sum_{m=1}^{N^{1/k}} e^{2\pi i m^k \alpha}$$

The number of solutions $R(N)$ is exactly the N-th Fourier coefficient of the function $f(\alpha)^s$:

$$R(N) = \int_0^1 (f(\alpha))^s e^{-2\pi i N\alpha} d\alpha$$

The Key insight of Hardy and Littlewood was to split this integral (the torus) into "Major Arcs" (near rational numbers with small denominators, where the sum behaves predictably) and "Minor Arcs" (where the sum oscillates and cancels out, similar to the Riemann-Lebesgue lemma).

5. *Spectral Geometry (Can You Hear the Shape of a Drum?):* The set of eigenvalues $\{\lambda_n\}$ of the Laplacian Δ on a compact Riemannian manifold (M, g) is called the *spectrum*. This is the set of "pure frequencies" the manifold can support. The Fourier Transform allows us to recover geometric data from this spectral data via the trace of the Heat Kernel:

$$\mathrm{Tr}(e^{t\Delta}) = \sum_{n=0}^{\infty} e^{-\lambda_n t} \sim \frac{1}{(4\pi t)^{d/2}} \left(\mathrm{Vol}(M) + \frac{t}{6}\int_M \mathrm{Scal}_g + O(t^2) \right)$$

As $t \to 0$ (high frequencies), the asymptotics of the heat kernel allow us to "hear" the dimension, the volume, and the total scalar curvature of the manifold.

Note: Milnor gave examples of isospectral tori (16-dimensional) that are not isometric, proving that you cannot hear the *entire* shape, but Fourier analysis recovers the local invariants.

1.3 Fourier Integral and Series

Let us now show that up to measure zero the Fourier transform is injective. Equivalently, let us now look into finding the inverse Fourier transform. Let $f \in L^1$. Then define:

$$f^\vee(x) = \hat{f}(-x) = \int f(\xi) e^{2\pi i \xi \cdot x} d\xi$$

We'll show that if $f, \hat{f} \in L^1$, then $(\hat{f})^\vee = f$. This cannot be done by appealing to Fubini's theorem as the integral:

$$(\hat{f})^\vee(x) = \int\int f(y) e^{2\pi i \cdot x} e^{2\pi i \cdot y} dy d\xi$$

is not in $L^1(\mathbb{R}^n \times \mathbb{R}^n)$ (it will a.e. diverge). Instead, we do the usual trick of sequences that converge to the desired values. The solution is the usual solution of taking a limit of functions that satisfy the property and insuring that the desired properties are preserved in the limit. First, a technical lemma that also has an amazing tangential property that simply must be mentioned:

Theorem 1.3.1: Fourier Transform Fixed Point

Let $f(x) = e^{-\pi a|x|^2}$ where $a > 0$. Then

$$\hat{f}(\xi) = a^{-n/2} e^{-\pi|\xi|^2/a}$$

In particular if $a = 1$ so that $f(x) = e^{-\pi|x|^2}$, then

$$\hat{f} = f$$

***Proof*:**

First consider the case $n = 1$. Since the derivative of $e^{-\pi a x^2}$ is $-2\pi a e^{-\pi a x^2}$, by proposition 1.2.13(e) we have

$$(\hat{f})'(\xi) = (-2\pi i x e^{-\pi a x^2})^\wedge(\xi) = \frac{i}{a}(f')^\wedge(\xi) = \frac{i}{a}(2\pi i\xi)\hat{f}(\xi) = -\frac{2\pi}{a}\xi\hat{f}(\xi).$$

Then:

$$\begin{aligned}
\frac{d}{d\xi}\left(e^{\pi\xi^2/a}\cdot\hat{f}(\xi)\right) &= \left[\frac{d}{d\xi}\left(e^{\pi\xi^2/a}\right)\right]\cdot\hat{f}(\xi) + e^{\pi\xi^2/a}\cdot\hat{f}'(\xi) \\
&= e^{\frac{\pi}{a}\xi^2}\cdot\left(\frac{2\pi}{a}\xi\right)\cdot\hat{f}(\xi) + e^{\pi\xi^2/a}\cdot\hat{f}'(\xi) \\
&= e^{\pi\xi^2/a}\left[\frac{2\pi}{a}\xi\hat{f}(\xi) + \hat{f}'(\xi)\right] \\
&= e^{\pi\xi^2/a}\left[\frac{2\pi}{a}\xi\hat{f}(\xi) + \left(-\frac{2\pi}{a}\xi\hat{f}(\xi)\right)\right] \\
&= e^{\pi\xi^2/a}[0] = 0
\end{aligned}$$

It follows that $(d/d\xi)(e^{\pi\xi^2/a}\hat{f}(\xi)) = 0$, so that $e^{\pi\xi^2/a}\hat{f}(\xi)$ is constant. To evaluate the constant, set $\xi = 0$. Then computing the integral:

$$\hat{f}(0) = \int e^{-\pi a x^2}dx = a^{-1/2}.$$

The n-dimensional case follows by Fubini's theorem, since $|x|^2 = \sum_1^n x_j^2$:

$$\begin{aligned}
\hat{f}(\xi) &= \prod_1^n \int \exp(-\pi a x_j^2 - 2\pi i\xi_j x_j)dx_j \\
&= \prod_1^n [a^{-1/2}\exp(-\pi\xi_j^2/a)] = a^{-n/2}\exp(-\pi|\xi|^2/a).
\end{aligned}$$

completing the proof.

Theorem 1.3.2: The Fourier Inversion Theorem

Let $f \in L^1$ and $\hat{f} \in L^1$. Then f agrees a.e. with a continuous function f_0, and

$$(\hat{f})^\vee = (f^\vee)^\wedge = f_0$$

***Proof*:**
Given $t > 0$ and fixed $x \in \mathbb{R}^n$, set

$$\varphi(\xi) = \exp(2\pi i\xi \cdot x - \pi t^2|\xi|^2).$$

We use the standard Gaussian $g(x) = e^{-\pi|x|^2}$ and the scaling notation $g_t(y) = t^{-n}g(y/t)$. Using the translation and dilation properties of the Fourier transform on the Gaussian, we have:

$$\hat{\varphi}(y) = t^{-n}\exp(-\pi|x-y|^2/t^2) = g_t(x-y).$$

Since $f, \varphi \in L^1$, Fubini's theorem justifies the multiplication formula $\int \hat{f}\varphi = \int f\hat{\varphi}$:

$$\int e^{-\pi t^2|\xi|^2}e^{2\pi i\xi\cdot x}\hat{f}(\xi)d\xi = \int \hat{f}(\xi)\varphi(\xi)d\xi = \int f(y)\hat{\varphi}(y)dy = f * g_t(x).$$

Now we take the limit as $t \to 0$:

1. Right hand side: since $\int g(y)dy = 1$, $\{g_t\}$ is an approximate identity. Thus $f * g_t \to f$ in the L^1 norm.
2. Left hand side: since $\hat{f} \in L^1$, the function is dominated by $|\hat{f}|$. By the Dominated Convergence Theorem, for every x:
$$\lim_{t\to 0}\int e^{-\pi t^2|\xi|^2}e^{2\pi i\xi\cdot x}\hat{f}(\xi)d\xi = \int e^{2\pi i\xi\cdot x}\hat{f}(\xi)d\xi = (\hat{f})^\vee(x).$$

Since the LHS converges pointwise to $(\hat{f})^\vee$ and the RHS converges to f in L^1 (which implies pointwise convergence of a subsequence a.e.), the limits must be equal almost everywhere.

It follows that $f = (\hat{f})^\vee$ a.e. Since $(\hat{f})^\vee$ is the Fourier transform of an L^1 function $(\hat{f})$, it is continuous, completing the proof.

Corollary 1.3.3: Injectivity of the Fourier Transform on L^1

If $f \in L^1$ and $\hat{f} = 0$, then $f = 0$ a.e.

Corollary 1.3.4: Fourier isomorphism on Schwartz Space

$\mathcal{F}$ is an isomorphism of $\mathcal{S}$ onto itself.

***Proof*:**
By corollary 1.2.16, $\mathcal{F}$ maps S continuously into itself, and hence so does $f \mapsto f^{\vee}$, since $f^{\vee}(x) = \hat{f}(-x)$. By the Fourier inversion theorem, these maps are inverse to each other.

Corollary 1.3.5: Fourier Transform Has Order 4

Let $\mathcal{F}$ represent the Fourier transform. Then:

$$\mathcal{F}^4 = I$$

and 4 is the smallest such number

***Proof*:**
First:

$$\begin{aligned}
\mathcal{F} \circ \mathcal{F}(f)(x) &= \mathcal{F}(\mathcal{F}(f)) \\
&= \int_{\mathbb{R}^d} \hat{f}(\xi) e^{2\pi i \xi \cdot x} d\xi \\
&= \int_{\mathbb{R}^d} \hat{f}(\xi) e^{2\pi i \xi \cdot (-x)} d\xi \\
&= f(-x) && \text{Fourier inversion formula}
\end{aligned}$$

Hence applying it it again gives the desired result.

We shall give a name to the representation of $f(x)$ in terms of its inversion:

Definition 1.3.6: Fourier Integral

Let $f \in L^1$ and $\hat{f} \in L^1$. Then the *Fourier integral* at x is:

$$f(x) = \int \hat{f}(\xi) e^{2\pi i \xi \cdot x} d\xi$$

An important result using the Fourier inversion theorem is the dual of proposition 1.2.14:

Corollary 1.3.7: Inverse Regularity and Decay

1. **Decay $\Rightarrow$ Smoothness:** If $(1+|\xi|)^k \hat{f}(\xi) \in L^1(\mathbb{R}^n)$ for some integer $k \geq 1$, then $f \in C^k(\mathbb{R}^n)$. The derivative is given by:

$$\partial^\alpha f(x) = \int_{\mathbb{R}^n} (2\pi i \xi)^\alpha \hat{f}(\xi) e^{2\pi i x\cdot\xi}\, d\xi$$

for all multi-indices $|\alpha| \leq k$.

2. **Smoothness $\Rightarrow$ Decay:** If $\hat{f} \in C^k(\mathbb{R}^n)$ and $\partial^\alpha \hat{f} \in L^1(\mathbb{R}^n)$ for all $|\alpha| \leq k$, then f decays rapidly:

$$|f(x)| \leq \frac{C}{(1+|x|)^k}$$

for some constant C.

***Proof*:**
The Inverse Fourier Transform is defined by:

$$f(x) = \int_{\mathbb{R}^n} \hat{f}(\xi) e^{2\pi i x\cdot\xi}\, d\xi.$$

This operator differs from the forward Fourier transform only by the sign of the exponential ($+i$ instead of $-i$). Since $|e^{i\theta}| = |e^{-i\theta}| = 1$, the absolute values used in the norm estimates and convergence arguments of proposition 1.2.14 remain unchanged.

Thus, the proof follows immediately by applying proposition 1.2.14 to the function $\hat{f}$ (with a trivial sign change in the derivative formula).

At last we are in a position to derive the analogue of the orthonormal basis as we have done in theorem 1.2.4:

Theorem 1.3.8: The Plancherel Theorem

If $f \in L^1 \cap L^2$, then $\hat{f} \in L^2$; and $\mathcal{F}|_{L^1\cap L^2}$ extends uniquely to a unitary isomorphism on L^2.

***Proof*:**
Let $X = \{f \in L^1 : \hat{f} \in L^1\}$. Since $\hat{f} \in L^1$ implies $f \in L^\infty$, we have $X \subset L^2$ by [13, L^p superset], and X is dense in L^2 because $\mathcal{S} \subset X$ and $\mathcal{S}$ is dense in L^2 by [13, smooth functions are dense in L^p].

Given $f, g \in X$, let $h = \overline{\hat{g}}$. By the properties of the Fourier transform under complex conjugation and inversion, we have $\hat{h} = \bar{g}$. Now, using the multiplication formula (Parseval's relation for L^1):

$$\int f\bar{g} = \int f\hat{h} = \int \hat{f}h = \int \hat{f}\overline{\hat{g}}.$$

Thus $\mathcal{F}|_X$ preserves the L^2 inner product. In particular, by taking $g = f$, we obtain $\|\hat{f}\|_2 = \|f\|_2$. Since $\mathcal{F}(X) = X$ by the inversion theorem, $\mathcal{F}|_X$ extends by continuity to a unitary isomorphism on L^2.

It remains only to show that this extension agrees with $\mathcal{F}$ on all of $L^1 \cap L^2$. But if $f \in L^1 \cap L^2$ and $g(x) = e^{-\pi|x|^2}$ as in the proof of the inversion theorem, we have $f * g_t \in L^1$ by Young's inequality and $(f * g_t)^\wedge \in L^1$ because $(f * g_t)^\wedge(\xi) = e^{-\pi t^2|\xi|^2}\hat{f}(\xi)$ and $\hat{f}$ is bounded. Hence $f * g_t \in X$; moreover, by [13, Good Kernels Like Mult. Identity], $f * g_t \to f$ in both the L^1 and L^2 norms. Therefore $(f * g_t)^\wedge \to \hat{f}$ both uniformly and in the L^2 norm, completing the proof.

From this, we upgrade The Hausdorff-Young Inequality over $\mathbb{R}^n$:

Theorem 1.3.9: The Hausdorff-Young Inequality

Suppose that $1 \leq p \leq 2$ and q is the conjugate exponent to p. If $f \in L^p(\mathbb{R}^n)$, then

$$\hat{f} \in L^q(\mathbb{R}^n) \qquad \text{and} \qquad \|\hat{f}\|_q \leq \|f\|_p$$

Folland goes on an interesting point about trying to make an $\mathbb{R}^n$ input function periodic which leads to the Poisson Summation Formula; I'm skipping this for now

(here is a theorem I want to add since it related to the analytic theorem)

Theorem 1.3.10: Fourier Decay and Analyticity

Let f be a smooth function such that its Fourier transform $\hat{f}$ satisfies the exponential decay estimate

$$|\hat{f}(\xi)| \leq Ce^{-R2\pi|\xi|}$$

for some constants $C, R > 0$. Then f has a global derivative growth rate α satisfying

$$\alpha \leq \frac{1}{R}.$$

Consequently, by the analyticity theorem of *Elementary Analysis* [2], f is analytic with radius of convergence at least R.

***Proof*:**
Recall that the Fourier transform diagonalizes the differentiation operator. Specifically, the Fourier transform of the k-th derivative is given by $\widehat{f^{(k)}}(\xi) = (2\pi i\xi)^k \hat{f}(\xi)$. By the Fourier inversion formula, we can express the k-th derivative as:

$$f^{(k)}(x) = \int_{-\infty}^{\infty} \widehat{f^{(k)}}(\xi)e^{2\pi i x\xi}\, d\xi = \int_{-\infty}^{\infty} (2\pi i\xi)^k \hat{f}(\xi)e^{2\pi i x\xi}\, d\xi.$$

We wish to bound the magnitude $M_k = \sup_x |f^{(k)}(x)|$. Applying the triangle inequality and the assumed decay bound on $\hat{f}$:

$$\begin{aligned} |f^{(k)}(x)| &\leq \int_{-\infty}^{\infty} |2\pi\xi|^k |\hat{f}(\xi)|\, d\xi \\ &\leq \int_{-\infty}^{\infty} (2\pi|\xi|)^k C e^{-2\pi R|\xi|}\, d\xi \\ &= 2C(2\pi)^k \int_0^{\infty} \xi^k e^{-2\pi R\xi}\, d\xi. \end{aligned}$$

The integral on the right is a standard form related to the Gamma function (specifically, $\int_0^\infty t^k e^{-t} dt = k!$). Making the substitution $u = 2\pi R\xi$:

$$\int_0^{\infty} \xi^k e^{-2\pi R\xi}\, d\xi = \frac{1}{(2\pi R)^{k+1}} \int_0^{\infty} u^k e^{-u}\, du = \frac{k!}{(2\pi R)^{k+1}}.$$

Substituting this back into our estimate for M_k:

$$M_k \leq 2C(2\pi)^k \frac{k!}{(2\pi R)^{k+1}} = \left(\frac{C}{\pi R}\right) k! \left(\frac{1}{R}\right)^k.$$

Now we compute the derivative growth rate α:

$$\begin{aligned} \alpha &= \limsup_{k\to\infty} \sqrt[k]{\frac{M_k}{k!}} \\ &\leq \limsup_{k\to\infty} \sqrt[k]{\frac{C}{\pi R}\left(\frac{1}{R}\right)^k} \\ &= \frac{1}{R}. \end{aligned}$$

Thus the derivative growth rate satisfies $\alpha \leq 1/R$, matching the statement, and f is analytic.

I don't know if I wanna continue this section for now, about Cecaro sums and etc

1.4 Point-wise Convergence of Fourier Series

We have established that for $f \in L^2(\mathbb{T}^n)$, the Fourier series converges to f in the L^2-norm. However, norm convergence does not imply point-wise convergence. We might ask: if we pick a specific point x_0, does the partial sum sequence $S_N(f)(x_0)$ converge to $f(x_0)$?

$$S_N(f)(x) = \sum_{|k|\leq N} \hat{f}(k) e^{2\pi i k\cdot x}$$

Surprisingly, for general integrable functions, the answer is often no. The mechanism governing this convergence is the *Dirichlet Kernel*.

Recall that convolution in the spatial domain corresponds to multiplication in the frequency domain. The partial sum $S_N(f)$ is the convolution of f with the inverse Fourier transform of the characteristic function of the range $[-N, N]$.

Definition 1.4.1: Dirichlet Kernel

The *Dirichlet Kernel* D_N on $\mathbb{T}^1$ is defined as:

$$D_N(x) = \sum_{k=-N}^{N} e^{2\pi i k x} = \frac{\sin((2N+1)\pi x)}{\sin(\pi x)}$$

Consequently, the partial sums of the Fourier series are given by:

$$S_N(f)(x) = (f * D_N)(x) = \int_{\mathbb{T}} f(x-t) D_N(t) dt$$

The equality in the above equation is a simple application of the geometric series with ratio $r = e^{2\pi i x}$:

$$\begin{aligned}\sum_{k=-N}^{N} r^k = r^{-N}\frac{1-r^{2N+1}}{1-r} &= \frac{r^{-N}-r^{N+1}}{1-r} \\ &= \frac{r^{-(N+1/2)}-r^{N+1/2}}{r^{-1/2}-r^{1/2}} \quad \text{(multiplying top and bottom by } r^{-1/2}\text{)} \\ &= \frac{e^{-2\pi i(N+1/2)x}-e^{2\pi i(N+1/2)x}}{e^{-\pi i x}-e^{\pi i x}} \\ &= \frac{-2i\sin((2N+1)\pi x)}{-2i\sin(\pi x)} = \frac{\sin((2N+1)\pi x)}{\sin(\pi x)}\end{aligned}$$

The convergence of $S_N(f)$ depends on whether D_N is a "good kernel" (an approximate identity). While $\int D_N = 1$, it fails the absolute integrability condition: $\int |D_N| \to \infty$ as $N \to \infty$. Specifically, $\|D_N\|_1 \sim \frac{4}{\pi^2}\log N$. This logarithmic growth allows for pathological behavior.

Theorem 1.4.2: Failure of Pointwise Convergence

1. (Du Bois-Reymond) There exists $f \in C(\mathbb{T})$ whose Fourier series diverges at a prescribed point. More precisely, the set of such f is residual in $C(\mathbb{T})$.
2. As a consequence, there exists $f \in C(\mathbb{T})$ whose Fourier series diverges at a dense set of points.
3. (Kolmogorov, 1923) There exists $f \in L^1(\mathbb{T})$ whose Fourier series diverges *everywhere*.

***Proof*:**
We prove (1) and (2) using the principle of uniform boundedness ([13, Principle Of Uniform Boundedness]); item (3) requires a separate and more delicate construction. This argument was previewed in [13, Divergence of Fourier Series].

Step 1: Setting up the functionals. Fix a point $x_0 \in \mathbb{T}$ (for concreteness, $x_0 = 0$). For each $N \in \mathbb{N}$, define the linear functional $\Lambda_N : C(\mathbb{T}) \to \mathbb{C}$ by:

$$\Lambda_N(f) = S_N f(0) = \int_{\mathbb{T}} f(t) D_N(t)\, dt$$

where we used the fact that D_N is even (definition 1.4.1). Each Λ_N is a bounded linear functional on $C(\mathbb{T})$ (equipped with the supremum norm $\|\cdot\|_\infty$), since:

$$|\Lambda_N(f)| \leq \int_{\mathbb{T}} |f(t)||D_N(t)|\,dt \leq \|f\|_\infty \cdot \|D_N\|_{L^1(\mathbb{T})}$$

Step 2: Computing the operator norms. We claim $\|\Lambda_N\| = L_N$ where $L_N = \|D_N\|_{L^1(\mathbb{T})}$ is the N-th *Lebesgue constant*. The upper bound $\|\Lambda_N\| \leq L_N$ was shown above. For the lower bound, let $g = \operatorname{sgn}(D_N)$. Since D_N is continuous with finitely many zeros, g is a piecewise constant function with finitely many jumps, and hence can be uniformly approximated by continuous functions f_ϵ with $\|f_\epsilon\|_\infty \leq 1$ and:

$$|\Lambda_N(f_\epsilon)| = \left|\int_{\mathbb{T}} f_\epsilon(t) D_N(t)\,dt\right| > L_N - \epsilon$$

Since $\epsilon > 0$ was arbitrary, $\|\Lambda_N\| = L_N$.

Step 3: The Lebesgue constants diverge. We show $L_N \to \infty$. Using $|D_N(x)| = \left|\frac{\sin((2N+1)\pi x)}{\sin(\pi x)}\right|$ and the bound $|\sin(\pi x)| \leq \pi|x|$:

$$L_N = \int_{-1/2}^{1/2} |D_N(x)|\,dx \geq 2\int_0^{1/2} \frac{|\sin((2N+1)\pi x)|}{\pi x}\,dx$$

Substituting $u = (2N+1)\pi x$:

$$L_N \geq \frac{2}{\pi}\int_0^{(N+1/2)\pi} \frac{|\sin u|}{u}\,du \geq \frac{2}{\pi}\sum_{k=1}^{N}\int_{(k-1)\pi}^{k\pi} \frac{|\sin u|}{u}\,du$$

On $[(k-1)\pi,\, k\pi]$, we have $u \leq k\pi$ and $\int_0^\pi |\sin u|\,du = 2$, giving:

$$L_N \geq \frac{2}{\pi}\sum_{k=1}^{N}\frac{2}{k\pi} = \frac{4}{\pi^2}\sum_{k=1}^{N}\frac{1}{k} \sim \frac{4}{\pi^2}\log N \xrightarrow{N\to\infty} \infty$$

Step 4: Applying the uniform boundedness principle. The space $C(\mathbb{T})$ is a Banach space and $\sup_N \|\Lambda_N\| = \sup_N L_N = \infty$. By the *contrapositive* of the principle of uniform boundedness, the pointwise boundedness hypothesis must fail: there cannot hold $\sup_N |\Lambda_N(f)| < \infty$ for every $f \in C(\mathbb{T})$. Hence there exists $f \in C(\mathbb{T})$ such that:

$$\sup_N |S_N f(0)| = \sup_N |\Lambda_N(f)| = \infty$$

and so the Fourier series of f diverges at $x_0 = 0$. By translation (replacing f with $\tau_{x_0} f$ and noting that $\|\tau_{x_0} f\|_\infty = \|f\|_\infty$), the same holds at any prescribed point $x_0 \in \mathbb{T}$.

In fact, the argument gives more. Inspecting the proof of the uniform boundedness principle ([13, Principle Of Uniform Boundedness]), the sets $E_n = \{f \in C(\mathbb{T}) \mid \sup_N |S_N f(0)| \leq n\}$ are closed in $C(\mathbb{T})$, and since $\sup_N \|\Lambda_N\| = \infty$, each E_n has empty interior. Hence $\bigcup_n E_n = \{f \mid \sup_N |S_N f(0)| < \infty\}$ is meager, and the set of f whose Fourier series diverges at 0 is *residual.* This proves (1).

Step 5: Divergence at a dense set. For (2), let $\{x_k\}_{k=1}^\infty$ be a countable dense subset of $\mathbb{T}$. By (1), for each k the set:

$$R_{x_k} = \left\{ f \in C(\mathbb{T}) \;\middle|\; \sup_N |S_N f(x_k)| = \infty \right\}$$

is residual. A countable intersection of residual sets is residual by the Baire Category Theorem ([13, Baire Category Theorem I]), so $\bigcap_k R_{x_k}$ is residual and in particular nonempty. Any f in this intersection has a Fourier series diverging at every point of the dense set $\{x_k\}$.

Notice the non-constructive nature of this argument: we proved the *existence* of a continuous function with divergent Fourier series without ever exhibiting one. The uniform boundedness principle, itself a consequence of the Baire Category Theorem, guarantees the existence of such pathological functions purely from the growth of the Lebesgue constants. The construction of an explicit example is considerably more involved (see the original work of Du Bois-Reymond, 1876).

The divergence of the Lebesgue constants has a second striking consequence, this time about the range of the Fourier transform as a map between Banach spaces. Recall that we denote by $c_0(\mathbb{Z})$ the Banach space of all sequences $(a_k)_{k\in\mathbb{Z}}$ of complex numbers satisfying $a_k \to 0$ as $|k| \to \infty$, equipped with the supremum norm $\|(a_k)\|_\infty = \sup_k |a_k|$.

For $f \in L^1(\mathbb{T})$, the bound $|\hat{f}(k)| \leq \|f\|_1$ shows that $\hat{f} \in \ell^\infty(\mathbb{Z})$. In fact, $\hat{f} \in c_0(\mathbb{Z})$: trigonometric polynomials are dense in $L^1(\mathbb{T})$ (by Stone-Weierstrass and the density of $C(\mathbb{T})$ in $L^1(\mathbb{T})$), their Fourier transforms are finitely supported, and $\|\hat{f} - \hat{p}\|_\infty \leq \|f - p\|_1$, so $\hat{f}$ is a uniform limit of sequences with finite support, hence decays to zero. We may thus view the Fourier transform as a bounded linear map $\mathcal{F} : L^1(\mathbb{T}) \to c_0(\mathbb{Z})$. It is natural to ask whether every sequence decaying to zero arises as the Fourier transform of some integrable function. The open mapping theorem answers this decisively:

Proposition 1.4.3: Non-Surjectivity of the Fourier Transform on L^1

The Fourier transform $\mathcal{F} : L^1(\mathbb{T}) \to c_0(\mathbb{Z})$ is injective, bounded, and has dense image, but is *not* surjective. In particular, $\mathcal{F}(L^1(\mathbb{T}))$ is a proper dense subspace of $c_0(\mathbb{Z})$ that is not closed.

***Proof*:**

Boundedness was established above ($\|\hat{f}\|_\infty \leq \|f\|_1$).

Injectivity. If $\hat{f}(k) = 0$ for all $k \in \mathbb{Z}$, then $\langle f, E_k\rangle = 0$ for all k, where $E_k(x) = e^{2\pi ikx}$. Since finite linear combinations of the E_k are uniformly dense in $C(\mathbb{T})$ (by Stone-Weierstrass, as used in theorem 1.2.4), it follows that $\int_{\mathbb{T}} f(x)\overline{g(x)}\,dx = 0$ for every $g \in C(\mathbb{T})$, whence $f = 0$ a.e.

Dense image. Every finitely supported sequence in $c_0(\mathbb{Z})$ is the Fourier transform of a trigonometric polynomial (which is in $L^1(\mathbb{T})$), and finitely supported sequences are dense in $c_0(\mathbb{Z})$.

Non-surjectivity. Suppose for contradiction that $\mathcal{F}$ is surjective. Since $\mathcal{F} : L^1(\mathbb{T}) \to c_0(\mathbb{Z})$ is then a continuous bijection between Banach spaces, the Banach isomorphism theorem ([13, Bijective And Continuous]) furnishes a bounded inverse. That is, there exists a constant $C > 0$ such that:

$$\|f\|_{L^1} \leq C\|\hat{f}\|_\infty \qquad \text{for all } f \in L^1(\mathbb{T})$$

Apply this to the Dirichlet kernel $D_N \in L^1(\mathbb{T})$. Since $\hat{D}_N(k) = 1$ for $|k| \leq N$ and $\hat{D}_N(k) = 0$ for $|k| > N$, we have $\|\hat{D}_N\|_\infty = 1$. The bound above then gives $\|D_N\|_{L^1} \leq C$ for all N. But as computed in the proof of theorem 1.4.2:

$$\|D_N\|_{L^1} \geq \frac{4}{\pi^2}\sum_{k=1}^{N}\frac{1}{k} \xrightarrow{N\to\infty} \infty$$

a contradiction. Hence $\mathcal{F}$ is not surjective, and since its image is dense but not all of $c_0(\mathbb{Z})$, the image is not closed.

This result illustrates a theme we will encounter repeatedly: the space L^1 is "too small" for the Fourier transform to be well-behaved. The failure of surjectivity means that there exist sequences decaying to zero that cannot arise as the Fourier coefficients of any integrable function. The constraints imposed by integrability on the Fourier side are strictly stronger than decay at infinity. Characterizing the precise image $\mathcal{F}(L^1(\mathbb{T})) \subsetneq c_0(\mathbb{Z})$ remains an open problem.

Note also that the open mapping theorem is used here in its contrapositive form: the non-openness of $\mathcal{F}$ (manifested through the unbounded ratio $\|D_N\|_1/\|\hat{D}_N\|_\infty$) forces the failure of surjectivity. The same Dirichlet kernel that produces divergent Fourier series (theorem 1.4.2) is the obstruction to surjectivity, a coincidence that is not accidental: both phenomena stem from the logarithmic growth of the Lebesgue constants.

This shows that continuity alone is *not* strong enough to guarantee that we can recover the function point-wise from its frequencies using the standard summation. To guarantee convergence, we need slightly more regularity than continuity. We look for conditions that cancel out the oscillations of the Dirichlet kernel.

Theorem 1.4.4: Pointwise Convergence Tests

Let $f \in L^1(\mathbb{T})$.

1. *Dini's Criterion (Local):* If f satisfies a local Hölder condition at x (or more generally, $\int_{-1/2}^{1/2} |\frac{f(x+t)-f(x)}{t}|dt < \infty$), then $S_N(f)(x) \to f(x)$.
2. *Dirichlet-Jordan Test (Global):* If f is of *Bounded Variation* on $\mathbb{T}$, then for every x:

$$\lim_{N\to\infty} S_N(f)(x) = \frac{f(x^+) + f(x^-)}{2}$$

***Proof*:**
We prove Dini's Criterion. We wish to show $S_N(f)(x) - f(x) \to 0$. Using the fact that $\int D_N(t)dt = 1$, we write:

$$S_N(f)(x) - f(x) = \int_{-1/2}^{1/2} (f(x-t) - f(x))D_N(t)dt$$

Substitute the explicit formula for $D_N(t)$:

$$= \int_{-1/2}^{1/2} \frac{f(x-t) - f(x)}{\sin(\pi t)} \sin((2N + 1)\pi t)dt$$

Define $g(t) = \frac{f(x-t)-f(x)}{t} \cdot \frac{t}{\sin(\pi t)}$. The term $\frac{t}{\sin(\pi t)}$ is bounded and smooth near $t = 0$. By the Dini condition, $\frac{f(x-t)-f(x)}{t}$ is integrable near 0. Thus $g(t) \in L^1(\mathbb{T})$. The expression becomes:

$$\int_{-1/2}^{1/2} g(t) \sin((2N + 1)\pi t)dt$$

This is simply the imaginary part of the $(2N + 1)$-th Fourier coefficient of $g(t)$. By the Riemann-Lebesgue Lemma, $\hat{g}(k) \to 0$ as $k \to \infty$. Thus the integral vanishes as $N \to \infty$.

If we cannot guarantee convergence for continuous functions, we can modify our summation method. Instead of partial sums, we take the average of partial sums (Cesàro means).

Theorem 1.4.5: Fejér's Theorem

Let $\sigma_N(f) = \frac{1}{N+1}\sum_{n=0}^{N} S_n(f)$ be the Cesàro means. Then $\sigma_N(f) = f * K_N$, where K_N is the *Fejér Kernel*:

$$K_N(x) = \frac{1}{N+1}\left(\frac{\sin((N+1)\pi x)}{\sin(\pi x)}\right)^2$$

Unlike the Dirichlet kernel, $K_N(x) \geq 0$ and $\int K_N = 1$. Consequently, if $f \in C(\mathbb{T})$, then $\sigma_N(f) \to f$ *uniformly.*

***Proof*:**
The formula for K_N follows from summing the geometric series for D_n. The key properties are: 1. $K_N(x) \geq 0$. 2. $\int_{-1/2}^{1/2} K_N(x)dx = 1$. 3. For any $\delta > 0$, $\int_{|x|\geq\delta} K_N(x)dx \to 0$ as $N \to \infty$ (Mass concentration).

Let f be continuous on $\mathbb{T}$. Since $\mathbb{T}$ is compact, f is uniformly continuous. Given $\epsilon > 0$, there exists $\delta > 0$ such that $|f(x-t) - f(x)| < \epsilon/2$ whenever $|t| < \delta$.

$$\begin{aligned}|\sigma_N(f)(x) - f(x)| &= \left|\int_{-1/2}^{1/2}(f(x-t) - f(x))K_N(t)dt\right| \\ &\leq \int_{-1/2}^{1/2}|f(x-t) - f(x)|K_N(t)dt\end{aligned}$$

Split the integral into region $A = \{|t| < \delta\}$ and $B = \{|t| \geq \delta\}$. On A: $|f(x-t) - f(x)| < \epsilon/2$. Since $\int K_N = 1$, the integral over A is $\leq \epsilon/2$. On B: $|f(x-t) - f(x)| \leq 2\|f\|_\infty$. The integral is bounded by $2\|f\|_\infty \int_{|t|\geq\delta} K_N(t)dt$. Using the explicit formula, for $|t| \geq \delta$, $K_N(t) \leq \frac{1}{N+1}\frac{1}{\sin^2(\pi\delta)}$. As $N \to \infty$, this goes to 0. Thus for large N, the second term is $< \epsilon/2$. The convergence is uniform because δ depended only on the uniform continuity of f, not on x.

1.4.6 Historical Note For a long time, it was unknown if $f \in L^2(\mathbb{T})$ converged pointwise. In 1966, Lennart Carleson proved that if $f \in L^2(\mathbb{T})$, then $S_N(f)(x) \to f(x)$ for *almost every* x. This was later extended by Hunt to all L^p for $p > 1$. This is a deep and difficult result, requiring a careful decomposition of the operator to avoid the logarithmic divergence of L^1.

1.5 *From Harmonic Analysis to Number Theory

The Fourier analysis of this book is the opening chapter of a long story that runs through modern number theory. The abelian Pontryagin duality of part II is the GL_1 case of the Langlands program, and its non-abelian and adelic generalizations organize much of arithmetic. Rather than develop these here, this starred section records the directions and where each is taken up; every entry is a forward pointer, and the pertinent volumes carry the development.

- *Unitary representations and the abstract Plancherel theorem* generalize the Fourier transform and Pontryagin duality to non-abelian locally compact groups, in a companion volume on unitary representation theory.
- *Tate's thesis* recasts the analytic continuation and functional equation of the Riemann zeta and Dirichlet L-functions as Fourier analysis on the adeles, a single global LCA group; it belongs to the theory of automorphic forms.
- The *Selberg and Arthur trace formulas* are the non-abelian replacement for the Poisson summation formula; the *Satake isomorphism*, *Eisenstein series and the spectral decomposition*, and *Langlands functoriality* are the further structures of the automorphic theory.
- The *Peter-Weyl theorem* and the *Plancherel formula* for compact and reductive groups are the harmonic analysis that makes the automorphic spectrum computable.

1.6 *Kakeya Sets and the Fourier Transform

The Kakeya problem originated in geometric measure theory, but its influence extends into the heart of harmonic analysis. The connection is through the Fourier transform: the geometric arrangement of tubes in physical space is dual to the arrangement of frequency caps on the sphere in frequency space, and the Kakeya conjecture places fundamental constraints on how efficiently these caps can be combined. This section develops this connection, starting with Fefferman's 1971 theorem on the ball multiplier (the result that first brought Besicovitch sets into Fourier analysis) and proceeding to the restriction conjecture and its relatives. The tools used here—the Fourier transform on L^1 and L^2 (definition 1.2.11, theorem 1.3.8), Young's inequality ([13, Young's convolution inequality]), and the L^p theory of the Hilbert transform (theorem 2.3.4)—are all developed earlier in this book.

In one dimension, the operation of truncating the Fourier transform to an interval is well-controlled: if $f \in L^p(\mathbb{R})$ for $1 < p < \infty$, then the function whose Fourier transform is $\hat{f} \cdot \chi_{[-R,R]}$ is again in L^p with norm bounded by $C_p\|f\|_p$. This is essentially the content of the L^p boundedness of the Hilbert transform (theorem 2.3.4), since the multiplier $\chi_{[-R,R]}$ can be expressed in terms of $\operatorname{sgn}(\xi)$ (the Hilbert transform multiplier) via $\chi_{[-R,R]}(\xi) = \frac{1}{2}(\operatorname{sgn}(\xi + R) - \operatorname{sgn}(\xi - R))$. The natural question is whether this extends to higher dimensions: if $f \in L^p(\mathbb{R}^n)$ for $n \geq 2$, is the function obtained by truncating $\hat{f}$ to a ball again in L^p?

Definition 1.6.1: Ball Multiplier

Let $R > 0$ and $n \geq 1$. The *ball multiplier* (or *disc multiplier* when $n = 2$) is the operator $S_R : L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$ defined by

$$\widehat{S_R f}(\xi) = \hat{f}(\xi) \cdot \chi_{B(0,R)}(\xi)$$

where $B(0,R) = \{\xi \in \mathbb{R}^n \mid |\xi| \leq R\}$ is the closed ball of radius R in frequency space. In physical space, $S_R f = f * \check{\chi}_{B(0,R)}$, where $\check{\chi}_{B(0,R)}$ is the inverse Fourier transform of the indicator function of the ball.

On $L^2(\mathbb{R}^n)$, the ball multiplier is bounded for all n with operator norm 1: by theorem 1.3.8, $\|S_R f\|_2 = \|\hat{f} \cdot \chi_{B(0,R)}\|_2 \leq \|\hat{f}\|_2 = \|f\|_2$. The question is whether S_R extends to a bounded operator on $L^p(\mathbb{R}^n)$ for $p \neq 2$. By duality (S_R is self-adjoint), boundedness on L^p implies boundedness on $L^{p'}$ where $1/p + 1/p' = 1$, so the question reduces to $1 < p < 2$ and its dual $2 < p < \infty$.

For $n = 1$, as noted above, S_R is bounded on $L^p(\mathbb{R})$ for all $1 < p < \infty$. This follows from the Hilbert transform connection and theorem 2.3.4. For $n \geq 2$, the situation is fundamentally different:

Theorem 1.6.2: Fefferman's Theorem on the Ball Multiplier

For $n \geq 2$, the ball multiplier S_R is bounded on $L^p(\mathbb{R}^n)$ if and only if $p = 2$.

The L^2 boundedness is immediate from Plancherel. The content of the theorem is that S_R is *un*bounded on L^p for every $p \neq 2$. The proof that S_R fails to be bounded uses the existence of Besicovitch sets ([14, Existence of Measure-Zero Besicovitch Sets]), establishing the first direct connection between the Kakeya problem and Fourier analysis.

***Proof Key ideas*:**
The proof constructs a sequence of functions $f_N \in L^p(\mathbb{R}^n)$ with $\|f_N\|_p = 1$ but $\|S_R f_N\|_p \to \infty$, using Besicovitch-type geometry.

Step 1: Angular decomposition. Decompose the ball $B(0, R)$ in frequency space into $N \sim \delta^{-(n-1)}$ thin sectors $\{\Omega_j\}_{j=1}^N$, each subtending a solid angle $\sim \delta^{n-1}$ and centered around a direction $e_j \in S^{n-1}$. Define the sector multiplier $S_j f = (\hat{f} \cdot \chi_{\Omega_j})^\vee$, so that $S_R f \approx \sum_j S_j f$ (up to errors from the boundary of the ball and the smooth decomposition).

Step 2: Wave packet interpretation. Each sector multiplier S_j localizes $\hat{f}$ to a thin region of frequency space oriented along e_j. By the uncertainty principle (proposition 1.2.14), a function whose Fourier transform is supported in a δ-neighborhood of a cap on S^{n-1} centered at e_j is spread in physical space along a tube of length $\sim 1/\delta$ and width ~ 1 in direction e_j. The operator S_j thus maps functions to "wave packets" concentrated along tubes in direction e_j. The physical-space geometry of these wave packets is controlled by the frequency-space geometry of the sectors, and the tube directions $\{e_j\}$ are δ-separated by construction.

Step 3: Constructing the test function. Choose a Besicovitch set $K \subseteq \mathbb{R}^n$ with $m(K) < \epsilon$ containing a unit segment in every direction ([14, Existence of Measure-Zero Besicovitch Sets]). For each j, let g_j be a wave packet concentrated on a δ-tube T_j in direction e_j that lies inside K_δ (the δ-neighborhood of K). The key properties are:

1. Each g_j has $\|g_j\|_p \sim \delta^{(n-1)/p}$ (the L^p norm of a function supported on a tube of volume $\sim \delta^{n-1}$).
2. The g_j have approximately disjoint Fourier supports (since the sectors Ω_j are disjoint), so $\|\sum_j g_j\|_2^2 \approx \sum_j \|g_j\|_2^2$ by Plancherel.
3. In physical space, the supports of g_j overlap inside K_δ, which has measure $\sim \epsilon$.

Step 4: Norm comparison. Let $f_N = \sum_j g_j$. By property (2) and the fact that the g_j have comparable magnitudes, $\|f_N\|_2^2 \sim N \cdot \delta^{n-1} \sim 1$. However, since $S_R f_N \approx f_N$ (the Fourier transforms of the g_j are already supported inside $B(0, R)$), the L^p norm of $S_R f_N$ is controlled from below by the concentration of f_N inside K_δ. For $p < 2$, Hölder's inequality gives

$$\|f_N\|_p \leq |K_\delta|^{1/p-1/2} \|f_N\|_2 \sim \epsilon^{1/p-1/2}$$

For this to be bounded while $\|S_R f_N\|_p = \|f_N\|_p$ stays comparable to $\|f_N\|_2 \sim 1$, the ratio $\|S_R f_N\|_p / \|f_N\|_p \gtrsim \epsilon^{1/2-1/p}$, which diverges as $\epsilon \to 0$. For $p > 2$, the dual argument gives the same conclusion.

This shows S_R cannot be bounded on $L^p(\mathbb{R}^n)$ for $p \neq 2$, since the operator norm would have to exceed $\epsilon^{1/2-1/p}$ for arbitrarily small ϵ.

The proof reveals the mechanism by which Besicovitch sets obstruct L^p boundedness: the wave packets g_j are concentrated in a set of small measure (the Besicovitch set), which inflates their L^p norm for $p \neq 2$ relative to their L^2 norm. The ball multiplier cannot "see" this concentration because it acts in frequency space, where the wave packets are spread out over disjoint sectors. The tension between frequency-space orthogonality and physical-space concentration is the core of the argument.

Example 1.6.3: Wave Packets and Tubes

In $\mathbb{R}^2$, let $\delta = 1/N$ and consider a function g whose Fourier transform is a smooth bump supported in a δ-cap on S^1 near direction $e = (1, 0)$. The Fourier support is a region of dimensions $\sim 1 \times \delta$ (radial $\times$ angular). By the uncertainty principle, g is concentrated in physical space on a tube of dimensions $\sim 1 \times 1/\delta = 1 \times N$ (perpendicular $\times$ parallel to e). The L^2 norm of g is $\sim \delta^{1/2}$ (since $\|\hat{g}\|_2 \sim (\text{area of Fourier support})^{1/2} \sim \delta^{1/2}$), while its L^p norm for $p < 2$ is $\sim \delta^{1/2} \cdot (\text{tube volume})^{1/p-1/2} \sim \delta^{1/p}$. The ratio $\|g\|_p / \|g\|_2 \sim \delta^{1/p-1/2}$, which grows as $\delta \to 0$ when $p < 2$: a single wave packet already shows the norm distortion, and summing over N wave packets in a Besicovitch configuration amplifies it.

Fefferman's theorem demonstrates that the Kakeya geometry imposes a hard limit on Fourier multiplier estimates: the ball multiplier fails for $p \neq 2$ precisely because Besicovitch sets exist. The theorem resolved a question that had been open since the work of Bochner in the 1930s, and it shifted the focus of harmonic analysis toward understanding how the geometry of frequency truncation interacts with L^p norms.

The failure of the ball multiplier does not mean that frequency truncation is impossible for $p \neq 2$; it means that the *sharp* truncation $\chi_{B(0,R)}$ is the wrong tool. Two natural modifications lead to active research programs. The first is to *smooth* the truncation, replacing the sharp indicator with a function that decays continuously near $|\xi| = R$; this leads to the Bochner–Riesz conjecture. The second is to *restrict* the Fourier transform to the boundary sphere S^{n-1} rather than the solid ball; this leads to the restriction conjecture. Both modifications exploit the curvature of the sphere (which the flat faces of a cube, for instance, would not provide), and both are intimately connected to the Kakeya conjecture.

The restriction problem asks: can the Fourier transform $\hat{f}$ of an L^p function be meaningfully restricted to a curved surface $\Sigma \subseteq \mathbb{R}^n$ (like the sphere S^{n-1})? For a flat surface (such as a hyperplane), this is impossible in general: $\hat{f}$ for $f \in L^p(\mathbb{R}^n)$ with $p > 1$ is an $L^{p'}$ function, not a continuous function, and restricting an $L^{p'}$ function to a set of Lebesgue measure zero is not meaningful. For curved surfaces, the situation is fundamentally different. The curvature of Σ causes the Fourier transform of the surface measure σ to decay at infinity (by stationary phase), and this decay translates into regularity of the extension operator $(g\,d\sigma)^\vee$. The decay rate governs the range of L^p spaces for which restriction is possible: stronger curvature gives faster decay, which in turn gives a larger range of admissible p.

Theorem 1.6.4: Tomas–Stein Restriction Theorem

Let σ denote the surface measure on $S^{n-1} \subseteq \mathbb{R}^n$. For $1 \leq p \leq \frac{2(n+1)}{n+3}$:

$$\|\hat{f}\|_{L^2(S^{n-1},d\sigma)} \leq C\|f\|_{L^p(\mathbb{R}^n)}$$

for all $f \in \mathcal{S}(\mathbb{R}^n)$ (and hence for all $f \in L^p(\mathbb{R}^n)$ by density).

The exponent $p_0 = 2(n+1)/(n+3)$ is called the *Tomas–Stein exponent*. For $n = 2$, $p_0 = 6/5$; for $n = 3$, $p_0 = 4/3$. The theorem says that the Fourier transform of an L^{p_0} function, despite being only an $L^{p_0'}$ function in general, has enough regularity to be restricted to the sphere as an L^2 function.

Proof Sketch:
The proof uses the TT^* method, a duality argument that reduces the estimate to a convolution bound.

Step 1: Duality. The restriction estimate $\|\hat{f}\|_{L^2(d\sigma)} \leq C\|f\|_{L^p}$ is equivalent, by duality, to the *extension estimate*:

$$\|(g\,d\sigma)^\vee\|_{L^{p'}(\mathbb{R}^n)} \leq C\|g\|_{L^2(S^{n-1},d\sigma)}$$

for all $g \in L^2(S^{n-1})$, where $(g\,d\sigma)^\vee(x) = \int_{S^{n-1}} g(\omega)e^{2\pi i x\cdot\omega}\,d\sigma(\omega)$ is the inverse Fourier transform of the measure $g\,d\sigma$.

Step 2: TT^* reduction. To bound $\|(g\,d\sigma)^\vee\|_{p'}$, consider the operator $T : L^2(S^{n-1}) \to L^{p'}(\mathbb{R}^n)$ defined by $Tg = (g\,d\sigma)^\vee$. The TT^* method reduces the $L^2 \to L^{p'}$ bound to an $L^p \to L^{p'}$ bound on the composition TT^*. A computation shows that $TT^*f = f * \hat{\sigma}$, where $\hat{\sigma}(x) = \int_{S^{n-1}} e^{2\pi i x\cdot\omega}\,d\sigma(\omega)$ is the Fourier transform of the surface measure.

Step 3: Decay of $\hat{\sigma}$. The key analytic input is the decay estimate $|\hat{\sigma}(x)| \lesssim (1 + |x|)^{-(n-1)/2}$, which holds by stationary phase (the curvature of S^{n-1} causes cancellation in the oscillatory integral). This decay estimate, combined with Young's inequality ([13, Young's convolution inequality]) for convolutions, gives

$$\|f * \hat{\sigma}\|_{p'} \leq \|\hat{\sigma}\|_r \|f\|_p$$

where $1+1/p' = 1/r+1/p$, and $\|\hat{\sigma}\|_r < \infty$ when $r(n-1)/2 > n$, i.e., $r > 2n/(n-1)$. Optimizing gives the exponent $p_0 = 2(n+1)/(n+3)$.

The Tomas–Stein theorem is sharp in its method (TT^* cannot give a better exponent) but is not expected to be sharp in its conclusion. The restriction conjecture predicts that the estimate should hold for a larger range of exponents:

Conjecture 1.6.5: Restriction Conjecture

For $n \geq 2$ and $1 \leq p < \frac{2n}{n+1}$, there exists $C = C(n,p) > 0$ such that

$$\|\hat{f}\|_{L^p(S^{n-1},d\sigma)} \leq C\|f\|_{L^p(\mathbb{R}^n)}$$

for all $f \in L^p(\mathbb{R}^n)$. More generally, the estimate is expected to hold with $L^p(S^{n-1})$ replaced by $L^q(S^{n-1})$ for appropriate (p,q) pairs.

The restriction conjecture implies the Kakeya set conjecture. The implication goes through wave packet decomposition: if the Kakeya conjecture failed (i.e., Besicovitch sets in $\mathbb{R}^n$ could have Hausdorff dimension $< n$), then one could construct families of wave packets concentrated on a low-dimensional set whose restriction norms would violate the conjectured bounds, using the same mechanism as in Fefferman's proof. The precise implication is: the restriction conjecture for the paraboloid $\left\{(\xi, |\xi|^2) \;\middle|\; \xi \in \mathbb{R}^{n-1}\right\} \subseteq \mathbb{R}^n$ implies the Kakeya set conjecture in $\mathbb{R}^n$. The converse is not known (and is likely false), so the restriction conjecture is strictly stronger.

The restriction conjecture fits into a hierarchy of conjectures, each implying the next. The hierarchy involves progressively "smoother" ways of truncating the Fourier transform near the sphere:

Definition 1.6.6: Bochner–Riesz Means

For $\alpha \geq 0$ and $R > 0$, the *Bochner–Riesz mean of order* α is the operator $S_R^\alpha : L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$ defined by

$$\widehat{S_R^\alpha f}(\xi) = \hat{f}(\xi) \left(1 - \frac{|\xi|^2}{R^2}\right)_+^\alpha$$

where $(t)_+ = \max(t, 0)$. When $\alpha = 0$, $S_R^0 = S_R$ is the ball multiplier.

The parameter α controls the smoothness of the frequency cutoff: $\alpha = 0$ gives the sharp indicator function (which fails for $p \neq 2$ by Fefferman's theorem), while increasing α makes the cutoff progressively smoother near $|\xi| = R$. The Bochner–Riesz conjecture asserts that sufficient smoothness restores L^p boundedness:

Conjecture 1.6.7: Bochner–Riesz Conjecture

For $n \geq 2$, S_R^α is bounded on $L^p(\mathbb{R}^n)$ for $\alpha > \max(n|1/p - 1/2| - 1/2, 0)$ and $1 < p < \infty$. The critical index $\alpha(p) = n|1/p - 1/2| - 1/2$ is sharp.

At the top of the hierarchy sits the local smoothing conjecture, which concerns solutions to the wave equation. The half-wave operator $e^{it\sqrt{-\Delta}}$ maps initial data f to the solution $u(x,t) = e^{it\sqrt{-\Delta}} f(x)$ of $i\partial_t u + \sqrt{-\Delta}\, u = 0$ (here $\sqrt{-\Delta}$ is the operator with Fourier multiplier $|\xi|$, so $\widehat{u}(\xi, t) = e^{2\pi i t|\xi|} \hat{f}(\xi)$). At any fixed time t, Sobolev embedding (theorem 2.5.3) gives $\|u(\cdot, t)\|_{L^p} \leq C\|f\|_{H^{s(p)}}$ where $s(p) = n(1/2 - 1/p)$. The local smoothing conjecture predicts that *averaging over a time interval* gains nearly $1/2$ of a derivative:

Conjecture 1.6.8: Local Smoothing Conjecture

For $n \geq 2$, $p > \frac{2n}{n-1}$, and every $\epsilon > 0$:

$$\left(\int_1^2 \|e^{it\sqrt{-\Delta}} f\|_{L^p(\mathbb{R}^n)}^p \, dt\right)^{1/p} \leq C_\epsilon \|f\|_{H^{s(p)-1/2+\epsilon}(\mathbb{R}^n)}$$

where $s(p) = n(1/2 - 1/p)$. The gain of $1/2 - \epsilon$ derivatives over the fixed-time estimate reflects the dispersive effect of time-averaging: oscillations in different directions cancel over time, producing additional regularity.

The conjecture was formulated by Sogge in 1991 and resolved in dimension $n = 2$ by Guth, Wang, and Zhang in 2020. It remains open for all $n \geq 3$.

The four central conjectures in this area form a chain of implications, each building on the geometric and analytic content of the previous one:

$$\text{Local Smoothing} \Rightarrow \text{Bochner–Riesz} \Rightarrow \text{Restriction} \Rightarrow \text{Kakeya}$$

The locl smoothing conjecture (**??** 1.6.8) sits at the top of the hierarchy: it is the most difficult and implies all the others. The chain of implications has been established over decades of work by Stein, Fefferman, Córdoba, Bourgain, Wolff, Tao, and many others. Each implication is nontrivial and passes through specific geometric-analytic mechanisms:

1. Restriction $\Rightarrow$ Kakeya: the restriction estimate for the sphere controls the L^p norm of superpositions of wave packets, and if Besicovitch sets of low dimension existed, one could construct wave packet sums violating this control (as in Fefferman's proof).
2. Bochner–Riesz $\Rightarrow$ Restriction: the Bochner–Riesz means S_R^α provide a smooth approximation to the restriction operator, and L^p bounds on S_R^α at the critical index imply restriction estimates by a limiting argument as $\alpha \to 0$.
3. Local Smoothing $\Rightarrow$ Bochner–Riesz: the local smoothing estimate for the wave equation controls the space-time L^p norm of wave propagation, and the Bochner–Riesz means can be recovered as time-averages of the wave solution.

The two-dimensional cases of the Kakeya, restriction, and Bochner–Riesz conjectures were settled decades ago (by Davies, Zygmund/Fefferman, and Carleson–Sjölin respectively), while the two-dimensional local smoothing conjecture was resolved only in 2020 by Guth, Wang, and Zhang. In three dimensions, the Wang–Zahl theorem ([14, Wang–Zahl Theorem]) settles the Kakeya conjecture at the bottom of the chain, but the restriction, Bochner–Riesz, and local smoothing conjectures remain open. The gap between the Kakeya set conjecture and the restriction conjecture is precisely the gap between the set estimate and the maximal function estimate discussed in [14, Kakeya Maximal Conjecture]: the maximal function version of Kakeya, not just the set version, is the input needed for the restriction implication.

1.6.9 Remark: Decoupling and Modern Developments In 2015, Bourgain and Demeter proved the "ℓ^2 decoupling conjecture" for the paraboloid. The decoupling theorem provides a way to decompose a function whose Fourier transform is supported near a curved surface into "decoupled" pieces with approximately independent L^p behavior. The theorem has had wide-reaching consequences: it implies the full restriction conjecture for the paraboloid in the $L^2 \to L^p$ range (improving on Tomas–Stein), resolves the main conjecture in Vinogradov's mean value theorem (a problem in analytic number theory open since the 1930s), and provides new bounds for exponential sums.

The proof of the decoupling theorem uses an "induction on scales" argument that is structurally analogous to the Wang–Zahl approach to the Kakeya conjecture: both reduce a problem at scale δ to a problem at scale $\sqrt{\delta}$ using a bilinear or multilinear decomposition. This structural parallel suggests that the techniques may eventually be unified into a common framework for problems involving the interaction of curvature and combinatorial geometry.

For a comprehensive treatment of decoupling, see Demeter, *Fourier Restriction, Decoupling, and Applications* (Cambridge, 2020).

The material in this section connects the geometric constructions of [14, Hausdorff Dimension and Kakeya Sets] to the analytic machinery of the Fourier transform. The central message is that the Kakeya problem is not an isolated question in geometric measure theory but a fundamental constraint on how the Fourier transform interacts with L^p spaces in multiple dimensions. The tools developed throughout this book—Lebesgue measure, Hausdorff dimension, the Fourier transform, L^p estimates, and Calderón–Zygmund theory—converge at this point, and the open problems (restriction, Bochner–Riesz, local smoothing) represent some of the central challenges in modern analysis.

Exercise 1.6.1

1. Verify that S_R is bounded on $L^2(\mathbb{R}^n)$ with operator norm 1 for all n. (*Hint*: use theorem 1.3.8.)
2. Show that for $n = 1$, the ball multiplier S_R is bounded on $L^p(\mathbb{R})$ for all $1 < p < \infty$ by expressing $\chi_{[-R,R]}(\xi)$ in terms of the Hilbert transform multiplier $-i\,\mathrm{sgn}(\xi)$ and applying theorem 2.3.4.
3. In the TT^* proof of the Tomas–Stein theorem, verify that $TT^*f = f * \hat{\sigma}$ where T is the extension operator and T^* is the restriction operator. (*Hint*: compute $\langle TT^*f, g\rangle$ using the definitions.)
4. Show that the Bochner–Riesz conjecture for $\alpha = 0$ reduces to the statement that S_R is bounded on L^p, recovering Fefferman's theorem as the assertion that the conjecture fails at the endpoint $\alpha = 0$ for $p \neq 2$.
5. For $n = 1$, verify that the Bochner–Riesz conjecture holds trivially: S_R^α is bounded on $L^p(\mathbb{R})$ for all $\alpha \geq 0$ and $1 < p < \infty$. (*Hint*: the multiplier $(1 - \xi^2/R^2)_+^\alpha$ is a bounded function with bounded variation on $\mathbb{R}$, so the result follows from the Marcinkiewicz multiplier theorem or directly from the L^p boundedness of S_R via subordination.)
6. Show that the restriction estimate $\|\hat{f}\|_{L^2(S^{n-1})} \leq C\|f\|_{L^p}$ cannot hold for $p > \frac{2n}{n+1}$ by testing on the function $f = \chi_{B(0,R)}$ and examining the behavior as $R \to \infty$. (*Hint*: compute $\hat{f}(\xi)$ for $\xi \in S^{n-1}$ using the stationary phase estimate $\hat{f}(\xi) \sim R^{(n+1)/2}$ on the sphere, and compare $\|\hat{f}\|_{L^2(S^{n-1})}$ with $\|f\|_{L^p} = |B(0,R)|^{1/p} \sim R^{n/p}$.)

2

Tempered Distributions and Sobolev Spaces

The foundational theory of distributions (test functions, distributions as continuous functionals, and the distributional derivative) is developed in the Functional Analysis part, [13, Distribution Theory]. This chapter adds the Fourier layer: tempered distributions and the Fourier transform, the H^s Sobolev scale, and singular integrals.

2.1 Tempered Distributions

In the Fourier transform chapter, we established the Schwartz space $\mathcal{S}(\mathbb{R}^n)$ as the natural domain for the Fourier transform: the map $\varphi \mapsto \hat{\varphi}$ is an isomorphism of $\mathcal{S}$ onto itself. We now extend this theory to distributions, obtaining a framework that allows us to take the Fourier transform of objects like δ, constant functions, and polynomials.

The key observation is that $C_c^\infty \subsetneq \mathcal{S}$, and so the dual of $\mathcal{S}$ is *smaller* than $\mathcal{D}'$: every continuous linear functional on $\mathcal{S}$ restricts to one on C_c^∞, but not every distribution extends continuously to $\mathcal{S}$ (since $\mathcal{S}$ has a stronger topology than C_c^∞). The distributions that do extend are precisely those of "at most polynomial growth", which is exactly the right condition for the Fourier transform to make sense.

Definition 2.1.1: Tempered Distribution

A *tempered distribution* is a continuous linear functional on $\mathcal{S}(\mathbb{R}^n)$. The space of all tempered distributions is denoted $\mathcal{S}'(\mathbb{R}^n)$, equipped with the weak* topology.

Concretely, $F \in \mathcal{S}'$ means: $F : \mathcal{S} \to \mathbb{C}$ is linear, and for each sequence $\varphi_j \to \varphi$ in $\mathcal{S}$ (meaning $\sup_x |x^\alpha \partial^\beta(\varphi_j - \varphi)(x)| \to 0$ for all α, β), we have $\langle F, \varphi_j \rangle \to \langle F, \varphi \rangle$. The analogue of Proposition [13, Distribution Continuity Check] holds: $F \in \mathcal{S}'$ if and only if there exist $C > 0$ and $N \in \mathbb{N}$ such that

$$|\langle F, \varphi \rangle| \leq C \sum_{|\alpha|,|\beta| \leq N} \sup_{x \in \mathbb{R}^n} |x^\alpha \partial^\beta \varphi(x)|$$

for all $\varphi \in \mathcal{S}$. The proof is analogous to the one for $\mathcal{D}'$ (using the Schwartz space seminorms $\|\varphi\|_{\alpha,\beta} = \sup |x^\alpha \partial^\beta \varphi|$ in place of the $C_c^\infty(K)$ seminorms).

The following is a nice simplification for testing for tempered distribution

Proposition 2.1.2: Tempered Distribution Criterion

Let F be a tempered distribution. Then there exists a positive integer $N \in \mathbb{N}$ such that:

$$|F(\varphi)| \leq c\|\varphi\|_N \qquad \forall \varphi \in \mathcal{S}$$

***Proof*:**
Suppose otherwise. Then it must be that for each $n \in \mathbb{N}$, there is a $\psi_n \in \mathcal{S}$ with $\|\psi_n\|_n = 1$ and $|F(\psi_n)| \geq n$. Let

$$\varphi_n = \frac{\psi_n}{\sqrt{n}}$$

Then when $n \geq N$, $\|\varphi_n\|_N \leq \|\varphi_n\|_n$, and so:

$$\|\varphi_n\|_N \leq n^{-1/2} \xrightarrow{n \to \infty} 0$$

but then:

$$|F(\varphi_n) \geq n^{1/2} \xrightarrow{n \to \infty} \infty$$

contradicting continuity.

Note this is not the same as order as the norm combines polynomial growth information and growth-order information.

Since $C_c^\infty(\mathbb{R}^n) \subseteq \mathcal{S}(\mathbb{R}^n)$ is dense, every tempered distribution restricts to a distribution, giving an inclusion $\mathcal{S}' \hookrightarrow \mathcal{D}'$. The inclusion is strict: for example, e^{x^2} defines a distribution (via $\varphi \mapsto \int e^{x^2}\varphi$, which converges for $\varphi \in C_c^\infty$ since φ has compact support) but not a tempered distribution (the integral $\int e^{x^2}\varphi$ may diverge for $\varphi \in \mathcal{S}$ that only decays rapidly).

Example 2.1.3: Tempered Distributions

1. *L^p functions.* Every $f \in L^p(\mathbb{R}^n)$ for $1 \leq p \leq \infty$ defines a tempered distribution via $\varphi \mapsto \int f\varphi$. Indeed, $|\int f\varphi| \leq \|f\|_p \|\varphi\|_q$ by Hölder, and $\|\varphi\|_q$ is controlled by finitely many Schwartz seminorms. More generally, any function f with $|f(x)| \leq C(1 + |x|)^N$ for some C, N (i.e., of polynomial growth) and locally integrable defines a tempered distribution.

2. *The delta function and its derivatives.* Since $|\partial^\alpha \varphi(0)| \leq \sup |\partial^\alpha \varphi| \leq \|\varphi\|_{0,\alpha}$, the distributions $\varphi \mapsto \partial^\alpha \varphi(x_0)$ are tempered.

3. *p.v.$(1/x)$.* We showed earlier that $|\langle \text{p.v.}(1/x), \varphi \rangle| \leq C(\|\varphi\|_\infty + \|\varphi'\|_\infty)$. Since these are

Schwartz seminorms (with $\alpha = 0$), p.v.$(1/x) \in \mathcal{S}'(\mathbb{R})$.

4. *Polynomials.* Any polynomial $p(x)$ defines a tempered distribution. The estimate is $|\int p\varphi| \leq C \sup |x^N \varphi(x)|$ for large enough N.

The Fourier transform on $\mathcal{S}$ satisfies $\int \hat{f} g = \int f \hat{g}$ for $f, g \in \mathcal{S}$ (this is immediate from Fubini's theorem and the definition $\hat{f}(\xi) = \int f(x) e^{-2\pi i x \cdot \xi}\, dx$). This identity is exactly of the transpose form we exploited in Section 3: the "transpose" of the Fourier transform is the Fourier transform itself (on $\mathcal{S}$). This motivates:

Definition 2.1.4: Fourier Transform On $\mathcal{S}'$

For $F \in \mathcal{S}'(\mathbb{R}^n)$, define $\hat{F} \in \mathcal{S}'(\mathbb{R}^n)$ by

$$\langle \hat{F}, \varphi \rangle = \langle F, \hat{\varphi} \rangle \qquad (\varphi \in \mathcal{S}).$$

This is well-defined because $\hat{\varphi} \in \mathcal{S}$ whenever $\varphi \in \mathcal{S}$ (by Corollary 1.3.4), and F is a continuous functional on $\mathcal{S}$. Moreover, $\hat{F}$ is continuous on $\mathcal{S}$ since $\varphi_j \to \varphi$ in $\mathcal{S}$ implies $\hat{\varphi}_j \to \hat{\varphi}$ in $\mathcal{S}$ (the Fourier transform is a continuous map on $\mathcal{S}$).

The operational rules of the Fourier transform extend immediately to tempered distributions by the transpose principle:

Proposition 2.1.5: Properties Of The Fourier Transform On $\mathcal{S}'$

Let $F \in \mathcal{S}'(\mathbb{R}^n)$. Then:

1. $\widehat{\partial^\alpha F}(\xi) = (2\pi i \xi)^\alpha \hat{F}(\xi)$ as tempered distributions.
2. $\widehat{x^\alpha F} = \left(\frac{-1}{2\pi i}\right)^{|\alpha|} \partial^\alpha \hat{F}$.
3. (Fourier inversion) If we define $\check{F}$ by $\langle \check{F}, \varphi \rangle = \langle F, \check{\varphi} \rangle$ where $\check{\varphi}(\xi) = \hat{\varphi}(-\xi)$, then $\check{\hat{F}} = \hat{\check{F}} = F$.
4. The Fourier transform is an isomorphism from $\mathcal{S}'$ to $\mathcal{S}'$.

***Proof*:**
Each identity is verified by testing against $\varphi \in \mathcal{S}$ and using the corresponding identity on $\mathcal{S}$.

For (1): $\langle \widehat{\partial^\alpha F}, \varphi \rangle = \langle \partial^\alpha F, \hat{\varphi} \rangle = (-1)^{|\alpha|} \langle F, \partial^\alpha \hat{\varphi} \rangle$. But by the properties of the Fourier transform on $\mathcal{S}$, $\partial^\alpha \hat{\varphi}(\xi) = \widehat{(-2\pi i x)^\alpha \varphi}(\xi)$, so:

$$= (-1)^{|\alpha|} \langle F, \widehat{(-2\pi i x)^\alpha \varphi} \rangle = (-1)^{|\alpha|} \langle \hat{F}, (-2\pi i x)^\alpha \varphi \rangle = \langle (2\pi i x)^\alpha \hat{F}, \varphi \rangle.$$

Replacing the dummy variable x by ξ, this reads $\widehat{\partial^\alpha F} = (2\pi i \xi)^\alpha \hat{F}$.

Part (2) is proved similarly, and (3) follows from Fourier inversion on $\mathcal{S}$ (theorem 1.3.2): $\langle \check{\hat{F}}, \varphi \rangle = \langle \check{F}, \hat{\varphi} \rangle = \langle F, \check{\hat{\varphi}} \rangle = \langle F, \varphi \rangle$, using $\check{\hat{\varphi}} = \varphi$. Part (4) follows from (3).

The payoff of tempered distribution is obvious: we can now compute Fourier transforms of more functions/distributions. Let us do so for the most important cases.

Example 2.1.6: Fourier Transforms Of Important Distributions

1. *The delta function.* For $\varphi \in \mathcal{S}$:
$$\langle \hat{\delta}, \varphi \rangle = \langle \delta, \hat{\varphi} \rangle = \hat{\varphi}(0) = \int \varphi(x)\,dx = \langle 1, \varphi \rangle.$$
Therefore $\hat{\delta} = 1$: the Fourier transform of a point mass is the constant function. By Fourier inversion, $\hat{1} = \delta$: the Fourier transform of the constant function 1 is the delta function. This encodes the principle that "maximally concentrated in space $\Leftrightarrow$ maximally spread in frequency".

2. *The sign function.* From $\operatorname{sgn}'(x) = 2\delta$ and the identity $\widehat{f'} = 2\pi i \xi \hat{f}$, we get $2\pi i \xi \cdot \widehat{\operatorname{sgn}} = 2\hat{\delta} = 2$, and so $\widehat{\operatorname{sgn}}(\xi) = \frac{1}{\pi i \xi}$ in $\mathcal{S}'(\mathbb{R})$. More precisely, $\widehat{\operatorname{sgn}} = \frac{1}{\pi i}\,\text{p.v.}(1/\xi)$.

3. *The Heaviside function.* Since $H = \frac{1}{2}(\operatorname{sgn} + 1)$:
$$\hat{H} = \frac{1}{2}\widehat{\operatorname{sgn}} + \frac{1}{2}\hat{1} = \frac{1}{2\pi i}\,\text{p.v.}\frac{1}{\xi} + \frac{1}{2}\delta.$$

4. *Polynomials.* From $\hat{1} = \delta$ and the identity $\widehat{x^\alpha f} = (2\pi i)^{-|\alpha|}(-\partial)^\alpha \hat{f}$, we get
$$\widehat{x^\alpha} = \left(\frac{-1}{2\pi i}\right)^{|\alpha|} \partial^\alpha \delta.$$
So the Fourier transform of a monomial is a derivative of δ.

2.2 Distributions With Compact Support

We introduced the notation $\mathcal{E}'(U)$ earlier for the set of distributions on U with compact support. This space turns out to have a very clean characterization: it is the dual of $C^\infty(U)$ (with *no* support restriction on the test functions).

Proposition 2.2.1: Characterization Of $\mathcal{E}'$

Let $F \in \mathcal{D}'(U)$. The following are equivalent:

1. F has compact support.
2. F extends (uniquely) to a continuous linear functional on $C^\infty(U)$.

Moreover, every continuous linear functional on $C^\infty(U)$ arises in this way.

***Proof*:**
(1) $\Rightarrow$ (2): Suppose $\mathrm{supp}(F) \subseteq K$ for some compact $K \subseteq U$. Choose $\chi \in C_c^\infty(U)$ with $\chi = 1$ on a neighborhood of K. For any $\psi \in C^\infty(U)$, define $\langle F, \psi \rangle := \langle F, \chi\psi \rangle$. This is well-defined since $\chi\psi \in C_c^\infty(U)$, and it is independent of the choice of χ: if χ' is another such cutoff, then $\chi - \chi' = 0$ on a neighborhood of $K = \mathrm{supp}(F)$, so $\langle F, (\chi - \chi')\psi \rangle = 0$. Continuity on $C^\infty(U)$ follows from the continuity estimate on $C_c^\infty(\mathrm{supp}(\chi))$.

(2) $\Rightarrow$ (1): Suppose F extends to a continuous functional on $C^\infty(U)$ but $\mathrm{supp}(F)$ is not compact. Then for each j, there exists $\varphi_j \in C_c^\infty(U)$ with $\mathrm{supp}(\varphi_j) \cap \{|x| > j\} \neq \emptyset$, $\mathrm{supp}(\varphi_j) \cap \mathrm{supp}(\varphi_k) = \emptyset$ for $j \neq k$ (by choosing supports far apart), and $\langle F, \varphi_j \rangle = 1$. But $\psi = \sum_j \varphi_j / j$ converges in $C^\infty(U)$ (since the supports are locally finite), and $\langle F, \psi \rangle = \sum_j \langle F, \varphi_j \rangle / j = \sum 1/j = \infty$, contradicting continuity.

The "moreover" part: given a continuous linear functional L on $C^\infty(U)$, its restriction to $C_c^\infty(U)$ is a distribution (since $C_c^\infty(K) \hookrightarrow C^\infty(U)$ is continuous for each compact K). The argument above shows this distribution has compact support, and L is its unique extension.

Proposition 2.2.2: Finite Order Of Compactly Supported Distributions

Every $F \in \mathcal{E}'(U)$ has *finite order*: there exists $N \in \mathbb{N}$ such that the continuity estimate

$$|\langle F, \varphi \rangle| \leq C \sum_{|\alpha| \leq N} \|\partial^\alpha \varphi\|_{L^\infty}$$

holds for *all* $\varphi \in C^\infty(U)$ (not just those supported in a particular compact set). The smallest such N is called the *order* of F.

***Proof*:**
Since $\mathrm{supp}(F) \subseteq K$ for some compact K, we have $\langle F, \varphi \rangle = \langle F, \chi\varphi \rangle$ for any cutoff $\chi \in C_c^\infty(U)$ with $\chi = 1$ near K. By the distribution continuity check (Proposition [13, Distribution Continuity Check]) applied on $\mathrm{supp}(\chi)$:

$$|\langle F, \varphi \rangle| = |\langle F, \chi\varphi \rangle| \leq C_{\mathrm{supp}(\chi)} \sum_{|\alpha| \leq N} \|\partial^\alpha (\chi\varphi)\|_{L^\infty}.$$

By the Leibniz rule, $\|\partial^\alpha(\chi\varphi)\|_{L^\infty} \leq C' \sum_{|\beta| \leq |\alpha|} \|\partial^\beta \varphi\|_{L^\infty(\mathrm{supp}(\chi))}$. The result follows.

The inclusion chain $\mathcal{E}'(\mathbb{R}^n) \subset \mathcal{S}'(\mathbb{R}^n) \subset \mathcal{D}'(\mathbb{R}^n)$ now has a clean interpretation:

1. $\mathcal{D}' = (C_c^\infty)'$: all distributions, no restrictions.
2. $\mathcal{S}' = (\mathcal{S})'$: "at most polynomial growth" distributions—these are the ones compatible with the Fourier transform.
3. $\mathcal{E}' = (C^\infty)'$: compactly supported distributions—these are the most "localized".

A fact we will use later: if $F \in \mathcal{E}'(\mathbb{R}^n)$, then $\hat{F}$ is not a tempered distribution but an actual smooth function. In fact, $\hat{F}(\xi) = \langle F_x, e^{-2\pi i x \cdot \xi} \rangle$ (where the subscript indicates the variable of the pairing) is well-defined for all ξ since $x \mapsto e^{-2\pi i x \cdot \xi}$ is in C^∞ and F extends to C^∞. Moreover, $\hat{F}$ extends to an *entire* function on $\mathbb{C}^n$, and satisfies the Paley–Wiener estimate $|\hat{F}(\xi + i\eta)| \leq C(1 + |\xi| + |\eta|)^N e^{2\pi R|\eta|}$ when $\text{supp}(F) \subseteq B_R$ (compare with Corollary 1.2.15, which established the analogous result for functions).

2.3 Singular Integrals And Calderón–Zygmund Theory

In Section [13, Distributional Derivatives of Classical Functions], we encountered the principal value distribution $\text{p.v.}(1/x)$. This object is the kernel an important operators in harmonic analysis: the *Hilbert transform*. In this section, we study the Hilbert transform from the distributional perspective and then generalize to the Calderón–Zygmund theory of singular integral operators.

Definition 2.3.1: Hilbert Transform

For $f \in \mathcal{S}(\mathbb{R})$ (or more generally $f \in L^p(\mathbb{R})$ for $1 \leq p < \infty$), the *Hilbert transform* of f is defined by

$$Hf(x) = \text{p.v.}\frac{1}{\pi}\int_{-\infty}^{\infty} \frac{f(y)}{x-y}\, dy = \frac{1}{\pi} \lim_{\varepsilon \to 0^+} \int_{|x-y|>\varepsilon} \frac{f(y)}{x-y}\, dy.$$

Equivalently, $Hf = f * K$ where $K = \frac{1}{\pi}\,\text{p.v.}(1/x)$ is the distributional convolution.

The Hilbert transform arises naturally in complex analysis (it relates the real and imaginary parts of a function holomorphic in the upper half-plane), in signal processing (it performs a 90ř phase shift), and in PDE theory (it is the prototypical example of a singular integral operator). Its distributional nature is essential: the kernel $1/(\pi x)$ is not locally integrable, so the convolution must be interpreted in the principal value sense.

The Fourier-analytic characterization of H is remarkably clean:

Theorem 2.3.2: Fourier Multiplier Characterization Of H

For $f \in \mathcal{S}(\mathbb{R})$:

$$\widehat{Hf}(\xi) = -i\,\text{sgn}(\xi)\,\hat{f}(\xi).$$

That is, H is a Fourier multiplier with symbol $m(\xi) = -i\,\text{sgn}(\xi)$.

***Proof*:**
Since $Hf = \frac{1}{\pi}\text{p.v.}(1/x) * f$ and convolution becomes multiplication under the Fourier transform (for tempered distributions), we have $\widehat{Hf} = \frac{1}{\pi}\widehat{\text{p.v.}(1/x)} \cdot \hat{f}$. From Example 2.1.6, $\widehat{\text{sgn}}(\xi) = \frac{1}{\pi i}\text{p.v.}(1/\xi)$. Taking the inverse Fourier transform of both sides: $\text{sgn}(x) = \frac{1}{\pi i}\,(\text{p.v.}\check{(1/\cdot)})(x)$, which gives $\widehat{\text{p.v.}(1/x)}(\xi) = -\pi i\,\text{sgn}(\xi)$. Therefore

$$\widehat{Hf} = \frac{1}{\pi}(-\pi i)\,\text{sgn}(\xi)\hat{f}(\xi) = -i\,\text{sgn}(\xi)\hat{f}(\xi).$$

Since $|-i\,\text{sgn}(\xi)| = 1$ for $\xi \neq 0$, an immediate consequence is:

Corollary 2.3.3: L^2 Boundedness Of H

The Hilbert transform is an isometry on $L^2(\mathbb{R})$: $\|Hf\|_{L^2} = \|f\|_{L^2}$. Moreover, $H^2 = -\mathrm{Id}$ on L^2, so $H^{-1} = -H$.

***Proof*:**
By Plancherel: $\|Hf\|_{L^2}^2 = \|\widehat{Hf}\|_{L^2}^2 = \int |\operatorname{sgn}(\xi)|^2 |\hat{f}(\xi)|^2 \, d\xi = \|\hat{f}\|_{L^2}^2 = \|f\|_{L^2}^2$. For the second claim: $\widehat{H^2 f} = (-i \operatorname{sgn})^2 \hat{f} = -\hat{f}$, so $H^2 f = -f$.

The result which lies at the heart of Calderón–Zygmund theory is that H extends to a bounded operator on L^p for all $1 < p < \infty$. This can be proved using the Riesz–Thorin interpolation theorem ([13, Riesz-Thorin Interpolation Theorem]) to interpolate between the L^2 bound and a suitable weak-type estimate.

Theorem 2.3.4: L^p Boundedness Of The Hilbert Transform

For $1 < p < \infty$, the Hilbert transform extends to a bounded operator $H : L^p(\mathbb{R}) \to L^p(\mathbb{R})$. That is, there exists $C_p > 0$ such that $\|Hf\|_{L^p} \leq C_p \|f\|_{L^p}$ for all $f \in L^p(\mathbb{R})$. The operator H is *not* bounded on L^1 or L^∞.

***Proof*:**
(Sketch) We already know H is bounded on L^2 with constant 1. For $p = 1$, one proves the *weak-type* $(1,1)$ *estimate*: $m(\{x : |Hf(x)| > \lambda\}) \leq C\|f\|_{L^1}/\lambda$ for all $\lambda > 0$. This is the delicate part, and it uses the Calderón–Zygmund decomposition of f into a "good" part (bounded by λ) and a "bad" part (concentrated on a set of measure $\lesssim \|f\|_{L^1}/\lambda$). Once this is established, the Marcinkiewicz interpolation theorem (or direct Riesz–Thorin interpolation between L^2 and the weak L^1 endpoint) gives boundedness on L^p for $1 < p \leq 2$. Duality (since $H^* = -H$) extends this to $2 \leq p < \infty$. The failure at $p = 1$ and $p = \infty$ can be seen from explicit examples (e.g., $H(\chi_{[0,1]})$ has a logarithmic singularity).

2.3.1 Calderón–Zygmund Kernels

The Hilbert transform is the one-dimensional prototype of a much larger class of operators. In $\mathbb{R}^n$ for $n \geq 2$, the analogous objects are the *Riesz transforms* $R_j f = c_n \, \mathrm{p.v.}(x_j/|x|^{n+1}) * f$, which are Fourier multipliers with symbol $-i\xi_j/|\xi|$. The general framework encompassing all of these is the Calderón–Zygmund theory.

Definition 2.3.5: Calderón–Zygmund Kernel

A *Calderón–Zygmund kernel* on $\mathbb{R}^n$ is a function $K : \mathbb{R}^n \setminus \{0\} \to \mathbb{C}$ satisfying:

1. (Size condition) $|K(x)| \leq C|x|^{-n}$ for all $x \neq 0$.
2. (Smoothness condition) $|\nabla K(x)| \leq C|x|^{-n-1}$ for all $x \neq 0$.
3. (Cancellation condition) $\sup_{0<r<R} \left| \int_{r<|x|<R} K(x)\, dx \right| < \infty$.

The associated *Calderón–Zygmund operator* is $Tf = \text{p.v.}(K) * f$, interpreted in the distributional sense.

The size condition says K has a singularity of order $|x|^{-n}$ at the origin—exactly on the boundary of local integrability in $\mathbb{R}^n$, which is why the principal value is necessary. The smoothness condition controls the regularity away from the origin, and the cancellation condition ensures that the principal value limit exists.

Theorem 2.3.6: Calderón–Zygmund Theorem

Let T be a Calderón–Zygmund operator on $\mathbb{R}^n$. If T is bounded on $L^2(\mathbb{R}^n)$, then T extends to a bounded operator on $L^p(\mathbb{R}^n)$ for all $1 < p < \infty$, and satisfies the weak-type $(1,1)$ estimate

$$m(\{x : |Tf(x)| > \lambda\}) \leq \frac{C}{\lambda}\|f\|_{L^1}.$$

The proof of this theorem is one of the landmark achievements of 20th-century harmonic analysis. The key tool is the *Calderón–Zygmund decomposition*, which splits a function into "good" and "bad" parts at a given height λ, and then estimates each part separately. The good part is handled by the assumed L^2 bound, while the bad part is estimated using the smoothness and cancellation conditions on the kernel K. The full proof can be found in Stein's *Singular Integrals and Differentiability Properties of Functions*, Chapter II. The connection to distribution theory is fundamental: the operators are defined by convolution with singular distributions, and their Fourier multiplier characterization (obtained via the Fourier transform on $\mathcal{S}'$) is the primary tool for establishing L^2 boundedness.

2.3.7 From Hilbert Transforms to the Langlands Program The Hilbert transform and Calderón–Zygmund theory may seem like purely analytic tools, but they have connections to number theory and representation theory. In the theory of automorphic forms, the intertwining operators that relate different representations of a reductive group are singular integrals of Calderón–Zygmund type, defined on the group manifold rather than $\mathbb{R}^n$. The L^p boundedness of these operators is closely related to the analytic continuation of Eisenstein series, a central construction in the Langlands program. In this way, the distributional framework developed here—principal values, Fourier multipliers, and singular kernels—extends far beyond classical analysis into the arithmetic world.

2.4 Homogeneous Distributions

These seem interesting, may cover them.

2.5 Sobolev Spaces

Sobolev spaces spaces quantify "how many derivatives a function has" in the L^2 sense, and they form the natural setting for weak solutions to differential equations. The theory of tempered distributions and the Fourier transform from the preceding sections will play a central role.

Definition 2.5.1: Sobolev Spaces (H^s)

For $s \in \mathbb{R}$, the *Sobolev space* $H^s(\mathbb{R}^n)$ is defined by

$$H^s(\mathbb{R}^n) = \left\{ f \in \mathcal{S}'(\mathbb{R}^n) : (1 + |\xi|^2)^{s/2} \hat{f}(\xi) \in L^2(\mathbb{R}^n) \right\},$$

equipped with the inner product

$$\langle f, g \rangle_{H^s} = \int_{\mathbb{R}^n} \hat{f}(\xi) \overline{\hat{g}(\xi)} (1 + |\xi|^2)^s \, d\xi$$

and corresponding norm $\|f\|_{H^s} = \langle f, f \rangle_{H^s}^{1/2}$.

The weight $(1 + |\xi|^2)^{s/2}$ is called the *Japanese bracket* and is sometimes written $\langle \xi \rangle^s$. The intuition is: requiring $(1 + |\xi|^2)^{s/2} \hat{f} \in L^2$ means that $\hat{f}$ must decay like $|\xi|^{-s}$ faster than a typical L^2 function. Since high-frequency components of $\hat{f}$ correspond to rapid oscillations (i.e., derivatives) of f, imposing more decay on $\hat{f}$ at high frequencies is equivalent to requiring f to have more derivatives. When s is a non-negative integer, this coincides with the classical requirement that f and its first s partial derivatives all lie in L^2.

Proposition 2.5.2: Properties of H^s

1. $H^s(\mathbb{R}^n)$ is a Hilbert space for every $s \in \mathbb{R}$.
2. $H^0(\mathbb{R}^n) = L^2(\mathbb{R}^n)$, with equality of norms (by Plancherel).
3. If $s > t$, then $H^s(\mathbb{R}^n) \hookrightarrow H^t(\mathbb{R}^n)$ continuously: $\|f\|_{H^t} \leq \|f\|_{H^s}$.
4. $\mathcal{S}(\mathbb{R}^n)$ is dense in $H^s(\mathbb{R}^n)$ for every s.
5. For $s \geq 0$ an integer, $f \in H^s(\mathbb{R}^n)$ if and only if $\partial^\alpha f \in L^2(\mathbb{R}^n)$ for all $|\alpha| \leq s$ (where derivatives are taken in the distributional sense). The norms are equivalent:
$$\|f\|_{H^s} \asymp \left(\sum_{|\alpha| \leq s} \|\partial^\alpha f\|_{L^2}^2 \right)^{1/2}.$$
6. (Duality) $(H^s)^* \cong H^{-s}$ via the L^2 pairing $\langle f, g \rangle = \int \hat{f}\overline{\hat{g}}$.
7. Differentiation is a bounded operator: $\partial^\alpha : H^s \to H^{s-|\alpha|}$ with $\|\partial^\alpha f\|_{H^{s-|\alpha|}} \leq C\|f\|_{H^s}$.

***Proof*:**

(1) The map $f \mapsto (1+|\xi|^2)^{s/2}\hat{f}$ is an isometry from H^s into L^2. Since L^2 is complete and the map is an isometry onto a closed subspace (in fact, onto all of L^2, since for any $g \in L^2$ the function f with $\hat{f} = g/(1+|\xi|^2)^{s/2}$ lies in H^s and maps to g), H^s is complete.

(2) is immediate from the definition: $(1+|\xi|^2)^0 = 1$, so $\|f\|_{H^0} = \|\hat{f}\|_{L^2} = \|f\|_{L^2}$.

(3) For $s > t$: $(1+|\xi|^2)^{t/2} \leq (1+|\xi|^2)^{s/2}$, so $\|f\|_{H^t} \leq \|f\|_{H^s}$.

(4) Given $f \in H^s$, let $\chi_R \in C_c^\infty$ with $\chi_R = 1$ on B_R and $0 \leq \chi_R \leq 1$. Define f_R by $\hat{f}_R = \chi_R \hat{f}$. Then $\hat{f}_R \in C_c^\infty \subset \mathcal{S}$, so $f_R \in \mathcal{S}$ (by the Fourier isomorphism on $\mathcal{S}$). And $\|f - f_R\|_{H^s}^2 = \int_{|\xi|>R} |\hat{f}|^2(1+|\xi|^2)^s \, d\xi \to 0$ as $R \to \infty$ by dominated convergence.

(5) For integer $s \geq 0$: $\widehat{\partial^\alpha f}(\xi) = (2\pi i\xi)^\alpha \hat{f}(\xi)$, so $\|\partial^\alpha f\|_{L^2}^2 = (2\pi)^{2|\alpha|} \int |\xi^\alpha|^2 |\hat{f}|^2 \, d\xi$. Since $\sum_{|\alpha|\leq s} |\xi^\alpha|^2 \asymp (1+|\xi|^2)^s$ (both are polynomials of degree $2s$ with the same leading behavior), the norm equivalence follows.

(6) The pairing $\langle f, g \rangle = \int \hat{f}\overline{\hat{g}}$ gives $|\langle f, g \rangle| \leq \|f\|_{H^s}\|g\|_{H^{-s}}$ by Cauchy–Schwarz. That every continuous linear functional on H^s is of this form follows from the Riesz representation theorem on the Hilbert space L^2 after conjugating by the isometry $f \mapsto \langle \xi \rangle^s \hat{f}$.

(7) $\|\partial^\alpha f\|_{H^{s-|\alpha|}}^2 = \int |(2\pi i\xi)^\alpha|^2 |\hat{f}|^2 (1+|\xi|^2)^{s-|\alpha|} \, d\xi \leq C \int |\hat{f}|^2 (1+|\xi|^2)^s \, d\xi = C\|f\|_{H^s}^2$, since $|\xi^\alpha|^2(1+|\xi|^2)^{-|\alpha|} \leq C$.

The most important result about Sobolev spaces is the embedding theorem, which says that if s is large enough, then functions in H^s are not just L^2 but actually continuous (and even classically differentiable).

Theorem 2.5.3: Sobolev Embedding Theorem

If $s > k + n/2$ for some non-negative integer k, then $H^s(\mathbb{R}^n) \hookrightarrow C^k(\mathbb{R}^n)$ (the space of k-times continuously differentiable functions that vanish at infinity). Moreover, the embedding is continuous: there exists $C = C(s, k, n) > 0$ such that

$$\sum_{|\alpha| \le k} \|\partial^\alpha f\|_{L^\infty} \le C\|f\|_{H^s}$$

for all $f \in H^s(\mathbb{R}^n)$.

***Proof*:**
It suffices to prove the case $k = 0$ (i.e., $s > n/2$ implies $H^s \hookrightarrow C^0$ with $\|f\|_\infty \le C\|f\|_{H^s}$); the general case follows by applying this to $\partial^\alpha f \in H^{s-|\alpha|}$ with $s - |\alpha| > n/2$ (which holds when $s > k + n/2$ and $|\alpha| \le k$).

By the Fourier inversion formula (applied to $f \in \mathcal{S}$, then extended by density): $f(x) = \int \hat{f}(\xi) e^{2\pi i x \cdot \xi}\, d\xi$, and so

$$|f(x)| \le \int |\hat{f}(\xi)|\, d\xi = \int |\hat{f}(\xi)|(1 + |\xi|^2)^{s/2} \cdot (1 + |\xi|^2)^{-s/2}\, d\xi.$$

By Cauchy–Schwarz:

$$|f(x)| \le \left(\int |\hat{f}(\xi)|^2 (1 + |\xi|^2)^s\, d\xi\right)^{1/2} \cdot \left(\int (1 + |\xi|^2)^{-s}\, d\xi\right)^{1/2} = \|f\|_{H^s} \cdot C_s.$$

The constant $C_s = \left(\int (1 + |\xi|^2)^{-s}\, d\xi\right)^{1/2}$ is finite precisely when $2s > n$, i.e., $s > n/2$: switching to polar coordinates, the integral behaves like $\int_0^\infty r^{n-1}(1 + r^2)^{-s}\, dr$, which converges if and only if $2s - n + 1 > 1$.

This gives $\|f\|_\infty \le C_s\|f\|_{H^s}$ for $f \in \mathcal{S}$. Since $\mathcal{S}$ is dense in H^s, every $f \in H^s$ is the H^s-limit of a sequence $f_j \in \mathcal{S}$, and the estimate $\|f_j - f_k\|_\infty \le C_s\|f_j - f_k\|_{H^s} \to 0$ shows that f_j converges uniformly. The uniform limit is a continuous function that agrees with f as an element of L^2. The vanishing at infinity follows from the density of C_c^∞ in H^s.

The threshold $s = n/2$ is sharp, as the following example shows.

Example 2.5.4: The Step Function (The Boundary Case)

On $\mathbb{R}$, the threshold for $H^s(\mathbb{R}) \hookrightarrow C^0(\mathbb{R})$ is $s > 1/2$. Consider the Heaviside function $H(x) = \mathbf{1}_{[0,\infty)}(x)$. We showed that $\hat{H}(\xi) = \frac{1}{2}\delta(\xi) + \frac{1}{2\pi i}\text{p.v.}(1/\xi)$, and more precisely, for a smooth truncation $H_R(x) = H(x)\chi_{[-R,R]}(x)$ (where χ is a smooth cutoff), one has $|\hat{H}_R(\xi)| \sim 1/|\xi|$ for large $|\xi|$.

Then $\|H_R\|_{H^s}^2 \sim \int |\xi|^{-2}(1 + |\xi|^2)^s\, d\xi \sim \int |\xi|^{2s-2}\, d\xi$, which converges near $\xi = 0$ (no issue) but diverges at infinity when $2s - 2 \ge -1$, i.e., $s \ge 1/2$.

Therefore $H \notin H^{1/2}(\mathbb{R})$, and a fortiori $H \notin H^s(\mathbb{R})$ for $s \ge 1/2$. The jump discontinuity at $x = 0$ prevents H from lying in H^s for s at or above the embedding threshold. This illustrates the sharpness of the Sobolev embedding theorem: at the boundary $s = n/2$, the embedding into C^0 fails, and discontinuous functions can sneak in.

The continuous embedding $H^{s+t}(\mathbb{R}^n) \hookrightarrow H^s(\mathbb{R}^n)$ of proposition 2.5.2 (3) is never compact, for any $s \in \mathbb{R}$ and $t > 0$. Translation $u \mapsto u(\cdot - y)$ multiplies $\hat{u}$ by the unimodular factor $e^{-2\pi i y\cdot\xi}$ (proposition 1.2.13 (1)) and so preserves every H^s norm, hence for a fixed nonzero $u \in \mathcal{S}(\mathbb{R}^n)$ and $y \neq 0$ the translates $u(\cdot - ky)$, $k \in \mathbb{N}$, are bounded in H^{s+t}. They have no H^s-convergent subsequence: writing $w = \langle\cdot\rangle^{2s}|\hat{u}|^2 \in L^1(\mathbb{R}^n)$, where $\langle\xi\rangle = (1+|\xi|^2)^{1/2}$ is the Japanese bracket introduced after definition 2.5.1,

$$\|u(\cdot - ky) - u(\cdot - ly)\|_{H^s}^2 = \int \left|1 - e^{-2\pi i(l-k)y\cdot\xi}\right|^2 w(\xi)\, d\xi = 2\|u\|_{H^s}^2 - 2\mathrm{Re}\, \hat{w}\big((l-k)y\big)$$

and $\hat{w} \in C_0(\mathbb{R}^n)$ by proposition 1.2.13 (7), so the distance between two translates tends to $\sqrt{2}\,\|u\|_{H^s} > 0$ as $l - k \to \infty$, whereas a Cauchy subsequence (k_j) would force it to zero along the pairs (k_j, k_{2j}), for which $k_{2j} - k_j \geq j$. Compactness therefore requires a spatial cutoff, and the cutoff is multiplication by a fixed $\chi \in C_c^\infty(\mathbb{R}^n)$. The proof rests on two preliminary facts: an inequality comparing $\langle\xi\rangle$ with $\langle\eta\rangle$ across the shift $\xi - \eta$, and the boundedness of multiplication by a Schwartz function on every H^s.

Lemma 2.5.5: Peetre's Inequality

For every $r \in \mathbb{R}$ and all $\xi, \eta \in \mathbb{R}^n$,

$$\langle\xi\rangle^r \leq 2^{|r|/2}\, \langle\xi - \eta\rangle^{|r|}\, \langle\eta\rangle^r$$

***Proof*:**
From $|\xi| \leq |\xi - \eta| + |\eta|$ and $(a+b)^2 \leq 2(a^2+b^2)$,

$$1 + |\xi|^2 \leq 1 + 2|\xi-\eta|^2 + 2|\eta|^2 \leq 2(1+|\xi-\eta|^2)(1+|\eta|^2)$$

the second step because $2(1+a)(1+b) = 2 + 2a + 2b + 2ab \geq 1 + 2a + 2b$ for $a, b \geq 0$. For $r \geq 0$, raising to the power $r/2$ preserves the inequality and gives the claim. For $r < 0$, the case already proved, with exponent $-r > 0$ and the roles of ξ and η exchanged, reads $\langle\eta\rangle^{-r} \leq 2^{-r/2}\langle\eta - \xi\rangle^{-r}\langle\xi\rangle^{-r}$. Multiplying through by $\langle\xi\rangle^r\langle\eta\rangle^r > 0$ and using $\langle\eta - \xi\rangle = \langle\xi - \eta\rangle$ yields $\langle\xi\rangle^r \leq 2^{|r|/2}\langle\xi-\eta\rangle^{|r|}\langle\eta\rangle^r$.

Proposition 2.5.6: H^s Is a Module over $\mathcal{S}$

Let $s \in \mathbb{R}$ and $\varphi \in \mathcal{S}(\mathbb{R}^n)$. For $u \in \mathcal{S}'(\mathbb{R}^n)$ the product φu, defined by $\langle \varphi u, \psi\rangle = \langle u, \varphi\psi\rangle$ for $\psi \in \mathcal{S}(\mathbb{R}^n)$, is a tempered distribution, and $u \mapsto \varphi u$ is a bounded operator $H^s(\mathbb{R}^n) \to H^s(\mathbb{R}^n)$ with

$$\|\varphi u\|_{H^s} \leq 2^{|s|/2}\, \|\langle\cdot\rangle^{|s|}\hat{\varphi}\|_{L^1}\, \|u\|_{H^s}$$

Moreover, for $u \in H^s(\mathbb{R}^n)$ the integral $(\hat{\varphi} * \hat{u})(\xi) = \int \hat{\varphi}(\xi - \eta)\hat{u}(\eta)\, d\eta$ converges absolutely for every $\xi \in \mathbb{R}^n$, and $\widehat{\varphi u} = \hat{\varphi} * \hat{u}$ almost everywhere. The convergence is quantitative: for every measurable Φ on $\mathbb{R}^n$ with $\langle\cdot\rangle^{|s|}\Phi \in L^2(\mathbb{R}^n)$, every $f \in H^s(\mathbb{R}^n)$, and every $\xi \in \mathbb{R}^n$,

$$\int |\Phi(\xi - \eta)||\hat{f}(\eta)|\, d\eta \leq 2^{|s|/2}\langle\xi\rangle^{-s}\, \|\langle\cdot\rangle^{|s|}\Phi\|_{L^2}\, \|f\|_{H^s} \tag{2.1}$$

***Proof*:**
Step 1: φu is tempered. For $\psi \in \mathcal{S}$ the Leibniz rule $\partial^\alpha(\varphi\psi) = \sum_{\beta\le\alpha}\binom{\alpha}{\beta}\partial^\beta\varphi\,\partial^{\alpha-\beta}\psi$ gives $\|\varphi\psi\|_{(N,\alpha)} \le \sum_{\beta\le\alpha}\binom{\alpha}{\beta}\|\partial^\beta\varphi\|_{L^\infty}\|\psi\|_{(N,\alpha-\beta)}$, so $\psi \mapsto \varphi\psi$ is a continuous linear map of $\mathcal{S}$ into itself and $\psi \mapsto \langle u, \varphi\psi\rangle$ is a continuous linear functional on $\mathcal{S}$.

Step 2: the product formula on $\mathcal{S}$. Let $f, g \in \mathcal{S}$. Since $h^\vee(x) = \hat{h}(-x)$, the convolution theorem proposition 1.2.13 (3) holds verbatim for the inverse transform: $(h * k)^\vee = h^\vee k^\vee$ for $h, k \in L^1$. Applied to $\hat{f}, \hat{g} \in \mathcal{S} \subseteq L^1$ (corollary 1.2.16) it gives $(\hat{f} * \hat{g})^\vee = (\hat{f})^\vee(\hat{g})^\vee = fg$, the last equality by theorem 1.3.2, as f and $\hat{f}$ lie in L^1 and f is continuous. Since $\hat{f} * \hat{g} \in \mathcal{S}$ by proposition 1.1.5, theorem 1.3.2 applies to it as well and yields

$$\widehat{fg} = \big((\hat{f} * \hat{g})^\vee\big)^\wedge = \hat{f} * \hat{g} \qquad (f, g \in \mathcal{S}) \tag{2.2}$$

Step 3: the estimate on $\mathcal{S}$. Let $u \in \mathcal{S}$. By (2.2) and lemma 2.5.5 with $r = s$,

$$\langle\xi\rangle^s|\widehat{\varphi u}(\xi)| \le \int \langle\xi\rangle^s|\hat{\varphi}(\xi-\eta)||\hat{u}(\eta)|\,d\eta \le \int 2^{|s|/2}\langle\xi-\eta\rangle^{|s|}|\hat{\varphi}(\xi-\eta)|\cdot\langle\eta\rangle^s|\hat{u}(\eta)|\,d\eta = (\Phi * v)(\xi)$$

where $\Phi = 2^{|s|/2}\langle\cdot\rangle^{|s|}|\hat{\varphi}|$ and $v = \langle\cdot\rangle^s|\hat{u}|$. Here $\Phi \in L^1$ because $\hat{\varphi} \in \mathcal{S}$ (for an integer $N \ge |s|$ the seminorm $\|\hat{\varphi}\|_{(N+n+1,0)}$ bounds Φ by a multiple of $(1+|\xi|)^{-n-1}$), and $v \in L^2$ with $\|v\|_{L^2} = \|u\|_{H^s}$. Young's inequality with $(p,q,r) = (1,2,2)$, [13, Young's Convolution Inequality], gives $\|\Phi * v\|_{L^2} \le \|\Phi\|_{L^1}\|v\|_{L^2}$, hence

$$\|\varphi u\|_{H^s} = \|\langle\cdot\rangle^s\widehat{\varphi u}\|_{L^2} \le C_\varphi(s)\,\|u\|_{H^s}, \qquad C_\varphi(s) = 2^{|s|/2}\|\langle\cdot\rangle^{|s|}\hat{\varphi}\|_{L^1}$$

Step 4: extension to H^s. Convergence in H^s implies convergence in $\mathcal{S}'$. Indeed, for $f \in H^s$ the distribution $\hat{f}$ is a function with $\langle\cdot\rangle^s\hat{f} \in L^2$, and for $\psi \in \mathcal{S}$, definition 2.1.4 with the test function $\psi^\vee$ and the inversion $(\psi^\vee)^\wedge = \psi$ give $\langle f, \psi\rangle = \langle\hat{f}, \psi^\vee\rangle = \int \hat{f}(\xi)\psi^\vee(\xi)\,d\xi$, so the Cauchy-Schwarz inequality yields

$$|\langle f, \psi\rangle| \le \|f\|_{H^s}\,\|\langle\cdot\rangle^{-s}\psi^\vee\|_{L^2} \tag{2.3}$$

with a finite second factor because $\psi^\vee \in \mathcal{S}$. Now let $u \in H^s$ and choose $u_k \in \mathcal{S}$ with $u_k \to u$ in H^s (proposition 2.5.2 (4)). By Step 3, $\|\varphi u_k - \varphi u_l\|_{H^s} \le C_\varphi(s)\|u_k - u_l\|_{H^s}$, so (φu_k) is Cauchy in H^s and converges in H^s to some w (proposition 2.5.2 (1)). For $\psi \in \mathcal{S}$, (2.3) applied to $\varphi u_k - w$ with test function ψ, and to $u_k - u$ with test function $\varphi\psi \in \mathcal{S}$, gives

$$\langle w, \psi\rangle = \lim_{k\to\infty}\langle\varphi u_k, \psi\rangle = \lim_{k\to\infty}\langle u_k, \varphi\psi\rangle = \langle u, \varphi\psi\rangle = \langle\varphi u, \psi\rangle$$

so $\varphi u = w \in H^s$ and $\|\varphi u\|_{H^s} = \lim_k\|\varphi u_k\|_{H^s} \le C_\varphi(s)\|u\|_{H^s}$.

Step 5: the estimate (2.1) and the pointwise formula on H^s. Let Φ and f be as in the statement. Lemma 2.5.5 with $r = -s$ reads $\langle\eta\rangle^{-s} \le 2^{|s|/2}\langle\eta-\xi\rangle^{|s|}\langle\xi\rangle^{-s}$, so

$$\begin{aligned}\int|\Phi(\xi-\eta)||\hat{f}(\eta)|\,d\eta &= \int|\Phi(\xi-\eta)|\langle\eta\rangle^{-s}\cdot\langle\eta\rangle^s|\hat{f}(\eta)|\,d\eta \\ &\le 2^{|s|/2}\langle\xi\rangle^{-s}\int\langle\xi-\eta\rangle^{|s|}|\Phi(\xi-\eta)|\cdot\langle\eta\rangle^s|\hat{f}(\eta)|\,d\eta\end{aligned}$$

and the Cauchy-Schwarz inequality with the translation invariance of the L^2 norm gives (2.1). With $\Phi = \hat{\varphi}$ and $f = u$ this is the absolute convergence of $(\hat{\varphi} * \hat{u})(\xi)$. With $\Phi = \hat{\varphi}$ and $f = u_k - u$

it shows $(\hat{\varphi} * \hat{u}_k)(\xi) \to (\hat{\varphi} * \hat{u})(\xi)$ for every ξ, where $\hat{\varphi} * \hat{u}_k = \widehat{\varphi u_k}$ by (2.2). On the other hand $\varphi u_k \to \varphi u$ in H^s says that $\widehat{\varphi u_k} \to \widehat{\varphi u}$ in $L^2(\langle \xi \rangle^{2s}\, d\xi)$, a measure with the same null sets as $d\xi$, so a subsequence converges almost everywhere. The two limits agree wherever both exist, whence $\widehat{\varphi u} = \hat{\varphi} * \hat{u}$ almost everywhere.

Theorem 2.5.7: Rellich Compactness on the Bessel Scale

Let $s \in \mathbb{R}$, $t > 0$, and $\chi \in C_c^\infty(\mathbb{R}^n)$. Then $u \mapsto \chi u$ is a *compact operator* $H^{s+t}(\mathbb{R}^n) \to H^s(\mathbb{R}^n)$: it is bounded, and for every sequence (u_k) bounded in $H^{s+t}(\mathbb{R}^n)$ the sequence (χu_k) has a subsequence converging in $H^s(\mathbb{R}^n)$.

***Proof*:**
Write $\sigma = s + t$ and let $\|u_k\|_{H^\sigma} \le M$ for all k. Since $C_c^\infty \subseteq \mathcal{S}$, proposition 2.5.6 at order s together with $\|u\|_{H^s} \le \|u\|_{H^\sigma}$ gives the boundedness, and at order σ it gives $\|\chi u_k\|_{H^\sigma} \le C_\chi M$ with $C_\chi = C_\chi(\sigma)$, as well as $\widehat{\chi u_k} = v_k$ almost everywhere, where

$$v_k(\xi) = \int \hat{\chi}(\xi - \eta)\hat{u}_k(\eta)\, d\eta, \qquad \hat{\chi} \in \mathcal{S} \text{ by corollary 1.2.16}$$

Step 1: uniform bound and equicontinuity on balls. Fix $R > 0$. For $|\xi| \le R$ one has $\langle \xi \rangle^{-\sigma} \le \langle R \rangle^{|\sigma|}$, so (2.1) at order σ with $\Phi = \hat{\chi}$ and $f = u_k$ gives

$$|v_k(\xi)| \le A_R M \quad (|\xi| \le R), \qquad A_R = 2^{|\sigma|/2}\langle R \rangle^{|\sigma|}\|\langle \cdot \rangle^{|\sigma|}\hat{\chi}\|_{L^2}$$

For $|\xi|, |\xi'| \le R$ the segment $\xi_\theta = \xi' + \theta(\xi - \xi')$, $0 \le \theta \le 1$, stays in the ball, and the fundamental theorem of calculus along it gives $|\hat{\chi}(\xi - \eta) - \hat{\chi}(\xi' - \eta)| \le |\xi - \xi'| \int_0^1 |\nabla \hat{\chi}(\xi_\theta - \eta)|\, d\theta$. Hence, by Tonelli and (2.1) at order σ with $\Phi = |\nabla \hat{\chi}|$ (each $\partial_j \hat{\chi}$ is in $\mathcal{S}$) and $f = u_k$,

$$|v_k(\xi) - v_k(\xi')| \le |\xi - \xi'| \int_0^1 \int |\nabla \hat{\chi}(\xi_\theta - \eta)||\hat{u}_k(\eta)|\, d\eta\, d\theta \le L_R M\, |\xi - \xi'|$$

where $L_R = 2^{|\sigma|/2}\langle R \rangle^{|\sigma|}\|\langle \cdot \rangle^{|\sigma|}|\nabla \hat{\chi}|\|_{L^2}$.

Step 2: Arzelà-Ascoli and a diagonal subsequence. By Step 1, on the compact metric space $\overline{B}_R = \{|\xi| \le R\}$ the real-valued functions $\mathrm{Re} v_k$ and $\mathrm{Im} v_k$ are continuous, bounded by $A_R M$, and Lipschitz with constant $L_R M$, all uniformly in k. Two applications of [6, Arzela-Ascoli Theorem], which is stated for real-valued functions, produce a subsequence along which first $\mathrm{Re} v_k$ and then $\mathrm{Im} v_k$ converge uniformly on $\overline{B}_R$, that is, v_k converges uniformly on $\overline{B}_R$. Doing this for $R = 1, 2, 3, \ldots$ in turn, each time inside the subsequence already extracted, and passing to the diagonal, yields a subsequence (v_{k_j}) converging uniformly on every ball.

Step 3: the tail. For $|\xi| \ge R$ one has $\langle \xi \rangle \ge \langle R \rangle$, and since $t > 0$, $\langle \xi \rangle^{2s} = \langle \xi \rangle^{-2t}\langle \xi \rangle^{2\sigma} \le \langle R \rangle^{-2t}\langle \xi \rangle^{2\sigma}$. As $v_k - v_l = \widehat{\chi u_k - \chi u_l}$ almost everywhere,

$$\int_{|\xi| \ge R} |v_k(\xi) - v_l(\xi)|^2 \langle \xi \rangle^{2s}\, d\xi \le \langle R \rangle^{-2t}\|\chi u_k - \chi u_l\|_{H^\sigma}^2 \le \langle R \rangle^{-2t}(2C_\chi M)^2 \qquad (k, l \in \mathbb{N})$$

This is the only place where $t > 0$ enters.

Step 4: conclusion. Let $\varepsilon > 0$ and choose $R \in \mathbb{N}$ with $\langle R\rangle^{-2t}(2C_\chi M)^2 < \varepsilon/2$. On the ball,

$$\int_{|\xi|\le R} |v_{k_j}(\xi) - v_{k_l}(\xi)|^2 \langle\xi\rangle^{2s}\, d\xi \le \sup_{|\xi|\le R} |v_{k_j}(\xi) - v_{k_l}(\xi)|^2 \int_{|\xi|\le R} \langle\xi\rangle^{2s}\, d\xi$$

and the last integral is finite while the supremum tends to zero as $j, l \to \infty$ by Step 2, so the left side is below $\varepsilon/2$ for all large j, l. Adding the two pieces, $\|\chi u_{k_j} - \chi u_{k_l}\|^2_{H^s} < \varepsilon$ for all large j, l, so (χu_{k_j}) is Cauchy in H^s and converges there by proposition 2.5.2 (1).

The compact operator is $u \mapsto \chi u$ and not the inclusion $H^{s+t} \hookrightarrow H^s$, which the translates in the opening paragraph show is never compact. For distributions supported in a fixed compact set, and $\chi = 1$ near that set, the two coincide.

Corollary 2.5.8: Rellich's Lemma for Compactly Supported Distributions

Let $K \subseteq \mathbb{R}^n$ be compact, $s \in \mathbb{R}$, and $t > 0$. Every sequence (u_k) that is bounded in $H^{s+t}(\mathbb{R}^n)$ and supported in K, meaning $\langle u_k, \varphi\rangle = 0$ for all $\varphi \in C_c^\infty(\mathbb{R}^n \setminus K)$, has a subsequence converging in $H^s(\mathbb{R}^n)$.

***Proof*:**
The smooth Urysohn lemma, [13, Smooth Urysohn's Lemma], applied to the compact set $\{x : \operatorname{dist}(x, K) \le 1\}$ inside the open set $\{x : \operatorname{dist}(x, K) < 2\}$, provides $\chi \in C_c^\infty(\mathbb{R}^n)$ with $\chi = 1$ on the open neighbourhood $W = \{x : \operatorname{dist}(x, K) < 1\}$ of K. Once $\chi u_k = u_k$ is known for every k, theorem 2.5.7 applied to $(u_k) = (\chi u_k)$ finishes the proof.

Fix k and $\psi \in \mathcal{S}$, and put $g = (1-\chi)\psi \in \mathcal{S}$, which vanishes on W. Choose $\theta \in C_c^\infty(\mathbb{R}^n)$ with $\theta = 1$ on $\{|x| \le 1\}$ and set $\theta_R(x) = \theta(x/R)$ for $R \ge 1$. Then $\theta_R g$ is smooth with compact support in $\mathbb{R}^n \setminus W \subseteq \mathbb{R}^n \setminus K$, so $\langle u_k, \theta_R g\rangle = 0$, and $\theta_R g \to g$ in $\mathcal{S}$: the derivatives of θ_R are bounded independently of $R \ge 1$, and $1 - \theta_R$ and all its derivatives vanish on $\{|x| < R\}$, so by the Leibniz rule

$$\|(1-\theta_R)g\|_{(N,\alpha)} \le C_\alpha \sup_{|x|\ge R} (1+|x|)^N \sum_{\beta\le\alpha} |\partial^\beta g(x)| \le \frac{C_\alpha}{1+R} \sum_{\beta\le\alpha} \|g\|_{(N+1,\beta)} \xrightarrow{R\to\infty} 0$$

with C_α depending only on α and θ. By continuity of u_k on $\mathcal{S}$, $\langle u_k, g\rangle = 0$, that is, $\langle \chi u_k, \psi\rangle = \langle u_k, \chi\psi\rangle = \langle u_k, \psi\rangle$, so $\chi u_k = u_k$.

Corollary 2.5.8 is the Euclidean statement that [7] localizes to coordinate charts in order to prove Rellich compactness on compact manifolds. With $s = 0$, $t = 1$, and $K = \overline{\Omega}$ it is also the mechanism behind the discrete Dirichlet spectrum presupposed by box 2.5.9: once the inverse of the Dirichlet Laplacian on a bounded domain Ω is known to map $L^2(\Omega)$ boundedly into functions of $H^1(\mathbb{R}^n)$ supported in $\overline{\Omega}$, Corollary 2.5.8 makes it a compact self-adjoint operator on $L^2(\Omega)$, and such operators are diagonalizable with eigenvalues tending to zero ([13, Normal Compact, Then Diagonalizable]).

2.5.9 Hearing the Shape of a Fractal Mark Kac's famous question "Can one hear the shape of a drum?" asks whether the eigenvalues of the Laplacian on a bounded domain Ω determine Ω up to isometry. In 1911, Hermann Weyl showed that the eigenvalue counting function $N(\lambda) = \#\{\lambda_j \leq \lambda\}$ satisfies the asymptotic law $N(\lambda) \sim c_n \operatorname{vol}(\Omega)\lambda^{n/2}$ as $\lambda \to \infty$—so one can at least "hear" the volume and dimension.

Sobolev spaces enter the picture through the connection between eigenvalue asymptotics and regularity. The Laplacian $-\Delta$ on Ω with Dirichlet boundary conditions is a self-adjoint operator on $L^2(\Omega)$, and its eigenfunctions lie in $H^s(\Omega)$ for all s (by elliptic regularity). The Sobolev embedding theorem then controls the L^∞ norms of eigenfunctions in terms of their H^s norms, which in turn controls the growth of the eigenvalue counting function.

When $\partial\Omega$ is a fractal (say, the von Koch snowflake), Weyl's law acquires a correction term involving the Minkowski content and dimension of the boundary. In Lapidus's theory of "fractal drums", the modified Weyl law takes the form $N(\lambda) = c_n \operatorname{vol}(\Omega)\lambda^{n/2} - c_{n-1}\mathcal{M}^d(\partial\Omega)\lambda^{d/2} + o(\lambda^{d/2})$, where d is the Minkowski dimension of $\partial\Omega$ and $\mathcal{M}^d$ is the Minkowski content. The spectral theory of the Laplacian, Sobolev regularity, and the fractal geometry of the boundary are thus intimately intertwined. In this way, distribution theory reaches from the abstract world of generalized functions all the way to the geometry of fractals.

2.6 Applications To Partial Differential Equations

Distributions were invented by Laurent Schwartz in the 1940s precisely to provide a rigorous framework for the generalized solutions that physicists had been using informally for decades. In this section, we develop the basic theory of constant-coefficient linear PDEs from the distributional perspective, culminating in the Malgrange–Ehrenpreis theorem: every such equation has a fundamental solution.

2.6.1 Fundamental Solutions

Consider a constant-coefficient linear differential operator

$$P(\partial) = \sum_{|\alpha|\leq m} a_\alpha \partial^\alpha, \qquad a_\alpha \in \mathbb{C}.$$

The associated *symbol* is the polynomial $p(\xi) = \sum_{|\alpha|\leq m} a_\alpha (2\pi i\xi)^\alpha$, chosen so that $\widehat{P(\partial)f} = p \cdot \hat{f}$ for $f \in \mathcal{S}'$. Given a source term g, we wish to solve the equation $P(\partial)u = g$.

Definition 2.6.1: Fundamental Solution

A *fundamental solution* (or *Green's function*) for $P(\partial)$ is a distribution $E \in \mathcal{D}'(\mathbb{R}^n)$ satisfying

$$P(\partial)E = \delta.$$

If such an E exists and g is a distribution for which the convolution $E * g$ is well-defined (for example, if $g \in \mathcal{E}'$ or $g \in L^p$ with compact support), then $u = E * g$ satisfies $P(\partial)u = P(\partial)(E * g) = (P(\partial)E) * g = \delta * g = g$, so u is a distributional solution. Thus, the problem of solving all constant-coefficient PDEs reduces to finding a single distribution E.

Example 2.6.2: Fundamental Solutions Of Classical Operators

1. *The Laplacian in $\mathbb{R}^n$ ($n \geq 3$).* For $\Delta = \sum_{j=1}^n \partial_j^2$, the fundamental solution is $E(x) = c_n|x|^{2-n}$, where $c_n = 1/(n(2-n)\omega_n)$ and ω_n is the volume of the unit ball. This is locally integrable (since $2-n > -n$ for $n \geq 3$), and the identity $\Delta E = \delta$ is verified by a direct computation: for $\varphi \in C_c^\infty$, $\langle \Delta E, \varphi\rangle = \langle E, \Delta\varphi\rangle = c_n \int |x|^{2-n}\Delta\varphi(x)\,dx$, and integrating by parts on $\{|x| > \varepsilon\}$ and letting $\varepsilon \to 0$ yields $\varphi(0)$ (the boundary terms on the sphere $|x| = \varepsilon$ contribute $\varphi(0)$ in the limit by the mean value property and the computation of the surface integral).
2. *The Laplacian in $\mathbb{R}^2$.* Here $E(x) = \frac{1}{2\pi}\log|x|$, and the verification is similar.
3. *The heat operator.* For $P(\partial) = \partial_t - \Delta_x$ on $\mathbb{R}^{n+1}$, the fundamental solution is the heat kernel $E(x,t) = (4\pi t)^{-n/2}e^{-|x|^2/4t}$ for $t > 0$, extended by 0 for $t < 0$.

2.6.2 The Malgrange–Ehrenpreis Theorem

The examples above might suggest that finding fundamental solutions requires ad hoc computations tailored to each operator. The remarkable theorem of Malgrange (1955) and Ehrenpreis (1954), proved independently, shows that this is not the case: *every* nonzero constant-coefficient operator has a fundamental solution. This is one of the great triumphs of distribution theory.

The proof uses the Fourier transform to convert the PDE $P(\partial)E = \delta$ into an algebraic problem: taking Fourier transforms gives $p(\xi)\hat{E}(\xi) = 1$, so we need to "divide by p" in $\mathcal{S}'$. The difficulty is that p may have real zeros. The key algebraic lemma shows that this division is always possible.

Lemma 2.6.3: Division By Linear Factors In $\mathcal{S}'(\mathbb{R})$

Let $a \in \mathbb{C}$. For every $f \in \mathcal{S}'(\mathbb{R})$, the equation $(\xi - a)u = f$ has a solution $u \in \mathcal{S}'(\mathbb{R})$.

***Proof*:**
If $a \notin \mathbb{R}$, then $1/(\xi - a)$ is a C^∞ function on $\mathbb{R}$ with all derivatives bounded by powers of $1/|\operatorname{Im}(a)|$, so $u = f/(\xi - a)$ (defined by $\langle u, \varphi\rangle = \langle f, \varphi/(\xi - a)\rangle$) is a well-defined tempered distribution.

If $a \in \mathbb{R}$, we may assume $a = 0$ by translation (replacing $\varphi(\xi)$ by $\varphi(\xi + a)$). We must solve $\xi \cdot u = f$, that is:

$$\langle u, \xi\varphi(\xi)\rangle = \langle f, \varphi(\xi)\rangle \qquad \text{for all } \varphi \in \mathcal{S}.$$

The map $\varphi \mapsto \xi\varphi$ sends $\mathcal{S}$ into itself, but is not surjective: its range is $\mathcal{S}_0 = \{\psi \in \mathcal{S} : \psi(0) = 0\}$, a closed subspace of codimension 1.

We define u on $\mathcal{S}_0$ as follows. For $\psi \in \mathcal{S}_0$, the function $\psi(\xi)/\xi$ is smooth on $\mathbb{R} \setminus \{0\}$ and extends smoothly to all of $\mathbb{R}$ (since $\psi(0) = 0$: write $\psi(\xi) = \xi \int_0^1 \psi'(t\xi)\,dt$, so $\psi(\xi)/\xi = \int_0^1 \psi'(t\xi)\,dt \in C^\infty$). Moreover, $\psi/\xi \in \mathcal{S}$: for large $|\xi|$, $|\xi^k\partial^j(\psi/\xi)| \lesssim |\xi|^{k-1}|\partial^j\psi| + \text{lower} \to 0$ since $\psi \in \mathcal{S}$, and near $\xi = 0$, smoothness was just established.

So for $\psi \in \mathcal{S}_0$, define $\langle u, \psi\rangle = \langle f, \psi/\xi\rangle$. This satisfies $\langle u, \xi\varphi\rangle = \langle f, \varphi\rangle$ for all $\varphi \in \mathcal{S}$ (since $\xi\varphi \in \mathcal{S}_0$ and $(\xi\varphi)/\xi = \varphi$).

To extend u to all of $\mathcal{S}$, fix any $\varphi_0 \in \mathcal{S}$ with $\varphi_0(0) = 1$. Every $\psi \in \mathcal{S}$ decomposes uniquely as

$\psi = (\psi - \psi(0)\varphi_0) + \psi(0)\varphi_0$, where $\psi - \psi(0)\varphi_0 \in \mathcal{S}_0$. Define

$$\langle u, \psi \rangle = \langle f, (\psi - \psi(0)\varphi_0)/\xi \rangle + c \cdot \psi(0)$$

for an arbitrary constant $c \in \mathbb{C}$. This is continuous on $\mathcal{S}$ (both $\psi \mapsto \langle f, (\psi - \psi(0)\varphi_0)/\xi \rangle$ and $\psi \mapsto \psi(0)$ are continuous), and satisfies $\xi u = f$ by construction. The constant c reflects the non-uniqueness: if u_1 and u_2 both solve $\xi u = f$, then $\xi(u_1 - u_2) = 0$, so $u_1 - u_2$ is supported at $\{0\}$, hence is a linear combination of δ and its derivatives—but the equation $\xi v = 0$ in $\mathcal{S}'(\mathbb{R})$ forces $v = c\delta$ for some c (since $\langle v, \xi\varphi \rangle = 0$ for all φ, meaning v vanishes on $\mathcal{S}_0$, and $\mathcal{S}_0$ has codimension 1 with the complementary functional being evaluation at 0, i.e., δ).

Lemma 2.6.4: Division By Polynomials In $\mathcal{S}'(\mathbb{R})$

Let $p(\xi)$ be a nonzero polynomial on $\mathbb{R}$. For every $f \in \mathcal{S}'(\mathbb{R})$, the equation $p(\xi)u = f$ has a solution $u \in \mathcal{S}'(\mathbb{R})$.

***Proof*:**
Factor $p(\xi) = c\prod_{j=1}^{m}(\xi - a_j)$ where $a_1, \ldots, a_m \in \mathbb{C}$ are the roots (counted with multiplicity). Apply lemma 2.6.3 inductively: first solve $(\xi - a_m)u_1 = f$, then $(\xi - a_{m-1})u_2 = u_1$, and so on. After m steps, we obtain $u = u_m/c$ with $p(\xi)u = f$.

To extend this to several variables, we use the following observation: division by a polynomial in $\mathbb{R}^n$ can be reduced to division in one variable by treating the remaining variables as parameters.

Lemma 2.6.5: Division By Polynomials In $\mathcal{S}'(\mathbb{R}^n)$

Let $p(\xi) = p(\xi_1, \ldots, \xi_n)$ be a nonzero polynomial on $\mathbb{R}^n$. For every $f \in \mathcal{S}'(\mathbb{R}^n)$, the equation $p(\xi)u = f$ has a solution $u \in \mathcal{S}'(\mathbb{R}^n)$.

***Proof*:**
We proceed by induction on n. The case $n = 1$ is lemma 2.6.4.

For $n \geq 2$, write $\xi = (\xi_1, \xi') \in \mathbb{R} \times \mathbb{R}^{n-1}$. After a linear change of coordinates (which preserves $\mathcal{S}'$), we may assume that p has degree m in ξ_1:

$$p(\xi_1, \xi') = a_m(\xi')\xi_1^m + a_{m-1}(\xi')\xi_1^{m-1} + \cdots + a_0(\xi')$$

where a_m is a nonzero polynomial in ξ' (this can always be arranged by a generic rotation).

We proceed in two stages. First, perform polynomial long division in ξ_1: using the Euclidean algorithm for polynomials in ξ_1 (with coefficients that are distributions in ξ'), we can reduce to the case where we divide by $a_m(\xi')$, which is a polynomial in $n-1$ variables, and then divide by a monic polynomial in ξ_1 (whose roots are the roots of p as a polynomial in ξ_1).

More precisely, here is the argument. Define the "partial Fourier transform" in ξ_1 and use the one-dimensional division lemma for each fixed ξ'. The subtlety is ensuring the result depends measurably (and in fact, in a tempered fashion) on ξ'.

We present the clean version. The polynomial p can be written as

$$p(\xi_1, \xi') = a_m(\xi') \prod_{j=1}^{m} (\xi_1 - r_j(\xi')),$$

where $r_j(\xi')$ are the roots in $\mathbb{C}$ (which are algebraic functions of ξ'). To solve $pu = f$, it suffices to solve $a_m(\xi')v = f$ (where v and f are tempered distributions on $\mathbb{R}^n$, and $a_m(\xi')$ acts by multiplication in the ξ' variables), and then solve $\prod_j(\xi_1 - r_j(\xi'))w = v$ by dividing by each linear factor in ξ_1 successively.

For the first step: $a_m(\xi')$ is a nonzero polynomial on $\mathbb{R}^{n-1}$. By the inductive hypothesis, the division $a_m(\xi')v = g$ can be solved in $\mathcal{S}'(\mathbb{R}^{n-1})$ for any tempered distribution g on $\mathbb{R}^{n-1}$. Extending this to $\mathbb{R}^n$ (treating ξ_1 as a parameter) solves $a_m(\xi')v = f$ in $\mathcal{S}'(\mathbb{R}^n)$.

For the second step: each factor $(\xi_1 - r_j(\xi'))$ involves division in the ξ_1 variable. When $r_j(\xi')$ is not real for a given ξ', the division is trivial (divide by a nonvanishing smooth function). When $r_j(\xi')$ is real, we apply the one-dimensional division lemma. The argument from lemma 2.6.3 extends to the parametric setting: the decomposition $\psi(\xi_1) = (\psi(\xi_1) - \psi(r_j(\xi'))\varphi_0(\xi_1)) + \psi(r_j(\xi'))\varphi_0(\xi_1)$ depends smoothly on the parameter ξ' (using the smooth dependence of roots on coefficients away from coalescence points, and handling the coalescence points by a partition of unity argument).

The full technical details require verifying that each step produces a tempered distribution (not a distribution), which follows from the polynomial growth of all quantities involved. We refer to theorem 7.3.2 in Hörmander's *The Analysis of Linear Partial Differential Operators I* for the complete argument in the parametric case.

With the division lemma in hand, the Malgrange–Ehrenpreis theorem is an immediate consequence.

Theorem 2.6.6: The Malgrange–Ehrenpreis Theorem

Let $P(\partial) = \sum_{|\alpha| \le m} a_\alpha \partial^\alpha$ be a nonzero constant-coefficient linear differential operator on $\mathbb{R}^n$. Then there exists a distribution $E \in \mathcal{D}'(\mathbb{R}^n)$ such that

$$P(\partial)E = \delta.$$

***Proof*:**

Let $p(\xi) = \sum_{|\alpha| \le m} a_\alpha (2\pi i \xi)^\alpha$ be the symbol of $P(\partial)$, so that $\widehat{P(\partial)f} = p \cdot \hat{f}$ for $f \in \mathcal{S}'$. Taking Fourier transforms, the equation $P(\partial)E = \delta$ becomes

$$p(\xi)\hat{E}(\xi) = \hat{\delta}(\xi) = 1.$$

By lemma 2.6.5, the equation $p(\xi)u = 1$ has a solution $u \in \mathcal{S}'(\mathbb{R}^n)$ (taking $f = 1$, which is a tempered distribution). Define $E = \check{u}$ (the inverse Fourier transform of u). Then $E \in \mathcal{S}' \subset \mathcal{D}'$ and

$$\widehat{P(\partial)E} = p \cdot \hat{E} = p \cdot u = 1 = \hat{\delta},$$

so $P(\partial)E = \delta$ by the injectivity of the Fourier transform on $\mathcal{S}'$.

Let us pause to appreciate what we have proved. The Malgrange–Ehrenpreis theorem says that the purely algebraic problem of dividing by a polynomial in $\mathcal{S}'$ translates, via the Fourier transform, into the analytical statement that every constant-coefficient PDE has a distributional fundamental solution. The entire proof rests on three results: the Fourier transform on $\mathcal{S}'$ (Section 2.1), the algebraic structure of polynomial division, and the one-dimensional division lemma (lemma 2.6.3), which itself is a simple consequence of the codimension-1 structure of the subspace $\{\psi \in \mathcal{S} : \psi(0) = 0\}$.

Corollary 2.6.7: Solvability Of Constant-Coefficient PDEs

Let $P(\partial)$ be a nonzero constant-coefficient operator and let $g \in \mathcal{E}'(\mathbb{R}^n)$ (a compactly supported distribution). Then the equation $P(\partial)u = g$ has a solution $u \in \mathcal{D}'(\mathbb{R}^n)$.

***Proof*:**
Let E be the fundamental solution from the Malgrange–Ehrenpreis theorem. Since $g \in \mathcal{E}'$, the convolution $u = E * g$ is well-defined as a distribution (the compact support of g ensures that $\langle E * g, \varphi\rangle = \langle E_x, \langle g_y, \varphi(x+y)\rangle\rangle$ converges for every $\varphi \in C_c^\infty$). Then

$$P(\partial)u = P(\partial)(E * g) = (P(\partial)E) * g = \delta * g = g.$$

Example 2.6.8: Revisiting The Laplacian

For $P(\partial) = \Delta = \sum_j \partial_j^2$ on $\mathbb{R}^n$, the symbol is $p(\xi) = -4\pi^2|\xi|^2$. The equation $-4\pi^2|\xi|^2\hat{E} = 1$ cannot be solved by simply writing $\hat{E} = -1/(4\pi^2|\xi|^2)$, since this function is not locally integrable near $\xi = 0$ for $n \leq 2$. However, the division lemma constructs a distributional solution. In this case, the solution can also be obtained by regularization: for instance, $\hat{E}(\xi) = \lim_{\varepsilon \to 0^+} -1/(4\pi^2(|\xi|^2 + \varepsilon))$, where the limit is taken in $\mathcal{S}'$. The inverse Fourier transform recovers the classical fundamental solution: $E(x) = c_n|x|^{2-n}$ for $n \geq 3$ and $E(x) = \frac{1}{2\pi}\log|x|$ for $n = 2$.

2.6.9 Distribution Theory in Context The Malgrange–Ehrenpreis theorem is an existence theorem: it guarantees a fundamental solution exists, but says nothing about its regularity, uniqueness (fundamental solutions are not unique in general—one can always add a solution of the homogeneous equation), or how it depends on the operator. For *elliptic* operators (those whose symbol satisfies $|p(\xi)| \geq c|\xi|^m$ for large $|\xi|$), the theory of elliptic regularity shows that distributional solutions are automatically smooth wherever the source term is smooth. For *hyperbolic* operators (like the wave operator), the fundamental solution has singularities propagating along characteristic surfaces. These deeper questions—regularity, propagation of singularities, and the fine structure of solutions—form the subject of microlocal analysis, which extends distribution theory by localizing in both position and frequency. The wavefront set of a distribution, a subset of the cotangent bundle $T^*\mathbb{R}^n$, provides the precise language for these phenomena, and opens the door to the modern theory of pseudodifferential operators and Fourier integral operators.

Part II

Abstract Harmonic Analysis on LCA Groups

The second part is the structural counterpart of the first. The Fourier transform of part I diagonalizes the translation-invariant operators on $\mathbb{R}^n$, and the same idea applies verbatim to any locally compact abelian group, with the frequencies $\xi \in \mathbb{R}^n$ replaced by the characters of the group. The dual group $\hat{G}$ collects these characters, the Fourier transform carries functions on G to functions on $\hat{G}$, and Pontryagin duality $\hat{\hat{G}} \cong G$ is the statement that the passage loses nothing. The non-abelian generalization, in which characters give way to higher-dimensional unitary representations, is the province of the companion theory of unitary representations, for which this part is the abelian model.

3

Abstract Harmonic Analysis on LCA Groups

This part carries the Fourier transform from Euclidean space to its natural home, a locally compact abelian group. The classical transform of Part I is the special case $G = \mathbb{R}^n$; the general theory replaces $\mathbb{R}^n$ by an arbitrary LCA group, its frequencies by the characters of that group, and reveals the structural fact hiding behind Fourier inversion and Plancherel, the Pontryagin duality between a group and its dual.

3.1 Locally Compact Groups and Haar Measure, Recalled

The two ingredients this theory rests on are developed in full elsewhere in the series; this section fixes notation and recalls what is used.

The point-set theory of topological and locally compact groups is [9]: a topological group is a group carrying a topology for which multiplication and inversion are continuous, and it is locally compact when the identity has a neighbourhood with compact closure. The standing examples are $\mathbb{R}^n$ and $\mathbb{R}^\times$, the circle $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ and its powers $\mathbb{T}^n$, the discrete groups $\mathbb{Z}^n$, and the p-adic groups $\mathbb{Z}_p$ and $\mathbb{Q}_p$; all are abelian, which is the standing hypothesis of this part.

The invariant measure that makes integration possible is Haar measure, constructed and proved unique up to a positive scalar in [14, Haar Measure]: every locally compact Hausdorff group carries a nonzero left-invariant Radon measure μ ([14, Existence of Haar Measure]), and any two differ only by scale ([14, Uniqueness of Haar Measure]). On an abelian group left and right translation coincide, so a Haar measure is automatically bi-invariant and the group is unimodular. The normalizations of the standard examples, all instances of the general theorem, are Lebesgue measure on $\mathbb{R}^n$; arc length of total mass 1 on $\mathbb{T}$; $d\mu(t) = dt/|t|$ on $\mathbb{R}^\times$; and counting measure on a discrete group. Fixing such a μ, the Haar integral $\int_G f\,d\mu$ is the invariant averaging on which the Fourier transform is built, and left- and right-uniform continuity of functions in $C_c(G)$ ([14, Left-Uniform Continuity]) is used without further comment.

3.2 Pontryagin Duality

In order to define the Fourier transformation on groups, we require the notion of *topological groups* which will allow us to define a Borel σ-algebra on which we will define a measure known as the *Haar measure*. With a measure, we may define the notion of *integrable function on a group G*. We next require the generalization of the domain of $\hat{f}$. When working with $\mathbb{T}^n$, the answer was "obvious" in that we had $\hat{f} : \mathbb{Z}^n \to \mathbb{R}$ since $\hat{f}$ enumerated the orthonormal basis of $L^2(\mathbb{T}^n)$. When we generlized the Fourier transform to functions $f \in L^p(\mathbb{R}^n)$ (for $1 \leq p \leq 2$), we had that the domain of $\hat{f}$ was $\mathbb{R}^n$, and in general the domain of $\hat{f}$ transformed like *functionals*. We will explore this concept when we define *Pontryagin Duality*.

After having introduced the relevant concepts, we shall introduce Fourier transforms on locally compact (Hausdorff) abelian group. Due to the Pontryagin duality, many classical results of Fourier Analysis are readily recovered. We shall not go over too much detail as this is more an essay than the beginning of a course in Harmonic Analysis, however some interesting highlights will be added.

The paper will end with some interesting uses and consequence of the Fourier transform and of the Pontryagin duality, including how it demonstrates compact abelian groups are just as hard to classify as abelian groups, how the Fourier Transform shows up in Representation Theory of finite groups, how Pontryagin Duality allows us to classify locally compact abelian groups through their characters.

A quick disclaimer that I will be assuming some 1st-year graduate analysis knowledge for this paper as I will have to touch on topics covered in those courses.

Let G be a locally compact (Hausdorff) abelian group. If G was also a topological vector space (i.e. a topological field k acts continuously on G so that G becomes a topological vector space), then we would have a natural notion of duality since G would be a vector space over some underlying field k. For general group, no such "natural dualizing" object exists. Instead, we may try to find some interesting group so that the contravariant functor between locally compact abelian groups (which we'll denote **LCA**):

$$\mathrm{Hom}(-, H) : \mathbf{LCA} \to \mathbf{LCA} \tag{3.1}$$

has properties similar to what we are used to in the dual theory we have in Functional Analysis (ex. reflexivity or properties of adjoints) and that we recover some classical results from Fourier Theory. Thanks to work by Lev Pontryagin, we have that a natural choice for H in equation (3.1) is the circle group with the usual subspace topology of $\mathbb{R}^2$. We shall henceforth label this group as $\mathbb{T}$.

Definition 3.2.1: Pontryagin Dual

Let G be a locally compact abelian group and $\mathbb{T}$ the circle group (with multiplication). Then

$$\widehat{G} := \mathrm{Hom}(G, \mathbb{T})$$

where $\mathrm{Hom}(G, \mathbb{T})$ is the collection of all continuous homomorphisms, is the *Pontryagin dual* of G endowed with the topology of uniform convergence of compact sets[a]. An element of $\widehat{G}$ is called a *character* of G.

[a] If you are familiar with complex analysis, this topology is commonly used for the convergence of holomorphic functions

For a better understanding of the topology on $\widehat{G}$, consider The collection of sets

$$W_K^V := \left\{ \chi \in \widehat{G} \;\middle|\; \chi(K) \subseteq V \right\}$$

where $K \subseteq G$ is a compact subset and $V \subseteq \mathbb{T}$ is a neighborhood containing the identity. Then the basis for the topology of $\widehat{G}$ is the collection of all W_K^V along with their translations. For examples of such convergence, recall power series centered at 0 ([1]) on $\mathbb{C}$ with radius R converge uniformly in $\overline{B_r(0)}$ when $r < R$, but it does not necessarily converge uniformly on $r = R$. Hence, power series technically converge in the *compact-open* topology[2].

Example 3.2.2: Pontryagin Dual

1. Let $G = \mathbb{Z}/n\mathbb{Z}$ with the discrete topology. Then:

   $$\widehat{\mathbb{Z}/n\mathbb{Z}} = \mathrm{Hom}(\mathbb{Z}/n\mathbb{Z}, \mathbb{T}) = \mathbb{Z}/n\mathbb{Z}$$

   To see this, notice that any $f \in \widehat{G}$ is determined by where it maps 1. Since G has finite order, $f(1)$ must have finite order. Necessarily, $f(1)$ maps to a root of unity $e^{\frac{2\pi i}{k}}$ where $1 \leq k \leq n$. Now it is easy to finish the computations to see that $\widehat{Z/n\mathbb{Z}} = \mathbb{Z}/n\mathbb{Z}$

2. Let $G = \mathbb{R}$. Then:

   $$\widehat{\mathbb{R}} = \mathrm{Hom}(\mathbb{R}, \mathbb{T}) = \mathbb{R}$$

   To see this, we will slightly modify a proof presented in Foland's *Real Analysis* Theorem 8.19. Let $\chi \in \widehat{\mathbb{R}}$. Let $a \in \mathbb{R}$ such that $\int_0^a \chi(t)dt \neq 0$. Such an a exists, or else by the Lebesgue Differentiation Theorem $\chi = 0$ a.e. Setting $A = \left(\int_0^a \chi(t)dt\right)^{-1}$, we get:

   $$\chi(x) = \chi(x)AA^{-1} = A\int_0^a \chi(x)\chi(t)dx = A\int_0^a \chi(x+t)dt = A\int_x^{x+a} \chi(t)dt$$

   Since the function inside the integral continuous, χ is in fact C^1 (in fact C^∞, but we will not need this fact). Now, notice that:

   $$\chi'(x) = A[\chi(x+a) - \chi(x)] = B\chi(x)$$

[1] they may be centered anywhere, but we will center them at zero for simplicity

[2] To be precise, they are locally uniformly convergent, which since $\mathbb{C}$ is locally compact is equivalent to uniform convergence on compact sets

where $B = A(\chi(a) - 1)$. This is now a simple ODE, which can be solved to show we get an exponential (and is unique by the uniqueness of a solution to an ODE). For an alternative approach, notice that

$$\frac{d}{dx}(e^{-Bx}\chi(x)) = 0$$

so $e^{-Bx}\chi(x)$ is constant. Since $\chi(0) = 1$, we have $\chi(x) = e^{Bx}$, and since $|\chi(x)| = 1$, B must be purely imaginary meaning $B = 2\pi i\xi$ for some $\xi \in \mathbb{R}$, completing the proof.

We may easily generalize this proof for when $G = \mathbb{R}^n$. In this case, the character's will be of the form $\chi(x) = e^{2\pi i\xi\cdot x}$ where $\xi \cdot x$ is the dot product.

3. Let $G = \mathbb{Z}$. Then:

$$\widehat{\mathbb{T}} = \operatorname{Hom}(\mathbb{T}, \mathbb{T}) = \mathbb{Z}$$

and similarly:

$$\widehat{\mathbb{Z}} = \operatorname{Hom}(\mathbb{Z}, \mathbb{T}) = \mathbb{T}$$

For the first, recall that $\mathbb{T} \cong \mathbb{R}/\mathbb{Z}$. Then looking at the above proof, we would consider χ to be periodic with period 1 and $e^{2\pi i\xi} = 1$ if and only if $\xi \in \mathbb{Z}$. A similar argument works for the second case.

Looking at this from a second perspective, from usual Fourier Analysis we know that the Fourier transform is a map $\mathcal{F} : L^2(\mathbb{T}) \to \ell^2(\mathbb{Z})$, and so it should be expected that $\mathbb{Z}$ and $\mathbb{T}$ are Pontryagin dual to each other.

It is natural to ask if it is always the case that that $\widehat{\widehat{G}} = G$ which demonstrates the dual spaces does not loose or introduce new information but re-paramaterizes what information G already has. We know that in general duality theory, this is not the case (the dual of the dual may be much larger than the original object, think of $C_b(X)$). If it is the case, that means that the dual $\widehat{G}$ of G contains all the required information about G. For Pontryagin duality, it is always reflexive:

Theorem 3.2.3: Pontryagin Duality Theorem

Let G be a locally compact abelian group. Then there is a natural isomorphism:

$$G \mapsto \widehat{\widehat{G}}$$

mapping $g \mapsto \operatorname{ev}_g$ where $\operatorname{ev}_g : \widehat{G} \to G$ is defined by $\operatorname{ev}_g(\chi) = \chi(g)$.

We delay the proof until the Fourier transform for groups is in hand; it is carried out in full in sections 3.2.2 and 3.2.3 and completed below. Note that in general $G \ncong \widehat{G}$ (think of $L^p(X) \ncong L^q(X) = (L^p(X))^*$), however if G is finite and abelian then $G \cong \widehat{G}$ (though not canonically). The essential take away from Pontryagin duality is that $\widehat{G}$ contains enough information on G to recover G.

3.2.4 Remark The map in theorem 3.2.3 can in fact be augmented to be a covariant functor (even an exact functor, i.e. it preserves short exact sequences). Hence, we may look at the study of extending group or the homology of groups in the Pontryagin dual.

The statement that the Fourier transform changes the basis from "positions" $\{\delta_x\}$ to "waves" $\{e^{ikx}\}$ hints at a broader duality: it changes the very nature of the "points" that constitute our space.

Let us define a function f by its values at geometric points. In the language of distributions, we can write any function as a continuous sum of Dirac deltas:

$$f(x) = \int_{\mathbb{T}^1} f(y)\delta_y(x)\, dy$$

Here, the building blocks are points $y \in \mathbb{T}^1$. A "point" is simply a linear functional that evaluates a function at a specific location: $\mathrm{ev}_y(f) = f(y)$.

Now consider the Fourier Transform. It maps a function f on the group G to a function $\hat{f}$ on the dual group $\widehat{G}$ (the set of characters). What are the "points" of this new space? They are the irreducible representations (the frequencies).

$$\hat{f}(\chi) = \int_G f(x)\overline{\chi(x)}\, dx$$

When we transform a geometric point δ_y, it becomes a function on the dual group:

$$\widehat{\delta_y}(\chi) = \overline{\chi(y)}$$

This reveals that the Fourier Transform is an isomorphism between two different kinds of spaces:

- The *Physical Space* G, whose points are geometric locations x.
- The *Spectral Space* $\widehat{G}$, whose points are irreducible representations χ.

This is a specific instance of a grander theme in mathematics known as *Gelfand Duality*. In this framework, any commutative C^*-algebra (like the algebra of continuous functions) corresponds to a topological space (its spectrum). The Fourier Transform is the map that identifies the algebra of functions under convolution ($L^1(G)$) with the algebra of functions under pointwise multiplication ($C_0(\widehat{G})$). Effectively, we trade the geometry of the group for the geometry of its representation theory.

3.2.1 Generalizing the Fourier Transform

We may now combine all we have defined into the following

Definition 3.2.5: L^1 Fourier Transform On Group

Let $f \in L^1(G)$. Then define the *Fourier Transform* of f to be $\widehat{f} : \widehat{G} \to \mathbb{C}$ via

$$\widehat{f}(\chi) = \int_G f(x)\overline{\chi(x)} d\mu(x)$$

This definition is the proper generalization of Fourier transform. Let's say $G = \mathbb{R}$ with the usual Lebesgue measure. Then as we saw in example 3.2.2, χ is the map $\xi \mapsto e^{-2\pi i \xi}$ so that

$$\widehat{f}(\chi) = \int_{\mathbb{R}} f(x)e^{2\pi i \xi} dx$$

3.2.6 Remark the Fourier transform $f \mapsto \widehat{f}$ is the *Galfand Transformation* of f. If we take $A(\widehat{G}) = \left\{ \widehat{f} \;\middle|\; f \in L^1(G) \right\}$, then $A(\widehat{G})$ is a separating sub-algebra of $C_0(\widehat{G})$. It is also not to hard to see that it is self-adjoint, and so by the Stone-Weiestrass theorem it is dense in $C_0(\widehat{G})$.

By our usual Fourier theory, we recover some important properties, namely:

Proposition 3.2.7: Properties Of Abstract Fourier Transform

Let $f, g \in L^1(G)$ for a locally compact abelian group G with Haar measure μ, and let $\chi \in \widehat{G}$. Then, writing $L_a f(x) = f(a^{-1}x)$ and $f^*(x) = \overline{f(x^{-1})}$:

1. $\widehat{f * g} = \widehat{f}\,\widehat{g}$;
2. $\widehat{L_a f}(\chi) = \overline{\chi(a)}\,\widehat{f}(\chi)$ for every $a \in G$;
3. $\overline{\widehat{f}} = \widehat{f^*}$;
4. $\widehat{f}$ is continuous with $\|\widehat{f}\|_\infty \le \|f\|_1$, and it vanishes at infinity (the Riemann-Lebesgue lemma, established in theorem 3.2.9).

***Proof*:**
For (1), Fubini's theorem and the multiplicativity $\overline{\chi(x)} = \overline{\chi(y)}\,\overline{\chi(y^{-1}x)}$ give

$$\widehat{f * g}(\chi) = \int_G \int_G f(y)\, g(y^{-1}x)\, \overline{\chi(x)}\, d\mu(y)\, d\mu(x) = \int_G f(y)\overline{\chi(y)}\Big(\int_G g(y^{-1}x)\overline{\chi(y^{-1}x)}\, d\mu(x)\Big) d\mu(y) = \widehat{f}(\chi)\,\widehat{g}(\chi).$$

For (2), the substitution $x \mapsto ax$ and invariance of μ give $\widehat{L_a f}(\chi) = \int_G f(a^{-1}x)\overline{\chi(x)}\, d\mu(x) = \int_G f(x)\overline{\chi(ax)}\, d\mu(x) = \overline{\chi(a)}\,\widehat{f}(\chi)$. For (3), using $\overline{\chi(x^{-1})} = \chi(x)$ and the invariance of μ under $x \mapsto x^{-1}$ (valid as G is unimodular),

$$\widehat{f^*}(\chi) = \int_G \overline{f(x^{-1})}\,\overline{\chi(x)}\, d\mu(x) = \int_G \overline{f(x)}\,\chi(x)\, d\mu(x) = \overline{\int_G f(x)\,\overline{\chi(x)}\, d\mu(x)} = \overline{\widehat{f}}(\chi).$$

Finally (4): $|\widehat{f}(\chi)| \le \int_G |f|\, d\mu = \|f\|_1$, continuity is dominated convergence, and the vanishing at infinity is the content of theorem 3.2.9.

3.2.2 The Group Algebra and Bochner's Theorem

The Fourier transform of definition 3.2.5 is best read not in isolation but as the Gelfand transform ([13, The Gelfand Transform]) of a commutative Banach algebra attached to G. Making this identification precise is what brings the deeper theorems of the subject - inversion, Plancherel, and duality itself - within reach. The algebra is the group algebra.

Proposition 3.2.8: The Group Algebra

Let G be a locally compact abelian group with Haar measure μ. Equipped with convolution

$$(f * g)(x) = \int_G f(y)\, g(y^{-1}x)\, d\mu(y)$$

and the involution $f^*(x) = \overline{f(x^{-1})}$, the space $L^1(G)$ is a commutative Banach $*$-algebra. It has a unit precisely when G is discrete, and always admits an approximate unit.

***Proof*:**
Tonelli's theorem ([14, Fubini-Tonelli Theorem]) and the invariance of μ give

$$\|f * g\|_1 \leq \int_G \int_G |f(y)|\, |g(y^{-1}x)|\, d\mu(y)\, d\mu(x) = \|f\|_1\, \|g\|_1,$$

so convolution is well defined and submultiplicative; associativity and the identities $\|f^*\|_1 = \|f\|_1$ and $(f * g)^* = g^* * f^*$ are further applications of Fubini and invariance. Commutativity is the substitution $y \mapsto xy^{-1}$, valid because G is abelian and μ invariant. A unit would act as the point mass at e, which lies in L^1 only when $\{e\}$ is open, i.e. when G is discrete; otherwise, for a neighbourhood basis (U_α) at e the normalized indicators $u_\alpha = \mu(U_\alpha)^{-1}\mathbf{1}_{U_\alpha}$ satisfy $u_\alpha * f \to f$ in L^1 by uniform continuity of translation ([14, Left-Uniform Continuity]), an approximate unit.

The characters of this algebra are exactly the characters of the group; this is what fuses the abstract Gelfand machinery with the concrete Fourier transform.

Theorem 3.2.9: The Spectrum of the Group Algebra

Let G be a locally compact abelian group. The assignment

$$\chi \longmapsto \omega_\chi, \qquad \omega_\chi(f) = \int_G f(x)\, \overline{\chi(x)}\, d\mu(x) = \widehat{f}(\chi),$$

is a homeomorphism of the dual group $\widehat{G}$ onto the Gelfand spectrum $\Delta(L^1(G))$. Under it the Gelfand transform of $L^1(G)$ is the Fourier transform $f \mapsto \widehat{f}$; consequently $\widehat{f} \in C_0(\widehat{G})$ with $\|\widehat{f}\|_\infty \leq \|f\|_1$ for every $f \in L^1(G)$, and the image $A(\widehat{G}) = \left\{\widehat{f} \;\middle|\; f \in L^1(G)\right\}$ is a dense $*$-subalgebra of $C_0(\widehat{G})$.

***Proof*:**
Each $\chi \in \widehat{G}$ gives a character: $\omega_\chi(f*g) = \widehat{f * g}(\chi) = \widehat{f}(\chi)\widehat{g}(\chi) = \omega_\chi(f)\omega_\chi(g)$ by proposition 3.2.7, and $\omega_\chi \neq 0$. Conversely a character $\omega \in \Delta(L^1(G))$ is a nonzero element of $L^1(G)^* = L^\infty(G)$ ([13, Characters, Maximal Ideals, and Compactness]), so $\omega(f) = \int_G f\,\overline{\psi}\, d\mu$ for a unique $\psi \in L^\infty(G)$ with $\|\psi\|_\infty \leq 1$. Writing $\omega(f * g) = \omega(f)\omega(g)$ as a double integral and comparing kernels forces $\psi(xy) = \psi(x)\psi(y)$ for almost every (x, y); convolving ψ with an approximate unit yields a continuous representative satisfying the identity everywhere, so ψ agrees a.e. with a continuous homomorphism $\chi\colon G \to \mathbb{C}^\times$. From $\|\psi\|_\infty \leq 1$ and $\chi(x)\chi(x^{-1}) = 1$ we get $|\chi| \equiv 1$, so $\chi \in \widehat{G}$

and $\omega = \omega_\chi$. A comparison of subbasic neighbourhoods matches the compact-open topology on $\widehat{G}$ with the weak-$*$ topology on $\Delta(L^1(G))$. Transporting the Gelfand transform of [13, The Gelfand Transform] through this homeomorphism turns $f \mapsto (\omega \mapsto \omega(f))$ into $f \mapsto \widehat{f}$, giving $\widehat{f} \in C_0(\widehat{G})$ and the norm bound; the range is a subalgebra by proposition 3.2.7, separates points, and satisfies $\overline{\widehat{f}} = \widehat{f^*}$, so it is dense in $C_0(\widehat{G})$ by the Stone-Weierstrass theorem ([6]).

In particular this is the rigorous form of the Riemann-Lebesgue lemma of proposition 3.2.7: $\widehat{f}$ vanishes at infinity because it is a Gelfand transform. We isolate next the class of functions that parametrizes the inversion theorem.

Definition 3.2.10: Positive-Definite Function

A function $\varphi\colon G \to \mathbb{C}$ is *positive-definite* if for every finite set $x_1, \dots, x_n \in G$ and every $c_1, \dots, c_n \in \mathbb{C}$,

$$\sum_{i,j=1}^{n} c_i\, \overline{c_j}\, \varphi(x_j^{-1} x_i) \geq 0.$$

The case $n = 1$ gives $\varphi(e) \geq 0$, and $n = 2$ gives $\varphi(x^{-1}) = \overline{\varphi(x)}$ and $|\varphi(x)| \leq \varphi(e)$, so a positive-definite function is bounded with $\|\varphi\|_\infty = \varphi(e)$. The basic examples are the Fourier-Stieltjes transforms $\check{\nu}(x) = \int_{\widehat{G}} \chi(x)\, d\nu(\chi)$ of finite positive measures ν on $\widehat{G}$: the defining sum is $\int_{\widehat{G}} |\sum_i c_i\, \chi(x_i)|^2\, d\nu(\chi) \geq 0$. Bochner's theorem is the converse.

Theorem 3.2.11: Bochner's Theorem

Let G be a locally compact abelian group. A continuous function $\varphi\colon G \to \mathbb{C}$ is positive-definite if and only if there is a unique finite positive Radon measure ν on $\widehat{G}$ with

$$\varphi(x) = \int_{\widehat{G}} \chi(x)\, d\nu(\chi) \qquad (x \in G).$$

***Proof*:**
The Fourier-Stieltjes transform of a finite positive measure is continuous and positive-definite by the remark above, and any representing ν then has total mass $\nu(\widehat{G}) = \check{\nu}(e) = \varphi(e)$; this is the "if" direction. For the converse, integrating definition 3.2.10 against $f \in C_c(G)$ in each variable gives the continuous Hermitian form

$$Q(f) = \int_G \int_G \varphi(y^{-1}x)\, f(x)\, \overline{f(y)}\, d\mu(x)\, d\mu(y) \geq 0. \qquad (*)$$

Cauchy-Schwarz for Q, applied against an approximate unit and iterated through the spectral-radius identity $\|\widehat{f}\|_\infty = r(f)$ of [13, The Gelfand Transform], yields the estimate

$$\left| \int_G f\, \overline{\varphi}\, d\mu \right| \leq \varphi(e)\, \|\widehat{f}\|_\infty \qquad (f \in L^1(G)). \qquad (**)$$

By $(**)$ the number $\int_G f\, \overline{\varphi}\, d\mu$ depends only on $\widehat{f}$ and is dominated by its sup-norm, so

$$L(\widehat{f}) := \int_G f(x)\, \overline{\varphi(x)}\, d\mu(x)$$

is a well-defined bounded functional on the dense subalgebra $A(\widehat{G}) \subseteq C_0(\widehat{G})$ (theorem 3.2.9), hence extends to $C_0(\widehat{G})$. It is positive: for $g \in L^1(G)$,

$$L(|\widehat{g}|^2) = L(\widehat{g^* * g}) = \int_G (g^* * g)\, \overline{\varphi}\, d\mu = Q(\overline{g}) \geq 0,$$

the last equality by expanding $g^* * g$ and substituting, and $Q(\overline{g}) \geq 0$ by $(*)$; such squares are dense in the nonnegative cone of $C_0(\widehat{G})$. The Riesz-Markov theorem ([14, Riesz-Markov Representation]) represents L as $L(h) = \int_{\widehat{G}} h\, d\nu$ for a finite positive Radon measure ν. Unwinding, for every $f \in L^1(G)$,

$$\int_G f\, \overline{\varphi}\, d\mu = \int_{\widehat{G}} \widehat{f}\, d\nu = \int_G f(x)\, \overline{\Big(\int_{\widehat{G}} \chi(x)\, d\nu(\chi)\Big)}\, d\mu(x),$$

so φ and the continuous function $\check{\nu}$ pair identically against all of $L^1(G)$ and therefore coincide. Uniqueness of ν is the density of $A(\widehat{G})$ in $C_0(\widehat{G})$.

Bochner already forces the dual group to be rich enough to see every point of G.

Corollary 3.2.12: Characters Separate Points

Let G be a locally compact abelian group. For every $x \in G$ with $x \neq e$ there is a character $\chi \in \widehat{G}$ with $\chi(x) \neq 1$.

***Proof*:**
Since G is Hausdorff there is an open neighbourhood U of e with $Ux \cap U = \emptyset$. Choose $h \in C_c(G)$, $h \neq 0$, supported in U, and set $\varphi = h^* * h$, a continuous positive-definite function with $\varphi(e) = \|h\|_2^2 > 0$. Substituting $z = y^{-1}$,

$$\varphi(x) = \int_G \overline{h(z)}\, h(zx)\, d\mu(z) = 0,$$

since $h(z)h(zx) = 0$ pointwise ($z \in U$ and $zx \in U$ are incompatible). By Bochner (theorem 3.2.11) $\varphi = \check{\nu}$ for a positive measure ν, so $\int_{\widehat{G}} \chi(x)\, d\nu(\chi) = \varphi(x) = 0 \neq \varphi(e) = \nu(\widehat{G}) = \int_{\widehat{G}} 1\, d\nu(\chi)$. Were $\chi(x) = 1$ throughout the support of ν the two integrals would agree; hence $\chi(x) \neq 1$ for some $\chi \in \widehat{G}$.

3.2.3 Fourier Inversion and Plancherel

One of the most important results in Fourier Analysis is the ability to recover f from $\widehat{f}$ under suitable conditions, that is the *Fourier Inversion Theorem*. This result naturally generalizes. Note that since $f \in L^1(G)$, we may treat f as being define μ-a.e., and so we may only recover f μ-a.e. in the following theorem unless f has more regularity:

Theorem 3.2.13: Fourier Inversion Theorem For Groups

Let G be a locally compact abelian group with Haar measure μ. Then there is a unique Haar measure $\widehat{\mu}$ on $\widehat{G}$ such that whenever $f \in L^1(G)$ and $\widehat{f} \in L^1(\widehat{G})$, then μ-a.e. we have:

$$f(x) = \int_{\widehat{G}} \widehat{f}(\chi)\chi(x)\, d\widehat{\mu}(\chi)$$

If f is continuous, then it is true for all $x \in G$.

***Proof*:**
Write $P^1(G)$ for the integrable continuous positive-definite functions and $\mathcal{B}(G)$ for their linear span, a dense subspace of $L^1(G)$ containing every $g^* * g$ with $g \in C_c(G)$. Bochner's theorem (theorem 3.2.11) attaches to each $u \in P^1(G)$ a finite positive measure ν_u on $\widehat{G}$ with $u = \check{\nu}_u$, and the closing step of its proof records

$$\int_{\widehat{G}} \widehat{k}\, d\nu_u = \int_G k\,\overline{u}\, d\mu \qquad (k \in L^1(G),\ u \in P^1(G)). \tag{†}$$

A common Haar measure. For $u, v \in P^1(G)$ the measures $\widehat{u}\, d\nu_v$ and $\widehat{v}\, d\nu_u$ coincide. Testing each against $\widehat{w}$ ($w \in L^1(G)$; recall $A(\widehat{G})$ is dense in $C_0(\widehat{G})$) and applying (†) with $k = w * u$ and with $k = w * v$,

$$\int_{\widehat{G}} \widehat{w}\,\widehat{u}\, d\nu_v = \int_G (w * u)\,\overline{v}\, d\mu = \langle w * u, v\rangle, \qquad \int_{\widehat{G}} \widehat{w}\,\widehat{v}\, d\nu_u = \langle w * v, u\rangle,$$

and the right-hand sides agree because a positive-definite function is self-adjoint ($u^* = u$, $v^* = v$, from definition 3.2.10) and $L^1(G)$ is commutative:

$$\langle w * u, v\rangle = \langle w, v * u^*\rangle = \langle w, u * v^*\rangle = \langle w * v, u\rangle.$$

For each $\chi \in \widehat{G}$ some $u \in P^1(G)$ has $\widehat{u}(\chi) > 0$ (as $A(\widehat{G})$ is dense in $C_0(\widehat{G})$, some $\widehat{g}$ is nonzero at χ, and $u = g^* * g$ has $\widehat{u}(\chi) = |\widehat{g}(\chi)|^2$). Hence the prescription $d\widehat{\mu} = d\nu_u/\widehat{u}$ on $\{\widehat{u} > 0\}$ is unambiguous and defines a positive Radon measure $\widehat{\mu}$ on $\widehat{G}$ with

$$d\nu_u = \widehat{u}\, d\widehat{\mu} \qquad (u \in P^1(G)).$$

Multiplying u by a character χ_0 keeps it positive-definite and translates both $\widehat{u}$ and ν_u by χ_0, so $\widehat{\mu}$ is translation-invariant: a Haar measure on $\widehat{G}$, unique up to the scale this construction fixes ([14, Uniqueness of Haar Measure]).

Inversion. For $u \in P^1(G)$, substituting $d\nu_u = \widehat{u}\, d\widehat{\mu}$ into $u = \check{\nu}_u$ gives $u(x) = \int_{\widehat{G}} \widehat{u}(\chi)\,\chi(x)\, d\widehat{\mu}(\chi)$. More generally, for any $f \in L^1(G)$ and $u \in P^1(G)$, writing $u = \check{\nu}_u$ and applying Fubini,

$$(f * u)(x) = \int_{\widehat{G}} \widehat{f}(\chi)\,\chi(x)\, d\nu_u(\chi) = \int_{\widehat{G}} \widehat{f * u}(\chi)\,\chi(x)\, d\widehat{\mu}(\chi),$$

so inversion holds for every convolution $f * u$. If now $\widehat{f} \in L^1(\widehat{G})$, take an approximate unit $u_\alpha = e_\alpha^* * e_\alpha \in P^1(G)$ ($e_\alpha \in C_c(G)$): then $f * u_\alpha \to f$ in $L^1(G)$ while $\widehat{f * u_\alpha} = \widehat{f}\,\widehat{u_\alpha} \to \widehat{f}$ in $L^1(\widehat{\mu})$, and passing to the limit gives $f(x) = \int_{\widehat{G}} \widehat{f}(\chi)\,\chi(x)\, d\widehat{\mu}(\chi)$ for μ-a.e. x, and everywhere when f is continuous.

The construction yields a fact used repeatedly below.

Corollary 3.2.14: Injectivity of the Fourier Transform

If $f \in L^1(G)$ has $\widehat{f} \equiv 0$, then $f = 0$.

***Proof*:**

For every $u \in P^1(G)$ the convolution $f * u \in L^1(G)$ has $\widehat{f * u} = \widehat{f}\,\widehat{u} \equiv 0 \in L^1(\widehat{G})$, hence $f * u = 0$ by inversion (theorem 3.2.13). Taking $u = u_\alpha = e_\alpha^* * e_\alpha$ an approximate unit gives $f = \lim_\alpha f * u_\alpha = 0$.

Another very important result is Fourier Analysis on L^2;

Theorem 3.2.15: Plancherel Inversion Theorem For Groups

Let G be a locally compact abelian group, μ a Haar measure on G, and $\widehat{\mu}$ the dual measure on $\widehat{G}$ given in theorem 3.2.13. If $f\colon G \to \mathbb{C}$ is continuous with compact support, then $\widehat{f} \in L^2(\widehat{G})$ and

$$\int_G |f(x)|^2 d\mu(x) = \int_{\widehat{G}} \left|\widehat{f}(\chi)\right|^2 d\widehat{\mu}(\chi)$$

In particular, the Fourier transform is an L^2_μ isometry from $C_c(G)$ to $L^2_{\widehat{\mu}}(\widehat{G})$.

***Proof*:**

For $f \in C_c(G)$ the function $f^* * f$ is continuous, integrable, and positive-definite, with transform $\widehat{f^* * f} = |\widehat{f}|^2$ and value $(f^* * f)(e) = \|f\|_2^2$ at the identity, so it lies in $\mathcal{B}(G)$. Evaluating the inversion theorem (theorem 3.2.13) at $x = e$ therefore gives

$$\|f\|_2^2 = (f^* * f)(e) = \int_{\widehat{G}} |\widehat{f}(\chi)|^2 \, d\widehat{\mu}(\chi) = \|\widehat{f}\|_2^2.$$

Thus $f \mapsto \widehat{f}$ is a linear isometry of $(C_c(G), \|\cdot\|_2)$ into $L^2(\widehat{G})$, which (as $C_c(G)$ is L^2-dense) extends uniquely to an isometry $\mathcal{F}\colon L^2(G) \to L^2(\widehat{G})$; it is onto, hence unitary, as shown next.

That the isometry

$$\mathcal{F}\colon L^2_\mu(G) \to L^2_{\widehat{\mu}}(\widehat{G})$$

is *onto* – a *unitary operator* – follows from the module structure of its range. Let $R = \mathcal{F}(L^2(G))$, closed because $\mathcal{F}$ is an isometry of a complete space. For $g \in L^1(G)$ and $\xi \in L^2(G)$, Young's inequality gives $g * \xi \in L^2(G)$ and $\mathcal{F}(g * \xi) = \widehat{g}\,\mathcal{F}\xi$, so $\widehat{g}\,R \subseteq R$; since $A(\widehat{G}) = \left\{\widehat{g} \;\middle|\; g \in L^1(G)\right\}$ is dense in $C_0(\widehat{G})$ (theorem 3.2.9), R is invariant under multiplication by all of $C_0(\widehat{G})$. A closed subspace of $L^2(\widehat{\mu})$ stable under multiplication by $C_0(\widehat{G})$ has the form $R = \mathbf{1}_E\, L^2(\widehat{\mu})$ for some measurable $E \subseteq \widehat{G}$: for $\psi \in R$ the products $\varphi\psi$ ($\varphi \in C_0(\widehat{G})$) are L^2-dense in $\mathbf{1}_{\{\psi \neq 0\}} L^2(\widehat{\mu})$, because $C_0(\widehat{G})$ is dense in $L^2(|\psi|^2\, d\widehat{\mu})$, a finite Radon measure. Every $\widehat{f} = \mathcal{F}f$ ($f \in L^1 \cap L^2$) lies in R, hence vanishes a.e. off E; were $\widehat{\mu}(\widehat{G} \setminus E) > 0$, a point χ_0 in the support of $\mathbf{1}_{\widehat{G}\setminus E}\widehat{\mu}$ would admit $f \in L^1 \cap L^2(G)$ with $\widehat{f}(\chi_0) \neq 0$ (take $f = f_0 * u$ with $\widehat{f_0}(\chi_0) \neq 0$ from theorem 3.2.9 and $u \in P^1(G)$ with $\widehat{u}(\chi_0) > 0$), so $\widehat{f} \neq 0$ on a neighbourhood of χ_0 meeting $\widehat{G}\setminus E$ in positive measure, a contradiction. Hence $E=\widehat{G}$ up to a null set, $R=L^2(\widehat{G})$, and $\mathcal{F}$ is unitary. When G is compact, $L^2(G)\subseteq L^1(G)$, so the L^1 transform already covers L^2.

The adjoint of this unitary is the inverse Fourier transform.

Theorem 3.2.16: Dual Group Fourier Transform

Let G be a locally compact abelian group with Haar measure μ and let $\mathcal{F}$ be the L^2 Fourier transform. Then the adjoint of the Fourier Transform restricted to continuous functions on compact support is the Inverse Fourier Transform

$$L^2_{\widehat{\mu}}(\widehat{G}) \to L^2_{\mu}(G)$$

***Proof*:**
For $f\in C_c(G)$ and $g\in C_c(\widehat{G})$, Fubini's theorem ([14, Fubini-Tonelli Theorem]) gives

$$\langle \mathcal{F}f, g\rangle_{\widehat{G}} = \int_{\widehat{G}} \widehat{f}(\chi)\,\overline{g(\chi)}\,d\widehat{\mu} = \int_G f(x)\,\overline{\int_{\widehat{G}} g(\chi)\,\chi(x)\,d\widehat{\mu}(\chi)}\,d\mu(x) = \langle f, \check{g}\rangle_G,$$

so $\mathcal{F}^*g(x)=\int_{\widehat{G}} g(\chi)\,\chi(x)\,d\widehat{\mu}(\chi)$ is the inverse Fourier transform of theorem 3.2.13, recognized as the adjoint of $\mathcal{F}$. Since $\mathcal{F}$ is unitary (theorem 3.2.15), this adjoint $\mathcal{F}^*=\mathcal{F}^{-1}$ is its two-sided inverse.

Notable example would be when $G=\mathbb{T}$ so that $\widehat{G}=\mathbb{Z}$ and the Fourier transform computes the Fourier series of periodic functions. If G is finite, then we will get the *discrete Fourier Transform* from this process.

3.2.4 Pontryagin Duality

The machinery now assembled proves theorem 3.2.3 in full: the canonical map $\alpha\colon G\to\widehat{\widehat{G}}$, $\alpha(x)(\chi)=\chi(x)$, is an isomorphism of topological groups.

***Proof*:**
That α is a continuous homomorphism is immediate: $\alpha(x)$ is the evaluation $\chi\mapsto\chi(x)$, a character of $\widehat{G}$, and $x\mapsto\alpha(x)$ is continuous for the compact-open topologies.

Injectivity. If $\alpha(x)=1$ then $\chi(x)=1$ for every $\chi\in\widehat{G}$, so $x=e$ by corollary 3.2.12. Hence α is injective.

Embedding. A comparison of subbasic neighbourhoods shows α is a homeomorphism onto its image, and since G is locally compact that image $\alpha(G)$ is a closed subgroup of $\widehat{\widehat{G}}$.

Surjectivity. The Plancherel theorem (theorem 3.2.15) makes both $\mathcal{F}_G\colon L^2(G)\to L^2(\widehat{G})$ and $\mathcal{F}_{\widehat{G}}\colon L^2(\widehat{G})\to L^2(\widehat{\widehat{G}})$ unitary. Pull the transform of $\psi\in L^2(\widehat{G})$ back along α: since $\alpha(x)(\chi)=$

$\chi(x)$,

$$(\mathcal{F}_{\widehat{G}}\psi)\big(\alpha(x)\big) = \int_{\widehat{G}} \psi(\chi)\,\overline{\chi(x)}\,d\widehat{\mu}(\chi) = \overline{\mathcal{F}^*_{G}(\overline{\psi})(x)},$$

the inverse transform of theorem 3.2.16. As $\mathcal{F}^*_G = \mathcal{F}^{-1}_G$ is unitary, $\left\|(\mathcal{F}_{\widehat{G}}\psi)\circ\alpha\right\|_{L^2(\mu)} = \|\psi\|_{L^2(\widehat{\mu})} = \|\mathcal{F}_{\widehat{G}}\psi\|_{L^2(\widehat{\widehat{\mu}})}$. The left-hand side is $\int_{\widehat{\widehat{G}}} |\mathcal{F}_{\widehat{G}}\psi|^2\,d(\alpha_*\mu)$, and as ψ runs through $L^2(\widehat{G})$ the transform $\mathcal{F}_{\widehat{G}}\psi$ runs through all of $L^2(\widehat{\widehat{\mu}})$; hence

$$\int_{\widehat{\widehat{G}}} |F|^2\,d(\alpha_*\mu) = \int_{\widehat{\widehat{G}}} |F|^2\,d\widehat{\widehat{\mu}} \qquad (F \in L^2(\widehat{\widehat{\mu}})),$$

so $\alpha_*\mu = \widehat{\widehat{\mu}}$. The pushforward $\alpha_*\mu$ is carried by the closed subgroup $\alpha(G)$, whereas the Haar measure $\widehat{\widehat{\mu}}$ charges every nonempty open set; thus $\widehat{\widehat{G}} \setminus \alpha(G)$ is open and null, hence empty. Therefore $\alpha(G) = \widehat{\widehat{G}}$, and α is a topological isomorphism.

3.2.5 Interesting Uses

Pontryagin has a couple of very interesting consequences, perhaps a good starting point is that the classification of compact abelian groups is just as difficult as the classification of general abelian groups using the following result:

Theorem 3.2.17: Compactness Criterion For LCA Group

Let G be a locally compact abelian group. Then G is compact if and only if $\widehat{G}$ is discrete. Conversely, G is discrete if and only if $\widehat{G}$ is compact.

***Proof*:**
The fact that G is compact implies $\widehat{G}$ is discrete and that G is discrete implies $\widehat{G}$ is compact comes straight from the topology of G and $\widehat{G}$ (namely that it is the compact-open topology). To get the converse of both, we simply apply Pontryagin duality.

Proposition 3.2.18: Unit In $L^1(G)$

Let G be a locally compact abelian group with Haar measure μ. Then $L^1(G)$ has a unit if and only if G is discrete.

***Proof*:**
We know that $L^1(G)$ always has an approximate identity (think of good kernels). The identity of $L^1(G)$ would be an element such that $f * \mathrm{id} = \mathrm{id} * f = f$, so $\widehat{f}\cdot\widehat{\mathrm{id}} = \widehat{f}$. We see through this formula that $\widehat{\mathrm{id}}$ would have to be the Dirac-delta distribution at 1 with norm 1. This distribution is a function if and only if G is discrete.

Representation Theory

Another place this comes in is within representation theory. Recall in [12, chapter 24.4] there was a quick word on Fourier transforms in representation theory. In classical representation theory of finite groups, we define the character to be $\chi : G \to \mathbb{C}^*$. Since G is finite, G maps to a root the roots of unity in $\mathbb{T}$, and hence the notion of character in classical representation theory aligns with our definitions. Then we may define the inner product of characters to be:

$$\langle \chi_1, \chi_2 \rangle = \frac{1}{|G|} \sum_{g \in G} \chi_1(g)\overline{\chi_2}(g)$$

which will give a way to decompose representations into irreducible components via irreducible characters. If G is abelian, then may translate the decomposition theorem for representations of finite abelian groups to the decomposition of$L^2(G)$ via

$$L^2(G) = \bigoplus_{i=1}^{n} E_n$$

and in the infinite case we would have take the closure

$$L^2(G) = \widehat{\bigoplus_{i \in \mathbb{N}} E_n}$$

where E_n are the subspaces given by the usual orthonormal basis for L^2 when doing Fourier analysis. Hence, representation theory and Fourier analysis are linked!!

Class Field Theory

For those how have seen some class field theory, they have seen the Artin Reciprocity map. Recall that the idele group is a locally compact group, and so it is precisely the context in which we expect Pontryagian duality to be applied! In the local case, we have the Artin map $K^* \cong \mathrm{Gal}_K(K^{\mathrm{ab}})$ between two locally compact groups; taking the Pontryagin dual will preserve this mapping and let us work with the characters of these groups! In the global case using the Adèle ring and Idele group, we shall not get an isomorphism in the Pontryagin dual but a perfect pairing.

4

What Next?

The organizing fact of this book was that the Fourier transform diagonalizes translation-invariant operators, and that the group being translated can be replaced. Both halves of that sentence lead somewhere.

Dropping commutativity

Part II needed G abelian at exactly one point, and everything rests on it: only for an abelian group are the irreducible unitary representations one-dimensional, which is what makes a character a function and $\hat{G}$ a group. Give that up and the characters become matrix-valued, $\hat{G}$ becomes a space with no group law, and inversion has to be rebuilt as a direct integral. *Unitary Representation Theory* [10] is that rebuilding, with $L^1(G)$ as a Banach algebra, the group C^*-algebra, Gelfand-Raikov, the SNAG theorem and the abstract Plancherel theorem, and it treats this book's Part II as the model case it generalizes. For compact groups the answer is clean and classical: Peter-Weyl, proved in *Lie Groups* [15]. Folland's *A Course in Abstract Harmonic Analysis* is the standard reference for the whole passage.

Equations

The singular integrals and Sobolev estimates of Part I exist to be used. *Linear Elliptic and Parabolic PDE* [4] uses them directly: L^p regularity theory is Calderón-Zygmund applied to the Newtonian potential, and the spectral theory of elliptic operators is where the H^s scale becomes a statement about eigenvalue asymptotics. On the nonlinear side, *Nonlinear Wave and Dispersive PDE* [8] runs on Littlewood-Paley theory and Strichartz estimates, both of which are frequency decompositions of the kind built here. *Microlocal Analysis* [7] refines the decomposition to localize in space and frequency at once, which is what a variable-coefficient operator requires. Evans's *Partial Differential Equations* and Tao's *Nonlinear Dispersive Equations* are the standard companions.

Number theory

Poisson summation relates a sum over a lattice to a sum over its dual, and it is the mechanism behind the functional equation of the Riemann zeta function: the theta function is its own transform, and ζ inherits the symmetry. *Analytic Number Theory* [1] develops that circle of ideas, with Dirichlet series, the explicit formula, and the prime number theorem as a statement about the zeros of a transform. Beyond it, the same machinery over a p-adic or adelic group is where the Langlands program begins. *Local Langlands for* $GL_2(F)$ [5] is the series' entry point. Iwaniec and Kowalski's *Analytic Number Theory* is the reference.

The geometry of the frequency side

The treatment of the Kakeya problem in Part I is a sample of a live research area in which the Fourier transform is used to measure sets rather than to solve equations: restriction and extension estimates, Bochner-Riesz summation, and the decoupling theorems that have recently settled long-standing conjectures. The measure-theoretic side, meaning Hausdorff dimension, Besicovitch sets and Frostman's lemma, is *Real Analysis* [14] and *Geometric Measure Theory* [3]. Wolff's *Lectures on Harmonic Analysis* is the classic short introduction to this frontier.

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